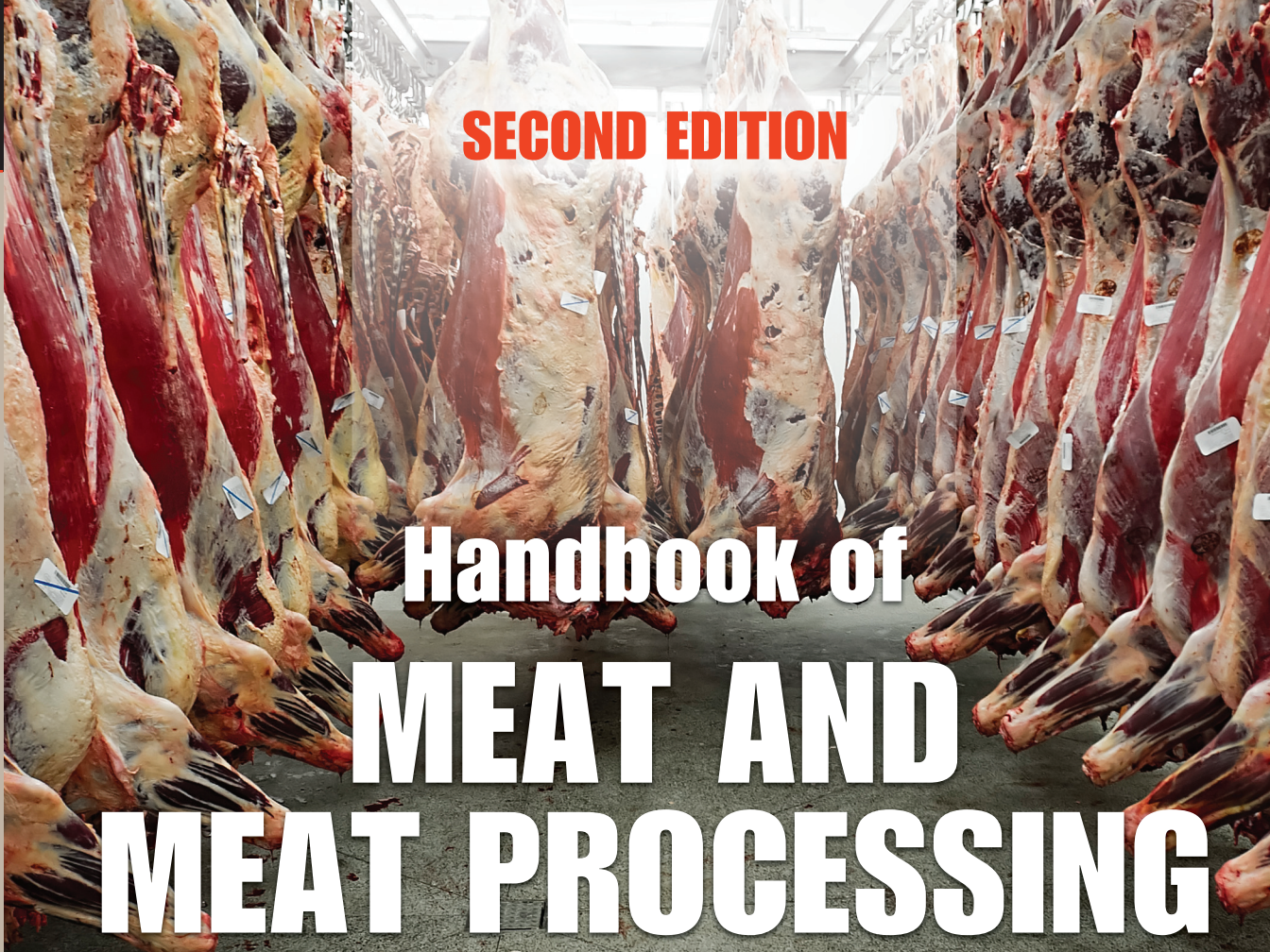




SECOND EDITION

Handbook of MEAT AND MEAT PROCESSING

Edited by Y. H. Hui, PhD

Associate Editors

J. L. Aalhus, PhD

L. Cocolin, PhD

I. Guerrero-Legarreta, PhD

L. M. Nollet, PhD

R. W. Purchas, PhD

M. W. Schilling, PhD

P. Stanfield, MS, RD

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Preface

The consumption of red meat and meat products has a long history in most cultures. Meat is a source of nutrients, as well as a sign of wealth in some countries. Various techniques have been developed in different parts of the world over the centuries to preserve meat for extended shelf life and enjoyment. Even nonedible parts of animals are used for various reasons. Thus, meat, meat products, and by-products are important to our daily life.

In the past three decades, many books on the science and technology of meats and meat products have been published. Many of these books are useful reference texts and well received by professionals in the meat industries, academia, and the government. *Meat Science and Applications* is one such example. It was published in 2001 by Marcel Dekker of New York. A decade later, CRC Press is publishing a second edition of the book with a new title, *Handbook of Meat and Meat Processing*. The change in title reflects the expansion of coverage in depth and breadth from the first edition, as illustrated in the table of contents of this book. Apart from updating materials in the first edition, the new edition contains 51 chapters instead of 27.

This second edition is divided into seven parts. Part I covers an overview of the meat-processing industries in the United States. Part II discusses the basic science of meat, including chapters on muscle biology, meat composition, and chemistry. Part III covers meat attributes and characteristics such as color, flavor, and analyses. Part IV describes the primary processing of meat, including antemortem and post-mortem slaughter, carcass evaluation, religious status, and so on. Part V discusses the principles and applications in the secondary processing of meat, for example, breading, curing, and fermentation. Part VI focuses on the manufacture of some processed meat products that include sausages, ham, and others. Part VII is devoted entirely to the safety of meat products and meat workers in the United States.

Of the new chapters, half is distributed among Parts I through VI. The remaining is located in Part VII. Apart from new developments in science and technology, the materials in Part VII should serve as excellent frames of reference for countries exporting meat to the United States. This book is unique in several aspects: it is an updated comprehensive source, it contains topics not covered in similar books, and its editors and contributors include experts from government, industry, and academia worldwide.

The editorial team is made up of eight associate editors from six countries. Each has unique contributions in the planning and execution of this enormous project, as illustrated by each person's professional status:

- Dr. Aalhus is a meat scientist with the Lacombe Research Centre, Alberta, Canada. She is an expert on meat tenderness, as documented by her numerous professional publications.
- Dr. Cocolin is the editor-in-chief of *International Journal of Food Microbiology*, with expertise in food fermentation. He is with the University of Turin, Italy.
- Dr. Guerrero, an established scientist in meat and poultry, is also a chemical engineer. She is a distinguished professor at the Metropolitan University Iztapalapa Campus, Mexico City. She is the editor of the two-volume texts *Handbook of Poultry Science and Technology*, Wiley, 2010.
- Dr. Nollet teaches at the Katholieke Universiteit Leuven, Belgium. He is the editor-in-chief of two books on meat quality and analysis. His research interest is reflected by the more than 20 professional reference texts on food and water analyses that he has edited.
- Dr. Purchas of Massey University, New Zealand, is the only academician in the team with extensive research activities in both animal science and meat science, with special interest in the characteristics of meat quality.
- Ms. Stanfield is a consultant to the health industry and has authored five books in allied health and many chapters in professional books in food science, technology, and engineering.

- Dr. Xiong of the University of Kentucky is a meat scientist. He is an elected IFT fellow and an associate editor for several food science/meat science journals.
- Dr. Schilling of Mississippi State University has expertise in meat science, sensory science, and food chemistry, as documented by his extensive professional publications. He is the recipient of the Achievement Award by the American Meat Science Association.

This book is a result of the combined efforts of nearly 100 highly qualified individuals from 16 countries: Belgium, Canada, China, Egypt, France, Germany, Hungary, Italy, Mauritius, Mexico, New Zealand, Singapore, Spain, Taiwan, Turkey, and the United States. Each contributor or editor was responsible for researching and reviewing subjects of immense depth, breadth, and complexity. Care and attention were paramount to ensure technical accuracy for each topic. This volume is unique as stated earlier, and it is our sincere hope and belief that it will serve as an essential reference on meat and meat products.

We wish to thank all the contributors for sharing their expertise throughout our journey. We also thank the reviewers for giving their valuable comments, leading to improvements in the content of each chapter.

The production of a volume of this size could not have been accomplished without the hard work and excellent suggestions of professionals in the production team assigned by CRC Press to manage the project. You are the best judge of the quality of their work.

Y. H. Hui

Editor

Y. H. Hui received his PhD from the University of California at Berkeley. He is currently a senior scientist with the consulting firm of Science Technology System at West Sacramento, California. He has authored, coauthored, edited, and coedited more than 35 books in food science, food technology, food engineering, and food laws, including *Handbook of Food Product Manufacturing* (2 volumes, Wiley, 2007), and *Handbook of Food Science, Technology, and Engineering* (4 volumes, CRC Press, 2006). He is currently a consultant for the food industries.

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Leo M. L. Nollet received his MS (1973) and PhD (1978) in biology from the Katholieke Universiteit Leuven, Belgium. Dr. Nollet is the editor and associate editor of numerous books. He edited for Marcel

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Part I

Meat Industries

1

*Meat Industries: Characteristics and Manufacturing Processes**

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1.1 Introduction

This chapter will give you an overview of the processes used and the products produced by the industries that we regulate. Each plant is unique, and the production processes used by plants have their own

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details. There are a wide variety of products produced and a number of different activities conducted by these establishments. The industry as a whole is dynamic, in that over time, production of products that are not favored by consumers are decreased or discontinued, and new products are created to meet consumer needs.

In the United States, the Food Safety Inspection Service (FSIS), a unit of the Department of Agriculture (USDA), is responsible for the safety of meat and meat produces. In accordance with the following, please note that, for this chapter:

1. The text and tables have been collated from multiple USDA public documents (legal, regulatory, and training), distributed in USDA'S website, <http://www.FSIS.USDA.gov>.
2. The figures were originally adapted from a series of USDA continuing educational documents divided into 3 categories:
 - a. *USDA HACCP Generic Models on Raw Ground and Not Ground Meat Products.*
 - b. *USDA HACCP Generic Models on Shelf-Stable Processed Meat Products.*
 - c. *USDA HACCP Generic Models on Not Shelf-Stable Processed Meat Products.*

They are available at USDA's website, www.FSIS.USDA.gov. It must be emphasized that, in this chapter:

1. All information has been designed by the USDA for training purposes only and is not meant to detail the actual manufacturing processes.
2. All abbreviations are standard ones used in law, science, and technology as issued by FSIS.
3. All terms with legal implications apply only in the United States.

We have organized these materials by what the FSIS calls *processing categories*. The nine different processing categories that we will cover include the following:

- Slaughter
- Raw product—not ground
- Raw product—ground
- Heat treated but not fully cooked—not shelf stable
- Fully cooked—not shelf stable
- Product with secondary inhibitors—not shelf stable
- Not heat treated—shelf stable
- Heat treated—shelf stable
- Thermally processed—commercially sterile

There are some processes and technologies that are not specifically addressed by a processing category in the Hazard Analysis and Critical Control Points (HACCP) regulations as listed below:

- Mechanically Separated and Advanced Meat Recovery
- Irradiation
- Egg Products

The Agency developed and distributed generic HACCP models that addressed most of these processing categories. We will be referring to some examples given in these models as we discuss the processing categories.

Every product produced by an establishment (when the hazard analysis reveals any food safety hazard that is likely to occur) must be produced according to a written *HACCP* plan. Many different products

may be grouped within a single processing category, as long as the food safety hazards, critical control points, and critical limits are essentially the same.

In this chapter, we will discuss both quality and safety issues. Both of these issues are important to both the agency and the industry. There are many quality issues, sometimes referred to as *nonfood safety consumer protection*, which would render the product adulterated. Some examples are products with low net weights or with water added above the allowed limits. Safety or *public health protection* issues are given an extremely high priority because of the potential to cause food-borne disease outbreaks. The most common hazard to public health is the presence of harmful bacteria. Throughout this module we will point out processes where quality or safety issues are important.

1.1.1 Definition of Terms

Before we go further, let us define a few terms:

- *Amenable*—accountable or liable to an order or regulation.
- *Product*—any carcass, meat, meat by-product, or meat food product, poultry, or poultry food product capable of use as human food.
- *Process*—is a procedure consisting of any number of separate, distinct, and ordered operations that are directly under the control of the establishment employed in the manufacture of a specific product, or a group of two or more products.
- *Process flow diagram*—process flow diagrams provide a simple description of the steps involved in the process. The flow diagram covers all steps in the process which are directly under the control of an establishment. It may also include steps in the food chain which are before and after the processing occurs.

We will discuss each of the nine categories of meat processing and meat products. However, one or two of the illustrations will be reused in another chapter discussing a specific category of meat product under the section on meat safety. This approach facilitates reading in those chapters, avoiding frequent referral to a specific illustration in this chapter.

1.2 Slaughter

Slaughter is the process whereby healthy, live animals are humanely stunned, bled, deheaded, dehaired and/or defeathered, and eviscerated. The resulting carcass may be split and/or fabricated in some fashion. During the process, inedible waste and products (e.g., products not used for human food such as the hides) are produced. Edible by-products (e.g., livers and gizzards) are also produced. The establishment must keep inedible materials separate from edible ones. We will cover by-products when discussing another processing category.

Slaughter includes all amenable red meat species, and all poultry classes. Some examples are beef and pork carcasses, and ready-to-cook whole chickens and turkeys. Some of the products, such as whole poultry, will be distributed for sale following the slaughter process. However, most products go for further processing.

1.2.1 Beef Slaughter Process

Look at the process flow diagram (Figure 1.1) for the beef slaughter process. They are only *examples* of the slaughter process. These processes will vary from plant to plant. The examples we use are more typical of large plants. Large plants (large volume/many employees) are typically highly mechanized and may process thousands of carcasses daily. Smaller plants follow many of the same steps, but with fewer employees and less automatic equipment.

Beef slaughter covers all market classes of cattle. Class is determined based on maturity and sex of the animal at the time of slaughter. The *classes* of beef carcasses are calves, steers, bulls, heifers, and cows.

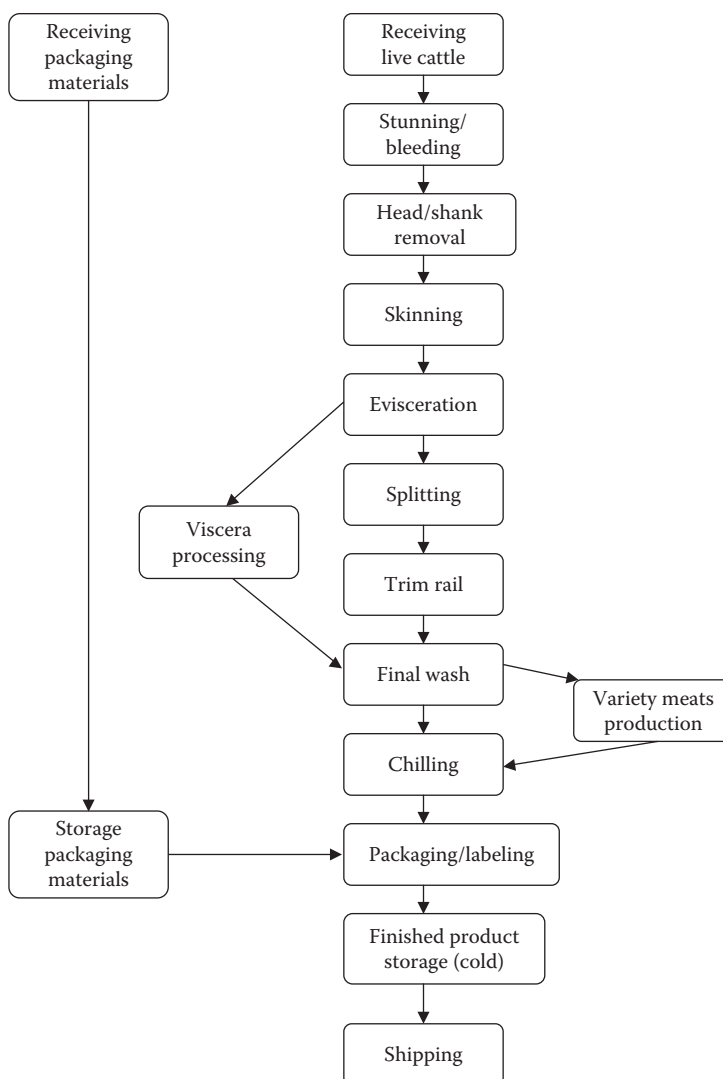


FIGURE 1.1 Flow chart for beef carcass slaughter. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

There are a number of quality issues that are of concern to the producer, such as marbling, color, and texture of the meat, that do not affect the slaughter process.

Cattle are received, unloaded from trucks, and held in pens. Prior to slaughter, packers will usually require that animals be kept off feed to facilitate the dressing procedures. The amount of time animals are kept off feed will vary by establishment. The animals must have access to water in all holding pens. This is a requirement of the Humane Slaughter Act.

A thorough inspection of the live animals before slaughter is called *ante mortem inspection*. The inspection is to identify any disease conditions in the cattle. Some disease conditions are unacceptable because they may affect human health. Others are unacceptable from an aesthetic standpoint.

Stunning is the first step in the slaughter procedure. This must be done in a way that complies with the Humane Slaughter Act. Most establishments use a mechanical method, such as a captive bolt, to render each animal completely unconscious with a minimum of excitement and discomfort. Because of the positive finding of Bovine Spongiform Encephalopathy (BSE) in the United States, the method of air injection stunning is prohibited.

Sticking is the second step in the process. A sharp blade is inserted into the neck, severing the carotid arteries and jugular vein, resulting in exsanguination and death. Typically, this is done while the animal is hanging head down from the rail. These overhead rails or tracks move at a controlled speed so that the carcass advances through the various slaughter processing steps.

The next step is *removing the hide*. This may be achieved through various methods, either using mechanical equipment or by hand (at small operations). After the head and hide are removed, many establishments use *antimicrobial interventions*. Antimicrobial interventions include, but are not limited to, hot and/or ambient temperature water wash, organic acid wash, and steam vacuuming. The steam vacuuming system is used to remove contamination from the dehided carcass either before or after evisceration. Additionally, some of these interventions are used as a multihurdle methodology. These interventions can be done before or after evisceration of the carcass for the removal of visible contamination.

Bunging, when the rectum (bung) is secured to prevent contaminating the carcass with fecal material, happens prior to evisceration.

Evisceration is done to separate the internal organs from the carcass. Even in highly mechanized plants, this is still done by hand. It is important that evisceration is done properly so as not to *contaminate* the carcass with the contents of organs such as the stomach or intestines. Fecal material or stomach contents (*ingesta*) contain many bacteria, and may possibly harbor certain harmful bacteria (*pathogens*) such as *E. coli* O157:H7, *Salmonella*, *Campylobacter jejuni*, and so on.

At this point the carcass receives *post mortem inspection*. Similar to ante mortem inspection, the carcass is examined for disease conditions that cause the carcass or parts to be unacceptable for human consumption.

Next, the carcass is *split* with a saw. At the trim rail, an inspection reveals whether the carcass is free from contamination or quality concerns that can be removed by *trimming*.

In the next step, the carcasses are weighed, *marked* with an official United States Department of Agriculture (USDA)-inspected and passed brand, and washed. They are then moved to a chill box or cooler and *chilled* to a specified temperature. The chilling step helps inhibit the growth of spoilage and harmful bacteria. There are various methods and equipment used for chilling the carcass.

Carcasses are typically *stored* in large refrigerated warehouses called coolers until they are shipped. It is important that an appropriate temperature, humidity, and air flow be maintained in coolers. Generally, the colder the temperature in the cooler, the slower the bacteria grow. Proper temperature is essential for preserving the quality and maintaining the safety of the product.

From this point, the carcass halves are ready to be *fabricated*, which means cut into parts. This *further processing* may happen at the slaughter facility in a processing or fabrication department, or the carcasses may be shipped to another plant.

Edible and inedible by-products will be covered when we discuss the Raw Product—Not Ground processing category. Some establishments may include edible and inedible by-products in their Slaughter HACCP plan.

1.2.2 Pork Slaughter Process

Look at the process flow diagram for pork slaughter (Figure 1.2). We will discuss the process for the skin-on pork carcass, which is most common. There are other pork slaughter processes, such as skinned, or hot boned.

Hog slaughter is somewhat similar to beef slaughter, but there are some major differences; we will highlight only the differences. Hogs are usually stunned with an electrical shock or using CO₂ that renders them unconscious and insensitive to pain. Sometimes hogs will be hanging during the sticking and bleeding step. However, sticking and bleeding can also be done while the hog is lying on its side or being held on its back. Instead of removing the hide, usually hogs are scalded and *dehaired*. Then, they are quickly singed with a flame to remove any remaining hairs.

The step called “*gambrelling*” refers to slicing the tendons on the back of the hock so that the carcass can be hung on a gambrel or special type of hook.

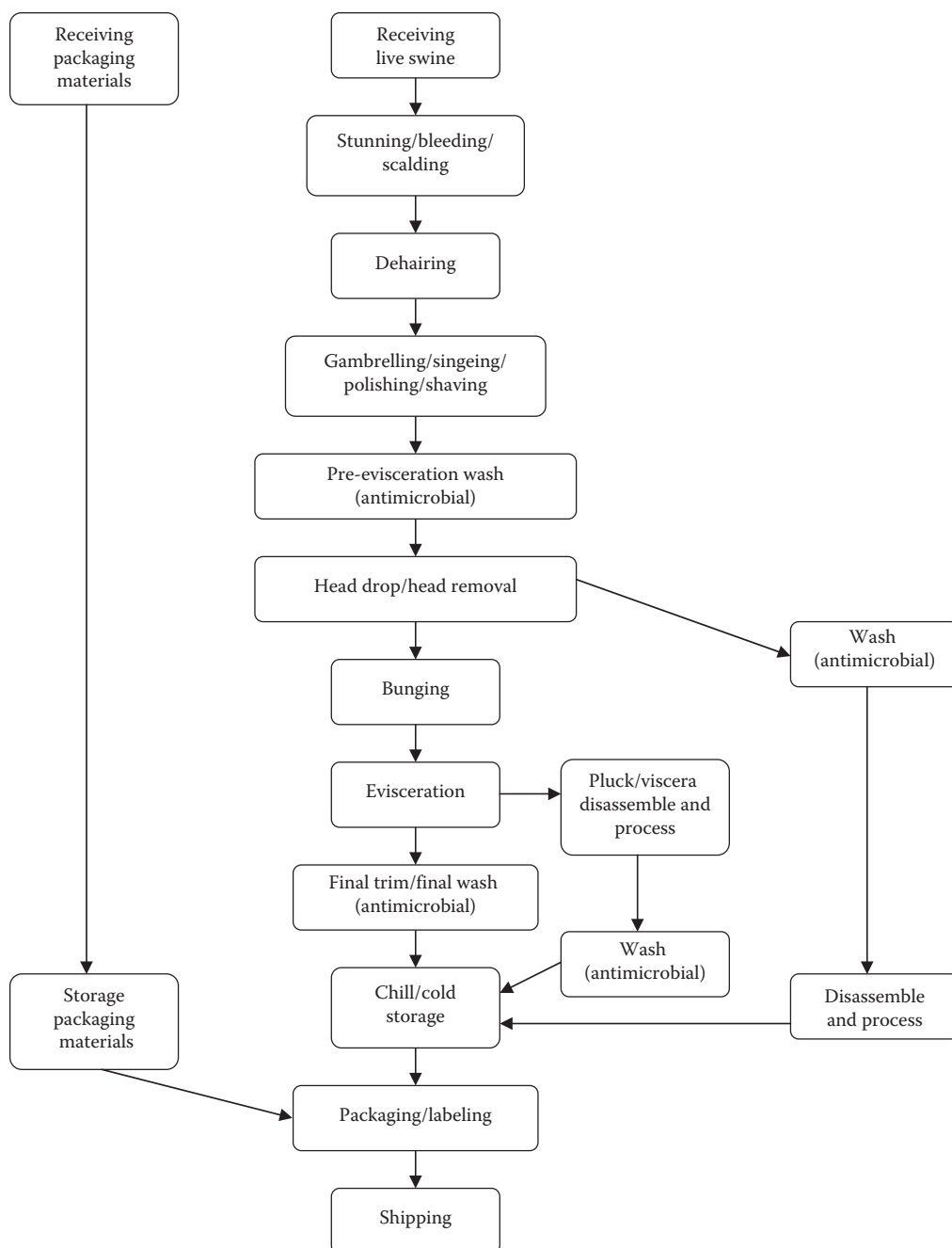


FIGURE 1.2 Flow chart for pork carcass slaughter. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

1.3 Raw Product: Not Ground

This processing category includes all raw products which are not ground in their final form. Some examples are beef trimmings, tenderized cuts (e.g., injected or marinated), steaks, roasts, chops, poultry parts, fabricated products, and edible by-products (e.g., livers, gizzards).

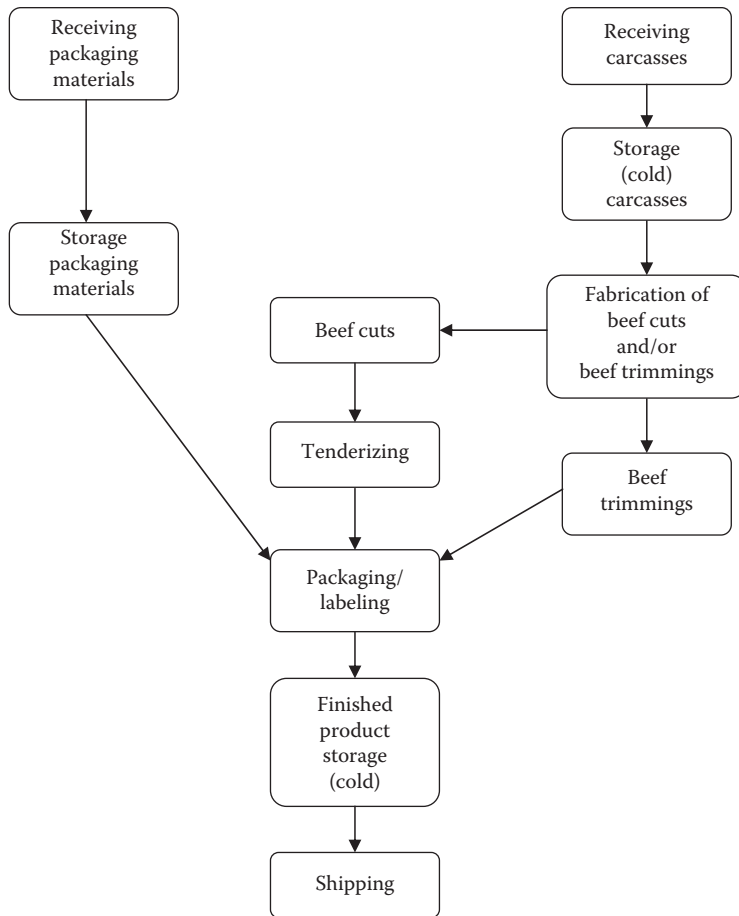


FIGURE 1.3 Flow chart for beef trimmings and tenderized beef cuts. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

The process flow diagram (Figure 1.3) for this category uses the example products beef trimmings and beef tenderized cuts. Notice that the first step is *carcass receiving*. Carcasses are chilled after slaughter for a specified period allowing them to become firm to facilitate a neat job of cutting.

1.3.1 Fabrication

Fabrication refers to creating the various cuts from the carcass to produce particular types of product. *Primal* or *wholesale* cuts are made first (Figure 1.4). Their names usually identify where the meat comes from on the animal, such as the loin, the shoulder, and so on. The plant typically uses large mechanized saws to fabricate the carcass into primal cuts.

As denoted in Figure 1.4, *retail cuts* describe what part of the primal cut the meat comes from, for example, rib roast or round steak. Retail cuts may be made with a saw, especially if they include bone. Often, primal parts are boned before cutting into retail cuts, in order to produce boneless items. Establishments that produce portion controlled retail cuts for hotels, restaurants, and institutions are often called *HRI* (Hotel, Restaurant, and Institution) operations.

Packaging materials (such as wax-treated paper or plastic film) protect the product from damage during refrigerated or frozen storage.

The final step is distribution, either to other departments in the same plant, other plants, or to retail markets.

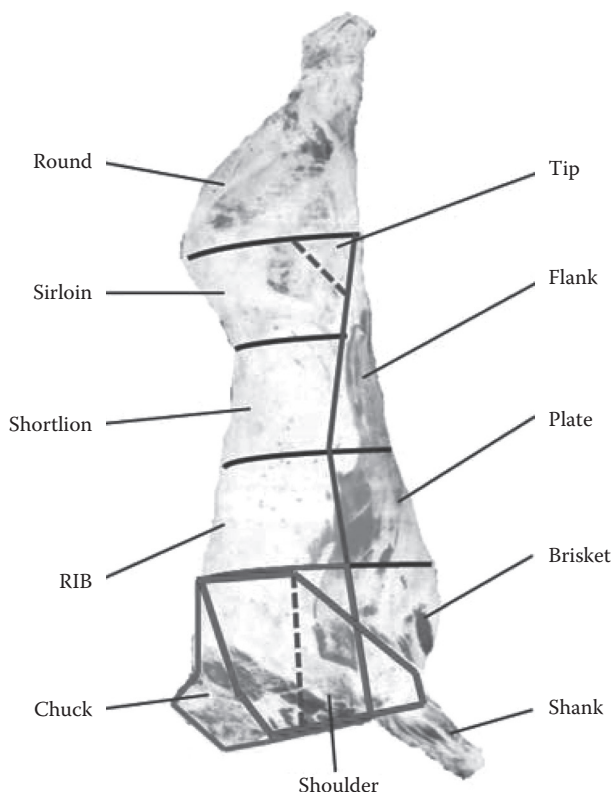


FIGURE 1.4 Beef retail cuts. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

1.3.2 Tenderization

Tenderization is another procedure used in some plants. All cuts can be tenderized, but this is typically used on cuts from lower quality grades and less tender cuts of higher graded carcasses. There are several methods for tenderizing meat. They include aging (natural chemical process), the use of enzyme solutions (artificial chemical process), and use of mechanical tenderizers. Mechanical tenderizers typically press many thin blades through the meat pieces, cutting the muscle fibers.

1.3.3 By-Products

The processing category of Raw Product—Not Ground includes edible by-products. Consumer demand has had an effect on the production levels of various by-products. For example, vegetable oils have replaced animal fats like lard and tallow in the frying industry. Technological developments have also had an impact on the demand for inedible by-products. Synthetic materials have been developed to make many items that were once made of animal products.

Edible by-products: Some of the edible by-products include tongues, brains, sweetbreads, hearts, livers, and kidneys. These are called variety meats. Because variety meats are more perishable than carcass meat, they must be chilled quickly after slaughter and processed or moved quickly into retail trade. They may be sold as fresh or frozen items, or used to make other processed foods. There are exceptions to the use of edible by-products derived from cattle 30 months of age and older, including brain, skull, eyes, trigeminal ganglia, spinal cord tissue, and dorsal root ganglia. As of January 12, 2004, new regulations and policies prohibited the use of the aforementioned materials for human consumption as a result of a positive case of BSE.

Some descriptions of the sources and uses of various edible by-products are as follows:

- *Casings* for sausages are sometimes made from sheep or hog intestines. (Note: the distal ileum of beef small intestines are prohibited regardless of age.)
- *Chitterlings* are made from thoroughly cleaned and cooked intestines of pigs, and consumed as a variety meat.
- *Blood* is used as an ingredient of certain specialty products.
- *Tripe* is obtained from the first (rumen) and second (reticulum) stomach compartments of cattle. It is consumed as a variety meat and used in specialty products.
- *Sweetbreads* are thymus glands obtained from the ventral side of the neck and inside the chest cavity of young cattle. They are used fresh or frozen.

Inedible by-products: The uses for inedible by-products are constantly changing based on the available technology and consumer interests. Next are some examples of the ways in which inedible by-products are used:

- *Hides, skins,* and pelts are used to make leather goods and glue.
- *Fats* are used to produce industrial oils, lubricants, soap, glycerin, and other cosmetic ingredients. Most inedible fats are processed by dry rendering.
- *Bones* are used to produce animal feed (except ruminant bones to feed ruminants) and fertilizer.
- Some *glands* are used to produce pharmaceuticals. For example, bovine ovaries yield estrogen. The pancreas glands yield insulin, which is used to treat diabetes.
- *Lungs* are used to produce pet foods.

These lists are by no means complete but give a few examples of the uses of edible and inedible by-products.

1.4 Mechanically Separated Product

Often, the industry searches for ways to yield the maximum edible, wholesome product from the meat or poultry carcass. The mechanical separation process is a technology that industry uses to obtain more usable product from bones from which the muscle has been removed. Often, you will see these products referred to as “*mechanically separated (species) or MS (species)*.” Any species can be used: veal, lamb, pork, chicken, or turkey. (Note: use of beef in MS product is prohibited.)

The process begins with bones (Figure 1.5). Bones for this process have usually already had most of the muscle tissues removed by hand boning, or they are bones, like neck bones, which are difficult to process. The bones are ground up, and the resulting mass is forced through a sieve. The softer muscle particles are thus separated from the hard bone particles, which remain behind the sieve. The resulting product has a paste-like consistency.

Great pressure is used to force the product through the sieve, and this results in a temperature rise in the product. Therefore, product must be processed quickly and the temperature immediately reduced, in order to prevent oxidation and microbial degradation of the product. Even with these precautions, this product will deteriorate quickly.

Although the MS product has many of the characteristics of meat and may be used as a meat ingredient in the formulation of quality meat food products, it is not meat, as defined in the regulations. In particular, the consistency of MS livestock product and its mineral content are different from those of meat. The bone marrow, spinal cord, and a certain amount of fine bone particles are included in the finished product. This actually provides a readily absorbed source of valuable nutrients like calcium and iron. However, there are specific limits on the quantity and size of the bone particles included in the final product. There are also limits on how much of the MS product can be used in meat or poultry products, and it must be identified in the ingredients statement of the label. As per regulations MS beef is prohibited for use as human food.

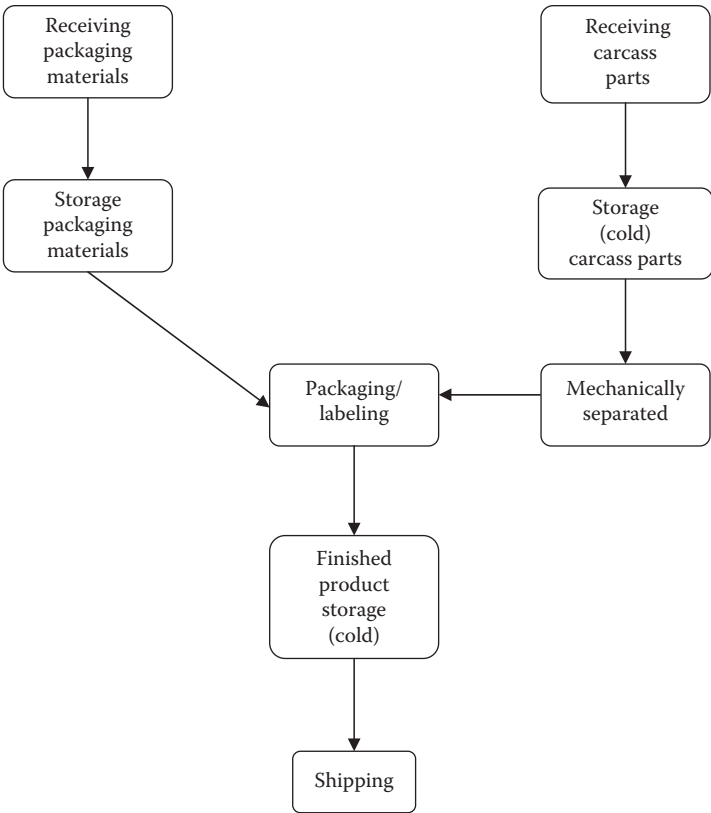


FIGURE 1.5 Flow chart for mechanically separated pork. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

A similar technology used by industry is called *advanced meat recovery (AMR)*. This process obtains the meat tissues from the bones without incorporating significant amounts of bone and bone products into the final product. The resulting product consists of distinct particles of meat, with the typical color and texture of the species used. There are no special limits on the use of this product. Nevertheless, the presence of central nervous system (CNS)-type tissue is evident in the product derived from AMR systems. Therefore, AMR-derived products from cattle that contain CNS tissue cannot be used as an ingredient of a meat food product. More specifically, Regulation 9 CFR 318.24 prohibits the use of skulls or vertebral columns (with exceptions) of cattle 30 months of age and older from use in AMR systems.

1.5 Raw Product: Ground

This processing category includes all raw products that are ground. Some of the common products are ground beef, hamburger, ground beef patties, ground pork, fresh sausage, Italian sausage, and ground poultry products. Beef, pork, veal, lamb, chicken, and turkey can all be ground and sold or used in other products. One of the favorite products served in this country is the hamburger patty, which is the example we will use in this section (Figure 1.6). However, the processing steps that are used to produce hamburger patties are also used for other products. Establishments differ in how they design their production processes, and you may see many variations of the basic processes that we illustrate.

Meat for use in ground products may come into the establishment from outside suppliers, or it may be produced within the establishment during fabrication and boning operations. Nonmeat ingredients and

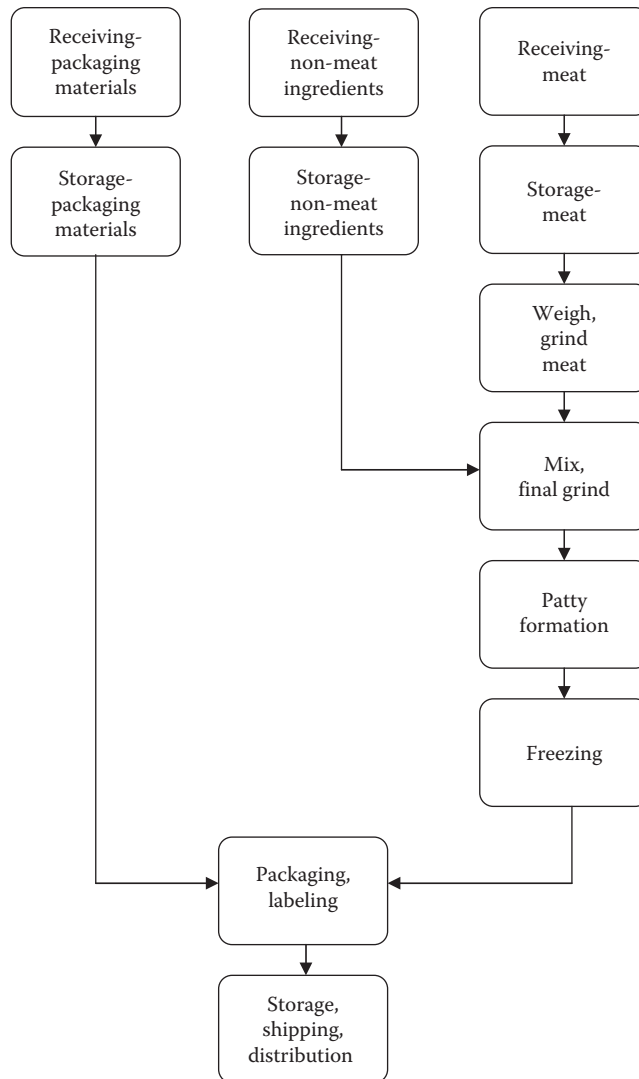


FIGURE 1.6 Flow chart for ground beef patties. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

packaging materials will come from outside suppliers. Many establishments use a combination of suppliers, depending on the cost and type of product available from each.

Written purchase specifications are developed by some establishments to ensure that a consistent product is received. Specifications are formal agreements between the supplier and the purchaser, and may include quality aspects, such as portions of lean and fat, and safety factors such as laboratory testing for pathogens.

After meat ingredients are received, they are stored in freezers or coolers until use. Meat products must be maintained at refrigeration temperatures adequate to prevent spoilage and growth of pathogens. *Refrigeration* achieves several purposes: it slows the growth of microorganisms, including spoilage bacteria and pathogens; slows the metabolic and enzymatic activities within the meat tissues that would lead to product deterioration; and also reduces moisture loss from the product.

Chiller or cooler temperatures in the range of 38–45°F will substantially retard most pathogen growth. However, certain types of bacteria, like *Yersinia* and *Listeria*, can grow at these temperatures and may

be a significant hazard. Chiller storage is temporary because even at these temperatures, the spoilage organisms will continue to grow, although at a very slow rate. Freezers, generally maintained at -100°F or below, halt the growth of all bacteria. Products kept at these temperatures will maintain safety and quality for longer periods of time.

Dry ingredients and packaging materials are also received and stored prior to use.

Ground products are often made from *trimmings*. Trimmings are the pieces that are removed from carcasses while producing higher quality retail cuts of meat. Grinding is a way that establishments can use lower quality products that would not be saleable to a retail consumer. In addition to trimmings, ground beef is also commonly made from flanks, short plates, shank meat, briskets, chucks, rounds, or sirloins. Meat ingredients used may be fresh or frozen, or a combination.

Often products contain *nonmeat ingredients*. Ground products are often seasoned with salt, sugar, spices, or other flavorings. Depending on the product being made, water may be added, and some product formulations include binders and extenders such as soy flour or nonfat dry milk.

Establishments use a specified recipe, called a *formulation*, to create a consistent product batch after batch. The formula lists the weights or percentages of ingredients to be used. Meats and other ingredients are weighed before use to ensure that the proper amount of each is added to the batch.

Comminution is the process of reducing the particle size of meats. Several different machines are used, including the grinder, the bowl chopper, and the flaker. Some producers use a combination of several of these in the production of a product.

The *grinder* consists of a hopper into which the meat chunks are placed. The meat then moves along an auger or screw, through a cylinder, at the end of which is a grinding plate and a knife. As the meat is pressed up against the plate the knife turns and cuts off small bits of the meat. The size of meat particle produced is determined by the size of the holes in the grinding plate.

Another method of reducing particle size is the *bowl chopper*. This machine consists of a metal bowl that revolves and a metal knife that rotates, cutting through the meat pieces in the bowl. The bowl chopper also mixes product as it chops it.

The *flaker* is used on large frozen blocks of meat or meat trimmings. Product is pressed against the knife blades, which shave off pieces of the still-frozen meat, enabling it to be used in formulation without thawing.

Sometimes meat ingredients go through several grinding processes. Often, fat and lean meat ingredients are ground separately and then combined.

After comminuting, products are mixed thoroughly. Often products are transferred to a separate piece of equipment, called a mixer or blender, in order to mix it. The *mixer* consists of a chamber that the ingredients are placed into, and blades or paddles that turn and mix the product, resulting in a uniform distribution of fat and lean particles. Nonmeat ingredients, if used, are added at this stage.

After comminuting and mixing, the ground meat mixture is often *shaped* into different forms. Fresh sausage may be extruded into a casing. Hamburger or ground beef is often shaped into patties using a patty machine. After formation, the patties may be frozen.

Because of the moving metal parts common in these operations, there is a possibility of metal chipping or breaking. Proper maintenance of equipment is essential to reduce this possibility. Some establishments use a *metal detector* to identify product that may be contaminated with metal fragments.

The final step for ground products at the processing establishment is *packaging and labeling*. Product may be packaged into retail size packages, into larger containers for institutional use, or into bulk containers for sale to other establishments for further processing. Although there are many different combinations of packaging materials in use, plastic liners and cardboard boxes are some of the materials commonly used. Labels must accurately reflect the product.

After packaging and labeling, products must be held at proper refrigeration temperature during storage, and throughout distribution to the customer.

Many variations of these steps are used to produce the different products available in the marketplace. Ground products may be marketed in bulk or in patty form. Fresh sausage products are also included in this category, and these products may be sold in bulk or in casings. Poultry products are common today, and ground poultry products are available in bulk, packaged into chubs (short plastic casings), or formulated into a variety of sausage products.

1.6 Irradiation

Food irradiation is a technology that exposes food to radiant energy in order to reduce or eliminate bacteria. Ionizing radiation will reduce, and in some circumstances eliminate, pathogenic microorganisms in or on meat and poultry (Figure 1.7). FSIS recognizes irradiation as an important technology for helping to ensure the safety of meat and poultry. FSIS has included ionizing radiation as an approved additive in pork carcasses and fresh, or previously frozen, cuts of pork that have not been cured or heat processed for the control of *Trichinella spiralis*, which causes trichinosis. Ionizing irradiation is also recognized as an approved additive in fresh or frozen, uncooked, packaged meat or poultry products for the purpose of reducing pathogenic microorganisms and extending shelf life.

Radiation is broadly defined as energy moving through space in invisible waves. Radiant energy has differing wavelengths and, hence, degrees of power. Forms of radiant energy include: microwave and infrared radiation, which heat food during cooking; visible light or ultraviolet light, which are used to dry food or kill surface microorganisms; and ionizing radiation, which penetrates deeply into food, killing microorganisms without raising the temperature of the food significantly. Food is most often irradiated commercially to reduce the numbers of pathogenic microorganisms, to extend shelf life, or to prevent reproduction of insects. Food irradiation for these purposes is practiced in many countries, including the United States.

Treating product with irradiation could result in significant *reduction or even the elimination of pathogens*. Ionizing radiation has been shown to be effective at eliminating *Salmonella*, *E. coli* O157:H7, *Clostridium perfringens*, *Staphylococcus aureus*, *Listeria monocytogenes*, and *Campylobacter jejuni*, among others. Irradiation also can significantly extend the shelf life of meat and poultry food products through the reduction of spoilage bacteria.

Irradiation *dose* is measured in kilo Gray (kGy); the maximum dose for use on meat products is 4.5 kGy. The radiation dose necessary to reduce the initial population of bacterial pathogens by 90% (the *D*-value, or 1-log_{10}) range from 0.1 to 1 kGy. Higher radiation doses (above 1 kGy) are needed to accomplish the same antimicrobial effect in a frozen food versus a nonfrozen food of the same type.

Irradiation does not significantly increase the temperature or change the physical, sensory, or nutritional characteristics of foods. Irradiation does not make food radioactive. During irradiation, the energy waves affect unwanted organisms but are not retained in the food. This is similar to the way that food cooked in a microwave oven does not retain those microwaves. Because irradiation does not raise product temperature, product is still raw and requires refrigeration.

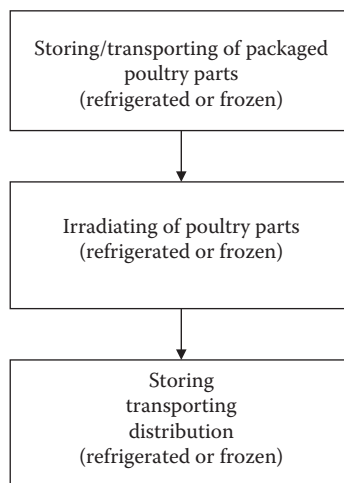


FIGURE 1.7 Flow chart for irradiation of poultry parts. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

The irradiation process requires a source of energy. The two types are radioisotopes (radioactive materials such as cobalt or cesium) or machines that produce high-energy beams. Specially constructed facilities are used to confine the beams so that the personnel will not be exposed.

The Food and Drug Administration (FDA) regulates all aspects of irradiation: what products it can be used on, allowable dose, and how those products are labeled. The USDA is responsible for the inspection and monitoring of irradiated meat and poultry products and for the enforcement of FDA regulations concerning those products.

The “*radura*” is an internationally recognized symbol identifying irradiated food. The FDA requires that both logo and a statement (“Treated with irradiation” or “Treated by irradiation”) must appear prominently on the label of packaged foods, and on bulk containers of unpackaged foods.



Irradiation can be used within an *HACCP system*. Establishments that irradiate product probably would establish critical limits such as radiation dosage. By ensuring that specific limits for these parameters are met, establishments could be reasonably sure that a predetermined reduction in pathogens has been achieved within the irradiated product.

1.7 Heat Treated but Not Fully Cooked: Not Shelf Stable

Up to now, all of the processing categories that we discussed dealt with raw product—product that had not been heat treated. This section covers a group of products that receive a heat treatment, but they are *not fully cooked*. These products still need to be thoroughly cooked in order to be safely consumed.

The products included in this category vary quite a bit from each other. What they have in common is that *they have received some type of heat treatment, but not sufficient heat treatment to result in a ready-to-eat (RTE) product*. One well-known product is bacon, a cured and smoked pork product. Another product is cold smoked sausage, a product that has been smoked to add flavor, but is still raw. Partially cooked battered and breaded poultry is included in this category; it has been cooked only enough to “set” the breading. Char-marked patties are similar; they have been cooked only enough to add distinctive char marks on the meat surface, but are still essentially raw. Low temperature rendered products are heat treated to melt and remove some of the fat in the meat tissues, but again, they are not fully cooked. As you can tell, there are many different types of products grouped into this category.

We are going to discuss the processes involved, focusing on some of the most common products. There will be many variations of these processes used by establishments, and this module will only provide an introduction to the procedures used.

1.7.1 Bacon

Bacon is an example of a product that is cured and smoked. Let us study the process flow chart (Figure 1.8) for this product. First, the raw meat ingredients are *received*, either from another establishment, or from the fabrication department within a large establishment. In this case, the raw meat ingredient used is the pork belly.

The nonmeat ingredients are weighed and combined. Bacon is a *cured* product, which means that *additives* are used to preserve the product and stabilize the color. Following are some of the most common additives:

- *Salt* is used for flavor and because it preserves the product by inhibiting bacterial growth.
- *Sugar* is sometimes used as a sweetener. It can counteract the harsh flavor of the high levels of salt used in some products.

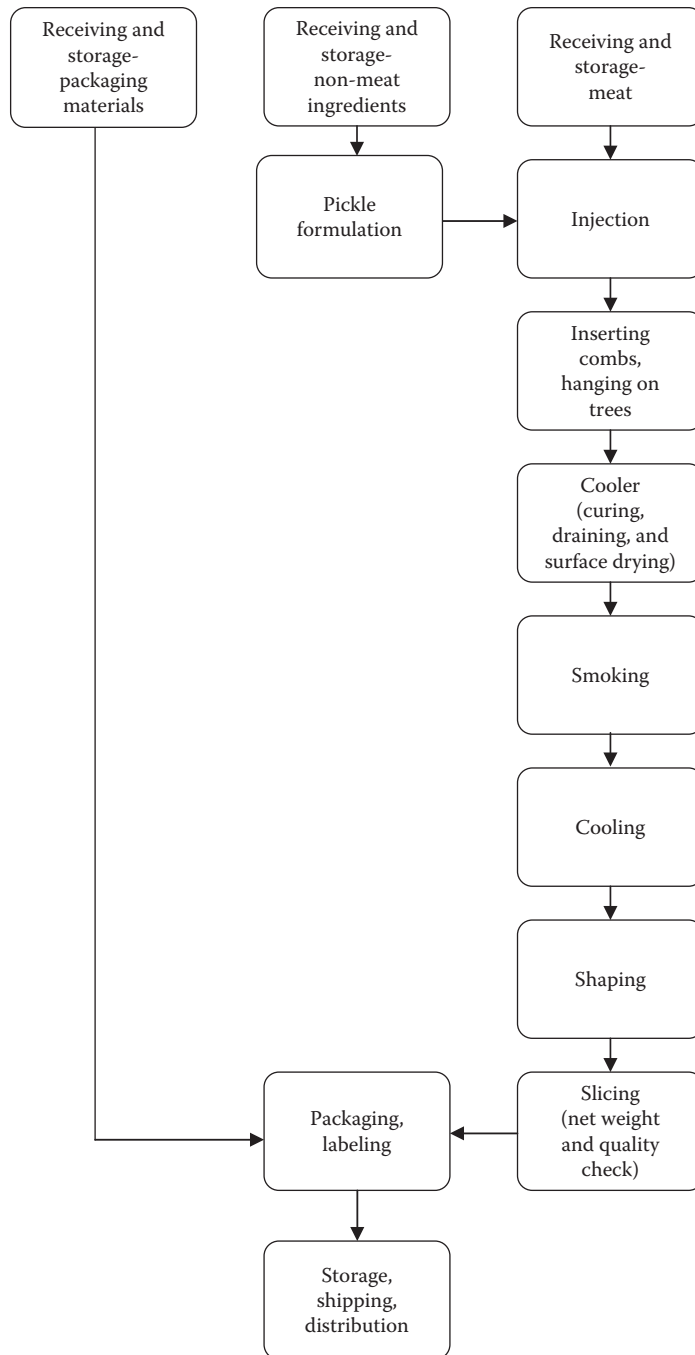


FIGURE 1.8 Flow chart for bacon. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

- *Nitrite (or less commonly used nitrates)*—stabilizes the color of the meat, contributes to the characteristic flavor of cured meat, inhibits the growth of both pathogens and spoilage micro-organisms, and retards rancidity (deterioration of the fat).

The amounts of nitrite and nitrate allowed are restricted by FSIS regulations. Additives that have regulatory limits are known as *restricted ingredients*. Because nitrates are reduced to nitrites and further

converted into nitric oxide which react with the amines present in muscle fibers to form *nitrosamines* (are known to cause cancer), the nitrite and nitrate levels must be closely monitored. Also, as nitrites can be very toxic to humans, the use of these ingredients is carefully controlled. Supplies of nitrites and mixtures containing them are kept securely under the care of a responsible employee of the establishment. Often, the nitrite is purchased premixed with salt and colored pink to prevent its accidental misuse.

Nitrite is most important because of its role in developing the cured meat color. Nitric oxide (the chemical reaction product of nitrite) reacts with *myoglobin*, a complex protein present in meat. Myoglobin is the pigment that is responsible for the red color of muscle tissue. A series of chemical reactions results in the formation of the stable pink color of cured meat.

These are just some of the most common curing ingredients. Many other ingredients are used by the industry, and will contribute to the variety of formulations that you may encounter.

These cure ingredients are sometimes mixed with water to form what is known as a *pickle*, or a curing solution. The solution is often injected into the meat using an *injector*. This equipment carries the meat past a series of needles that pierce it and force the pickle solution into the interior of the meat pieces. This process is called *pumping*. This results in a fast and even distribution of the pickle. There are many other means of introducing the pickle; sometimes meat pieces are simply placed into a barrel or vat of the pickle. This is a much slower process than injection.

After injection, the meat pieces are hung onto racks called *trees* or *cages*, which hold the meat while it is further processed. The meat hangs in a cooler for a period of time to ensure that the cure ingredients have time to react with the meat, and to allow some of the solution to drain out, if necessary.

The next step is the *smoking* process. The racks of meat are loaded into a smokehouse. The establishment operator carefully controls the smokehouse. Time, temperature, and humidity are parameters that affect the product. These parameters are usually carefully monitored to ensure that the smoking step proceeds as designed by the plant.

One common type of monitoring equipment used is the dry bulb/wet bulb thermometer. This device monitors the temperature inside the smokehouse with two thermometers set right next to each other, one dry, and one inside a moist piece of cloth. The difference between the two temperature readings is used to calculate the humidity of the environment.

The operator sets the smokehouse controls to run through a series of processes in which the addition of steam and smoke will change the conditions inside the smokehouse. Although bacon receives some heat treatment in the smokehouse, it is not fully cooked. The smokehouse treatment is primarily designed to deposit the smoke onto the surface of the meat.

Smoke has several important effects on the meat product:

- It develops the characteristic smoke flavor.
- It results in a color change (browning effect) on the surface of the meat.
- It has some preservative effect.
- It protects the meat from *oxidation*, which is the development of off-flavors.

Smoke is a complex mix of chemical compounds, including phenols, alcohols, organic acids, carbonyls, hydrocarbons, and gases. The phenols and carbonyls produce the color and flavor of smoke. Smoke has a *bactericidal* action; that is, it kills some of the bacteria present. This is due to the combined effects of heating, drying, and depositing the chemical components of the smoke. Smoke is often produced from hardwood sawdust in a *smoke generator*. Liquid smoke is also used.

After the product has been smoked according to the establishment's desired process standards, it must be cooled down to safe product storage temperatures. This is often done initially in a *blast cooler* for maximum cooling effect. This cooler forces cold air at a very high velocity around the bacon pieces, quickly cooling them.

After the product is properly chilled, it may be sold in the bulk form. Most bacon, however, is sold as sliced product. The meat pieces are usually shaped in a press, or *blocked*, in order to produce uniform slices. The product is sliced and packaged. *Net weights* are checked by establishment personnel to ensure that the net weight statement on the label is accurate. Other quality checks are often performed by the establishment on the finished product. The product is now ready for final distribution.

1.7.2 Other Products in this Category

Cold smoked sausage is similar to bacon in that the product is smoked primarily for appearance and flavor. The process is called “cold-smoked” because the smoking does not result in high enough temperatures to cook the product. The smoke used is not actually cold; it is usually 90–120°F. The product must be quickly cooled to prevent bacterial growth.

Partially cooked battered, breaded poultry products are another product in this category. The raw poultry pieces are coated with *batter*, a liquid mixture of flour, egg, milk, or water; or with *breeding*, a powder or granular mixture of cereal products, like breadcrumbs; or they are both battered and breaded. The pieces are then heat treated to “set” or precook this coating, usually in hot oil. The poultry product inside is still uncooked. The products are cooled, usually in a special *IQF* (*individually quick frozen*) freezer, and packaged.

Char-marked patties are also included in this category. These products receive a heat treatment on the outside surface that produces a “char-mark” which imitates the marks created from cooking the product on a grill. The product is still essentially uncooked, and it is important that product labeling distinguish this product from RTE products. It is crucial that this product be fully cooked by the final user, to ensure safety.

Low-temperature rendered products are derived from the low-temperature rendering of fresh meat. The products are usually ground, heated, then treated to a process that separates some of the fat from the lean portion. The temperature used must not exceed 120°F. The product is then cooled quickly to limit potential growth of bacteria at these warm temperatures. The heat treatment is not sufficient to eliminate pathogens or to result in a cooked appearance. The rendered product is frozen and used in further processing operations. If the raw meat trimmings had at least 12% lean meat prior to rendering, the resulting product is called *Partially Defatted Chopped (species)*. If the fatty trimmings used as raw materials contain less than 12% lean meat, the resulting products are called *Partially Defatted (species) Fatty Tissue*.

There are, of course, many other products that you may encounter that would fall into this category. The common characteristic is that these products receive some heat treatment, but not enough to result in a fully cooked, RTE product.

1.8 Fully Cooked: Not Shelf Stable

This processing category includes all food items that have been fully cooked, but are not shelf stable. *Fully cooked* means that these products have been sufficiently cooked so that they are safe to eat as they are, with no further preparation required by the consumer. This is also known as “ready-to-eat.” Note, however, that many of these products are customarily eaten hot, and cooking instructions may be included on the label. This does not affect the classification of these products into this processing category. An example is the hot dog. This product receives sufficient heat treatment to be fully cooked, and does not necessarily need to be reheated by the customer. Most customers do, however, heat this product, and cooking instructions may be included on the label of the product. Another parameter that defines this category is that the products are not shelf stable. *Shelf stable* means that the product has received a treatment that renders it safe to store without refrigeration. This does not apply to this category. These products, although fully cooked, are not shelf stable, and must be kept refrigerated or frozen in order to maintain safety and quality. Again, the hot dog is a great example. It must be kept refrigerated by the consumer until it is eaten.

There are many different types of products that fall under this category. There are some major groups that we will closely examine, but there will be many other products that you will encounter in the industry. Keep in mind that what they all have in common is that they are fully cooked, but they are not shelf stable. Some examples that we will discuss are the cooked and smoked sausages, cooked deli meats such as ham, roast beef, pastrami, corned beef, cooked chicken roll, and smoked turkey breast. Other meat and poultry products that fall under this classification are salads, such as chicken or ham salad, and frozen entrees.

1.8.1 Cooked and Smoked Sausage

One of the major product groupings that fall under this category is the cooked and smoked sausage. There are many different types of these sausages made; some common examples are bologna, cooked salami, polish sausage, and hot dogs. Let us take a closer look at hot dogs as an example of how these products are produced (Figure 1.9).

The first steps are the same as we have previously covered: meat and/or poultry, other ingredients, and packaging materials are received and stored in the establishment until ready to use. Many establishments carefully control the quality of the incoming ingredients through purchasing specifications. Meat

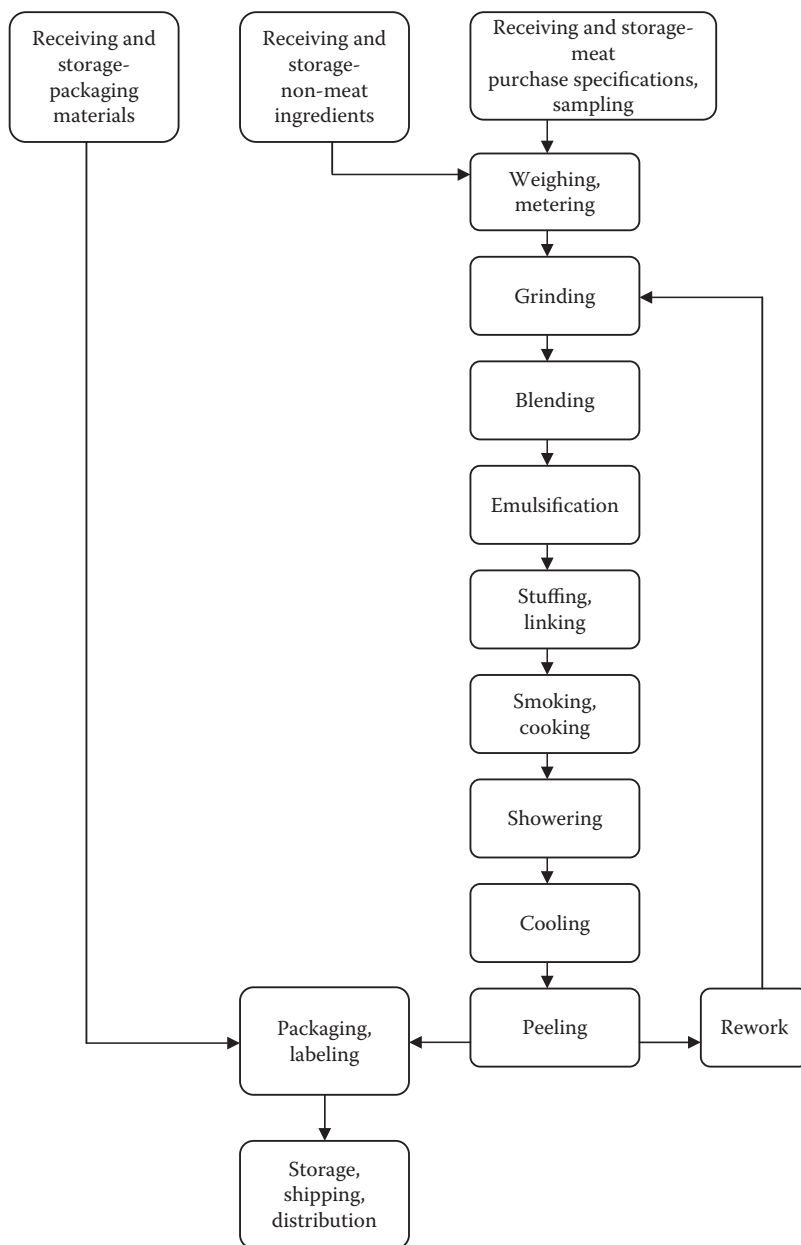


FIGURE 1.9 Flow chart for hot dogs. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

ingredients may have quality specifications such as percent fat, moisture, and protein. These are parameters that will affect the final quality of the product.

Raw meat ingredients used in these products will depend on the type of finished product desired. Not long ago, most hot dogs were either a combination of pork and beef, or they were all beef. Today, establishments still make these products, but many more combinations of ingredients are used. Many formulations include at least some *poultry* products, (turkey or chicken), and some products are made exclusively with poultry.

Many larger volume establishments use a system called *least cost formulation*. This is a computerized program that allows the processor to determine the specific allocation of ingredients required for a given product at a minimum cost. The product can be manufactured subject to the ingredients available. These establishments carefully analyze samples of each batch of ingredients and enter the data into the computer program. The program determines how many pounds of each ingredient to use, in combination, to produce the desired product. Theoretically, each finished batch of product will then be identical to the other batch. Of course, the final retail label must have a list of ingredients in the correct *order of predominance*, despite any variations caused by the least cost formulation system.

The first step in the formulation process is weighing or measuring the meat and/or poultry ingredients. They are ground and mixed or blended with the nonmeat ingredients. Often establishments will *preblend*, that is, they will grind and mix the meats with water and salt, and sometimes with the nitrite, and let it stand for a period of time in a cooler.

We have already discussed the most common nonmeat ingredients used in hot dogs: water, salt, curing agents like sodium nitrite, and sugar. Let us take a look at some of the other ingredients that may be used, depending on the formulation.

Binders and extenders, such as dry milk powder, cereal flours, and soy protein, have a number of uses in a sausage formulation. They increase the overall yield, improve binding qualities, and add certain flavor characteristics.

Cure accelerators such as ascorbates and erythorbates are used to speed up the curing process. They also stabilize the color of the final product.

Phosphates are used to improve the water-binding capacity of the meat, and contribute to the flavor and color of the product.

Spices and flavorings are used to add flavor to the sausage. The wide range of available spices, seasonings, and flavorings is a primary reason for the variety available in sausages in the marketplace.

- *Spices* are any aromatic vegetable substance that is intended to function as contributing flavor to food, rather than as a nutritional substance. The active aromatic or pungent properties of spices that contribute the most to the flavoring effect are present in the volatile oils, resins, or oleoresins of the spice. Spices may be used whole or ground. White pepper, paprika, and nutmeg are common spices used to produce the characteristic flavor of the hot dog. Because paprika also adds color and makes meat look brighter red, it must be listed as “paprika” on labels.
- *Flavorings* are substances that are extracted from a food, and contribute flavoring, such as spice extracts.

After the nonmeat ingredients are blended with the ground meats, the mixture is *emulsified*. This is done in an emulsifier, which further reduces the size of the meat particles to achieve a very fine texture. Fat, protein, salt, and water are mixed and combined into a semifluid emulsion. The meat muscle protein, myosin, is *solubilized*, or released from the muscle fibers, by salt. The solubilized protein and water combine and surround the fat globules, and suspend the fat particles within the mixture.

Careful control of the amount of each ingredient is essential to the quality of the final product. The manufacturer must select a mix of raw meat materials with the appropriate binding characteristics. Different meats vary in their ability to bind. Lean beef, for example, bull, cow, and shank meat, has high binding ability. Regular pork or beef trimmings with more fat, and poultry, have medium binding ability. Low binding meats contain high levels of fat, such as jowls and briskets. Organ meats have no binding qualities. The binding capabilities are directly proportional to the myosin (red) in the muscles. Thus, the paler the muscle, the less bind it contributes to the mixture.

Control of the emulsification process is also essential. Product defects result from too much chopping or from an increase in temperature during the process. Over-chopping makes the protein fibers too short. It also creates heat from friction that melts fat. This results in product defects such as pockets of fat in the final product.

After emulsification, the mixture (or “batter”) is *stuffed* into casings, usually artificial plastic casings that allow moisture to cook out and smoke flavors to penetrate. Natural casings such as sheep small intestines may also be used.

Following stuffing, the product is *linked* by pinching and twisting the casing to form separate units of sausage. The sausages are still held together by the casing. These lengths of casings are then placed on racks or trees, and are ready to be loaded into the smokehouse. Some establishments load trees into individual smokehouses, however, some large volume establishments use continuous smokehouses.

The smokehouse parameters that must be controlled are temperature, time, and humidity. The product must be exposed to a high enough temperature in order to produce a fully cooked, RTE product. The temperature inside the smokehouse, and the internal temperature of the sausage, may be monitored by the establishment in order to verify that the critical limits are met. Cooking is a very important step, because it is here that any pathogens that may be in the product will be eliminated and the numbers of spoilage bacteria will be lowered to an acceptable level.

After the product has reached the desired final temperature, the cooling process begins. This product is often showered with cold water inside the smokehouse. This removes some of the heat from the product, and immediately halts the cooking process. The shower is usually not sufficient to complete the cooling process. Usually the product is moved to another chiller or cooler to finish cooling. Some establishments use very cold water as a chilling medium, sometimes with salt added to lower the temperature below the normal freezing point of water. This is called a *brine chiller*. Other establishments may use cold air, and some use a combination of methods.

The cooling process is also known as *stabilization*. There are two types of bacterial contamination that must be addressed by the stabilization process:

- Spore-forming bacteria (*Clostridium perfringens* and *Clostridium botulinum*) can survive cooking when in the heat-resistant spore form, and these organisms need to be considered as the products are chilled. Growth (sometimes referred to as “*outgrowth*”) of these bacteria is slowed by rapid cooling. Cooling rates, or time/temperature relationships, must be carefully controlled in order to ensure that product does not remain at warm temperatures that would support the outgrowth.
- Recontamination with bacteria (e.g., *E. coli*, *Salmonella*, *L. monocytogenes*) must be considered as cooked products are exposed to the environment, food contact surfaces, or cross-contamination with raw product prior to final packaging. Proper chilling and cold storage temperatures are essential to limit the growth of these bacteria.

After product has been chilled to the desired temperature, it is removed from the artificial casing in a machine called a *peeler*. This equipment quickly runs the sausage through a tunnel that has a tiny blade that slices the casing. Steam or air is then used to blow the casing away from the sausage. The sausage links are now separate. If you closely examine the outside of a hot dog, you might see where the casing had been cut. This blade is a potential source of contamination, since it contacts every hot dog!

Sometimes a product that has partially or fully completed the production cycle is not sellable but is still wholesome, and can be used for food. For example, the casing of some sausages may split during the cooking or smoking cycle. Manufacturers may reuse these edible but unsaleable products by removing the casing and adding the contents to the grinder to include in another run of the same product. This is called *rework*. Since the proteins are coagulated from cooking, rework has no bind capabilities. Of course, the ingredients of the rework must be compatible with the ingredients of the batch to which they are added.

The final steps are packaging, labeling, and storage. The product is ready for distribution to retail stores, restaurants, or institutions.

1.8.2 Deli Meats

Deli meats such as ham, roast beef, and smoked turkey breast all have very similar processes. These products are produced by adding a solution of ingredients to the raw meat ingredient. Cured products, such as ham, turkey ham, and corned beef, have nitrite in the solution. Other products, like roast beef or chicken roll, may have only salt and seasonings used. The solution is often added with an injector, but products may also simply be immersed in the solution.

Traditionally, these products were produced with whole muscle pieces, such as bone-in hams, pieces of beef round, whole briskets, or whole boneless turkey breasts. Today, many products are made with chunks of meats of various sizes, to make chopped and formed products.

Products may be *tumbled* or *massaged*, which increases both yield and tenderness. In this procedure, meat and solution are added to a chamber with baffles. The chamber or the baffles rotate, which subjects the meat pieces to a gentle beating process. This produces muscle fiber disruption, with a corresponding release of salt-soluble protein, which in turn coats the meat pieces. The protein is then coagulated by cooking to form a matrix between the individual pieces, thus giving the product an intact muscle appearance.

The meat pieces are often formed into uniform shapes. This can be done by stuffing them into nets, casings, or molds. The product takes on the shape of the mold when cooked.

The cooking and cooling of these products is similar to the cooked sausage procedure. Some of these products, however, are cooked in a water bath or in a steam chamber.

After chilling, many of the products are packaged as whole roasts, for the retail deli market. Other products are sliced and packaged in retail consumer-sized portions. Most are vacuum packed, which helps to protect the product quality and increases the shelf life.

1.8.3 Salads

Another type of product in the fully cooked—not shelf stable category is the meat salad. Ham and chicken salad are some of the common salads produced. The establishment starts with fully cooked product. The fully cooked meat is chopped or ground, and mixed with other RTE ingredients such as mayonnaise, salt, spices, onions, celery, or pickle relish. The finished salad is packed into containers, and may be distributed fresh or frozen.

These products are rarely reheated; most consumers eat them cold. Therefore, there is no chance that any pathogens present will be eliminated by consumer reheating. Whatever bacteria are in the product when mixed, or contamination of the product during mixing, will remain in the product when eaten. The temperature of this product must be carefully controlled to ensure that bacteria in the mixture do not have a chance to grow.

1.8.4 Fresh or Frozen Entrees

Another group of products that fall within this category are the fresh and frozen entrées. These range from precooked chicken pieces and barbecue beef to prepared dinners with meat or poultry along with rice, pasta, sauce, and vegetables. There are many of these convenience-type items produced today, and new products are introduced almost daily.

The processing procedure for all of these products is very similar. Fully cooked meat or poultry portions are combined with sauces, vegetables, pasta, or other ingredients. Each of the ingredients is individually weighed or portioned, to result in the desired finished proportions.

Most of these products are designed to be reheated by the customer. Most packages will include instructions for reheating. This does offer some degree of safety to the consumer, in that the reheating may eliminate some bacteria if present. However, this reheating must not be depended on, as consumers may vary greatly, how well they reheat the product. These products are intended to be fully cooked and must be safe to eat without the reheating step.

These are some of the major product groups that fall within the Fully Cooked—Not Shelf Stable category. There are other products not mentioned here that you might encounter in the marketplace or being produced in an establishment. These products all have some things in common: they are fully cooked

and RTE by the consumer; and they require refrigeration or freezing in order to maintain product safety and quality.

1.9 Product with Secondary Inhibitors: Not Shelf Stable

Finished products produced under this regulatory processing category can be not-ready-to-eat (NRTE) or ready-to-eat (RTE) meat and poultry products that have been processed in a manner that utilizes strategies which produce results that will inhibit secondarily the growth of pathogenic bacteria. Finished products in this regulatory processing category may or may not have had heat applied to the product. The finished products in this category are *not shelf stable products* and require special handling to maintain their wholesome condition. In other words, the product may be heat treated, but not fully cooked, and a secondary inhibitor gives a cumulative effect (heat plus a food additive that affects the product) so that the product is RTE, yet it would *not* be RTE in the *absence* of the secondary inhibitor. The NRTE products must be kept refrigerated or frozen to maintain product quality and safety. Refrigeration, therefore, is still the primary inhibitor of the growth of pathogens and spoilage bacteria in NRTE products. RTE products need to be kept refrigerated to inhibit growth of spoilage organisms that are still present and capable of growth at ambient, nonrefrigerated temperatures.

Secondary inhibitors are usually ingredients or processes such as fermentation or drying that when used, in combination or alone, assists in inhibiting, or slowing the growth of possibly harmful bacteria. Primary microbial growth inhibitors include lowered water activity (a_w) and higher acidity. Salt or sugar in quantities that effectively lower the water activity of the finished product is an example of a secondary inhibitor. We will discuss each of the inhibitors and then concentrate on one product example—country-style ham.

Water activity (a_w): Microorganisms in food need water in order to live and grow. The water must be in a form that is available to the microorganisms. Water activity is a measurement of how much water is available in a product. The water activity can be reduced by removing water (drying) or by increasing the concentration of solutes dissolved in the water (adding salt or sugar).

Acidity: Most bacteria grow best in a medium that is neutral or slightly acidic, and the growth of most bacteria is significantly inhibited in very acidic foods. The ionic hydrogen concentration (pH) is measured on a scale from 1 to 14, with 7 being neutral; pH levels above 7 are basic or alkaline, while those below 7 are acidic. Foods that are highly acidic are seldom the vehicles for pathogens. Many foods are acidified to prevent the growth of undesirable microbes. This may be done by adding acidic ingredients, like tomatoes, or by adding the acid directly, like vinegar. The acidity of products may also be increased by the process of fermentation.

The process of using secondary inhibitors is a very complex system. Often several different inhibitors are used, each depending on the others in order to result in a safe product. You will learn more about this topic in the Food Safety Regulatory Essentials (FSRE) training.

Some examples of products that may fall into this processing category include products that are uncooked, cured, fermented, dried, salted, or brine treated, which are not shelf stable but can be RTE or NRTE, such as sliced country style ham, salt pork, and semidry fermented sausage. The product standards, processing methods, and labeling are all factors that must be considered in determining the regulatory processing category for the 03I products.

Let us look at the example of the perishable sliced country-style ham (Figure 1.10).

1.9.1 Country-Style Ham or Shoulder

Sliced country-style ham or shoulder is a cured, dried product, traditionally made from a single piece of raw meat from a pork shoulder. The example shown is for an NRTE, cured product (Figure 1.10).

Dried whole muscle products are mostly dry cured. An initial process for manufacturing whole muscle products consists of dry mixing the nonmeat ingredients with the meat. Curing is the addition of salt, saltpeter, nitrites, sugars, spices, and flavorings. Nitrate and nitrite contribute to the characteristic cured

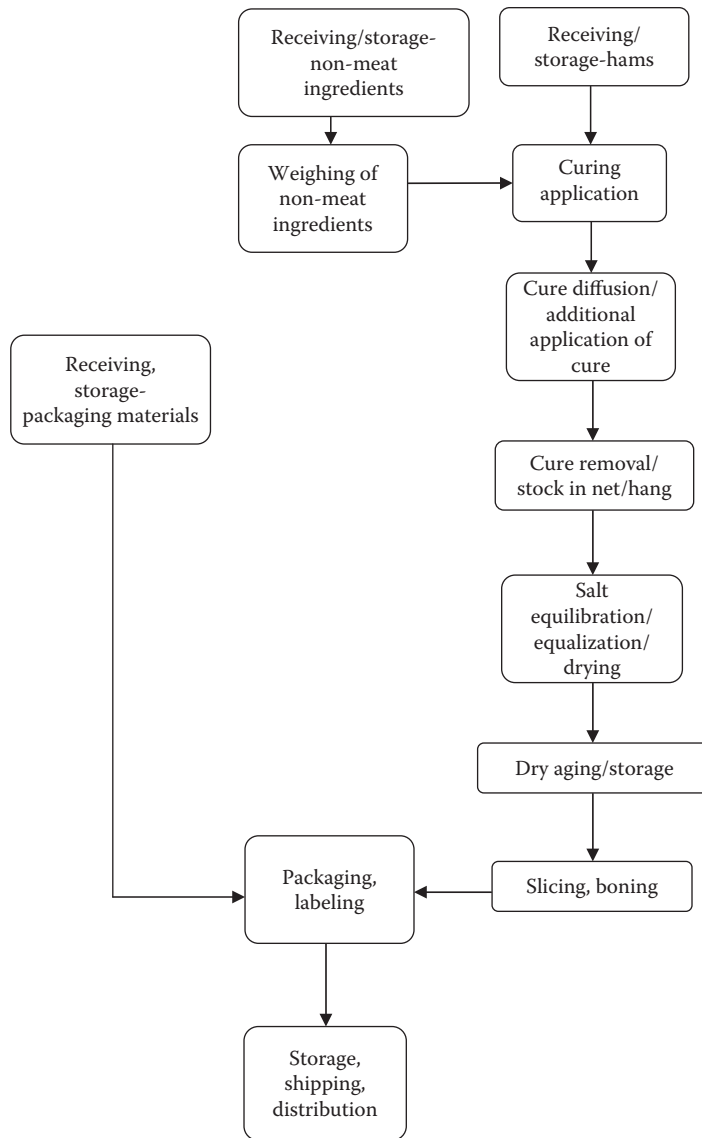


FIGURE 1.10 Flow chart for sliced country-style ham. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

flavor and reddish-pink color of the cured pork. All ingredients added are carefully weighed in order to conform to the product formula.

The entire exterior of the ham or pork shoulder is coated or rubbed by the dry application of salt combined with the other ingredients. Additional salt or dry-cure mixture of salt may be reapplied to the product as necessary to insure complete penetration. The high salt level and the colder temperatures are the only measures protecting against the growth of spoilage and pathogenic microorganisms.

After the initial salting, the product is held for some period of time at refrigeration temperatures (at 40°F) for the salt mix penetration and equilibration (“burning” period). This period often takes many weeks (at least 28 days) to achieve uniform salt distribution to greater than 4.5% with a water activity below 0.96. The goal is to eventually lower the water activity sufficiently to inhibit microorganisms to a point at which the temperature can be elevated. The product undergoes a maturation period (during this stage, the product is held at elevated temperatures for drying and flavor development), air drying, and

smoking (if desired), and storage. During these periods at higher temperatures, the humidity and air circulation is lowered, with further moisture loss. This final step in the process can be from 3 to 12 months in duration. The final product is then sliced and vacuum packaged for sale.

This is an example of a system of inhibitors. The nitrite and the salt are both inhibitors, they work together to achieve a certain amount of preservation of the product. This also extends the shelf life of the product, because spoilage organisms are inhibited. The lethality of the process for pathogens achieved in a salt-cured product will depend on the interaction of salt content, time and temperature of curing, drying, and aging. In addition, this product does not receive the amount of drying or reduction in water activity needed to make it shelf-stable. Therefore, the product is not completely preserved, and still requires refrigeration for safe storage. Products that are shelf stable will be covered in other sections.

1.10 Not Heat Treated: Shelf Stable

This processing category includes products controlled by water activity, pH, freeze dried, and dehydrated product, such as salami, pepperoni, or prosciutto. What defines this category is that the product is shelf stable, and while heat may be applied, it is not the *primary* means of achieving lethality. Many processors of products, such as pepperoni, include a low temperature heat treatment step in order to safely produce this type of product. We will examine the processing steps that are necessary for this type of product.

The process flow diagram (Figure 1.11) for this category represents salami/pepperoni. These are dry sausages that are RTE without any further cooking. Pepperoni was traditionally made with pork, while salami was made of pork along with some beef. Today, of course, many combinations of meat ingredients are used.

Many of the processing steps are similar to the processes we have already discussed. Raw meat ingredients are ground and mixed with nonmeat ingredients. The meat mixture is stuffed into casings. Let us take a look at some of the unique aspects of these processes.

Starter cultures: Bacterial fermentation is used to produce the lactic acid that results in the tangy flavor associated with this type of sausage. The acid inhibits bacterial growth. The resulting lower pH causes the proteins to release water, which assists in the drying process, and further inhibits bacterial growth in the finished product. Producers typically use a commercial lactic acid bacteria starter culture. Commercial starter cultures consist of a blend of harmless bacterial strains such as *Lactobacillus*. Simple sugars, such as dextrose or corn syrup, are added. They help promote lactic acid bacterial growth by serving as food to the bacteria during fermentation.

Fermentation: This step is an important one for pathogen control. During this step, the pH level of the product is reduced by the starter culture activity and by appropriate time/temperature factors. The starter culture is added to the meat mixture along with the sugar. The mixture is stuffed, and held in an environment optimum for their growth. These rooms are sometimes called *green rooms*. The temperature and humidity are carefully monitored. The starter culture bacteria actively reproduce, and as they do they give off lactic acid, which decreases the pH of the product. It is important that lactic acid is produced quickly, because it inhibits undesirable bacteria, like the toxin-producing *Staphylococcus*. The pH is monitored over time to determine when the process is complete. During fermenting, the establishment will probably want to achieve a pH of 5.0 or less within a certain time.

Most microorganisms thrive on pH near neutral (7.0) although there are exceptions. Meat processors can control pH to limit microbial growth and give meat a longer shelf life. Muscle tissues are close to 7 in the live animal. At slaughter, lactic acid builds up and the pH is lowered. The pH of fresh meat ranges between 5.3 and 6.4. At pH between 6.0–6.4 meat spoils faster than meat in the lower pH range (5.3–5.7), because the spoilage bacteria are more active at the higher pH.

Heating (optional): Both *Salmonella* and *E. coli* O157:H7 have been isolated from fermented sausage products. Consequently, many dry/semidry fermented sausages, particularly in the United States, have a significant “heat step” in the process to assure lethality of high numbers of bacterial pathogens.

Drying: At this stage, the salami/pepperoni is hung in a *dry room* to dry. Again, temperature and humidity are controlled. One of the factors that affect microbiological growth is moisture. Bacteria

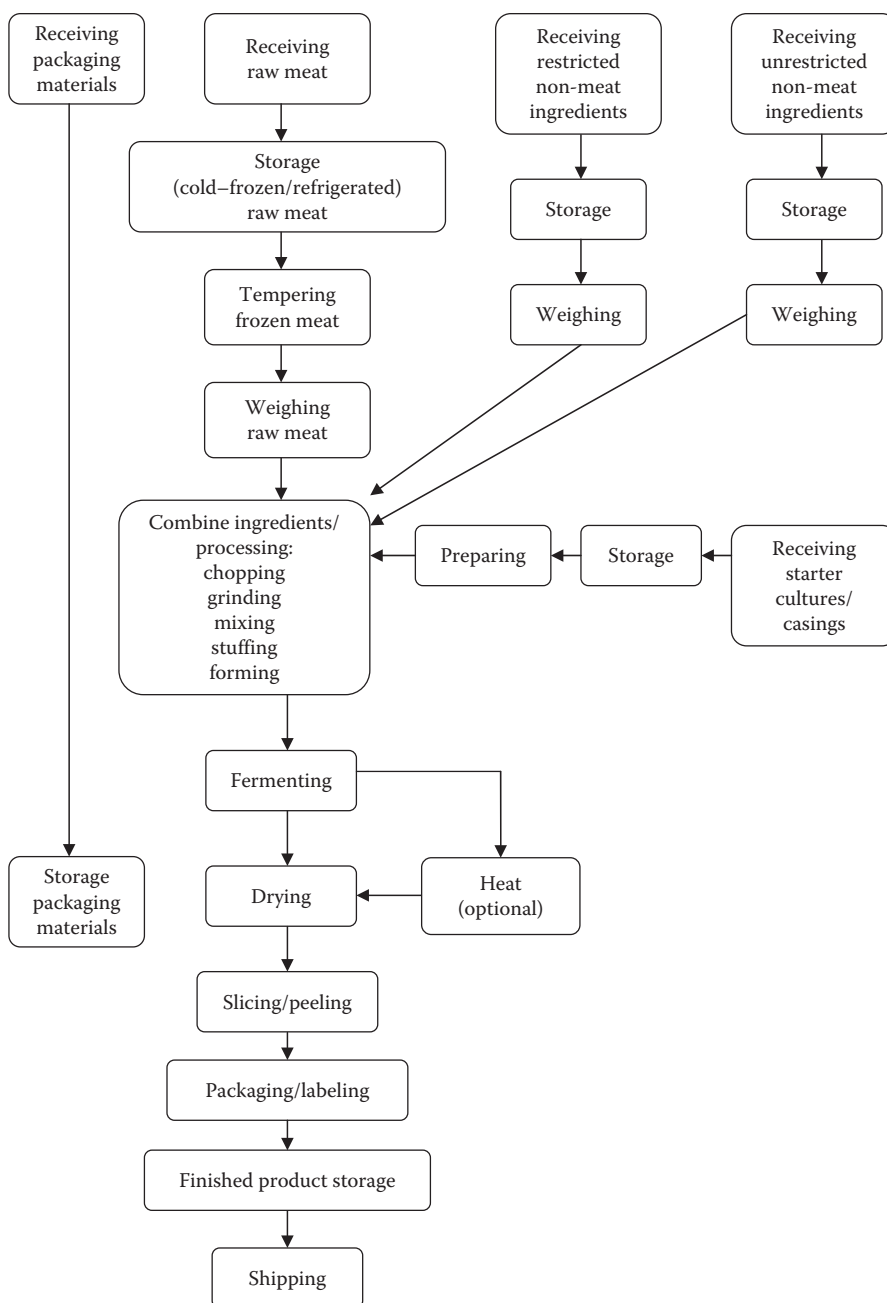


FIGURE 1.11 Flow chart for pepperoni. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

need moisture to survive and grow. Drying is the simple process of dehydration in which osmosis withdraws water from the cell of the spoilage organisms, shriveling or inactivating the cells. The product must be dried to the point at which bacteria are inactive or destroyed in order to create a safe, shelf stable product.

FSIS product standards state that dry sausage must have a *Moisture Protein Ratio (MPR)* of 1.9:1 or less, in order to qualify as a shelf stable product. This is a calculation that compares the percentage of moisture to the percentage of protein. For comparison, fresh meat has an MPR of about 4.0:1.

At this point the finished sausage sticks are dry and ready to be packed for storage and distribution. Some product is *sliced* so that it can be used for purposes such as sandwiches, pizza, or salads. This step is another one where the potential for cross-contamination of product must be controlled by the establishment.

Validation for E. coli: In light of food-borne outbreaks of *E. coli* O157:H7 linked to dry fermented RTE beef sausage products, FSIS strongly recommends that all procedures for dry and semidry fermented sausages be validated to show a 5-log reduction of *E. coli* O157:H7. Full documentation is required.

This process category contains all products that are shelf stable, which may or may not have been heat treated. These products are rendered safe by a combination of processes, such as fermentation, heating, and drying.

1.11 Heat Treated: Shelf Stable

This processing category includes rendered products, popped pork skins, bacon bits, snack sticks or jerky, summer sausage, kippered beef, and pickled sausages. These products are considered RTE, meaning they can be consumed as packaged. This category contains products that are shelf stable, and have received a full lethality treatment. Cooking is generally the primary method for achieving all or most of the lethality in these products.

1.11.1 Rendering

Rendering refers to the extraction of edible and inedible fats and oils from meat after slaughter. The rendering process can be either wet (usually through steam) or dry. It yields products such as tallow and lard.

1.11.2 Summer Sausage

Summer sausage is a semidry sausage. It is not as dry as the “hard” sausages such as salami and pepperoni. This product is made RTE by a combination of fermentation (pH), smoking, cooking, and drying. Just one or two of these steps is not enough to produce a safe, shelf stable product.

The formulation for summer sausage typically includes lean beef, lean pork, sage or ground mustard seed, salt, sugar, pepper, and sodium nitrite. It may include a starter culture. The meat ingredients are ground, and seasonings and preservatives are added. Then the mixture is stuffed into casings. Next it is placed in a cooking chamber and heated at a low temperature (100–110°F) to quickly ferment the product. Often a smokehouse is used, although generally no smoke is applied. When the required pH is reached, the temperature is raised, and product is heated to a desired temperature for lethality. This also kills off the lactic acid bacteria and stops the fermentation. The product is then placed in a drying room until it dehydrates to the specified level. Sometimes a smokehouse is used as the drying room.

1.11.3 Snack Sticks and Jerky

Different types of snack sticks are made (Figure 1.12). Some are fermented, with either a low or high moisture level. Some are not fermented, with a very low moisture level. They are usually ground, and are similar to a dry sausage. Jerky is not fermented; it is often beef, but turkey and other products are also found in the marketplace. Jerky may be made from either solid pieces of muscle or chopped product. Processing steps are similar to dry sausage and summer sausage:

- *Acidifiers* such as citric acid, lactic acid, and glucono delta-lactone may be used to reduce the pH of the product.
- *Antioxidants* such as butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), or propyl gallate may be used to prevent oxidation and rancidity.

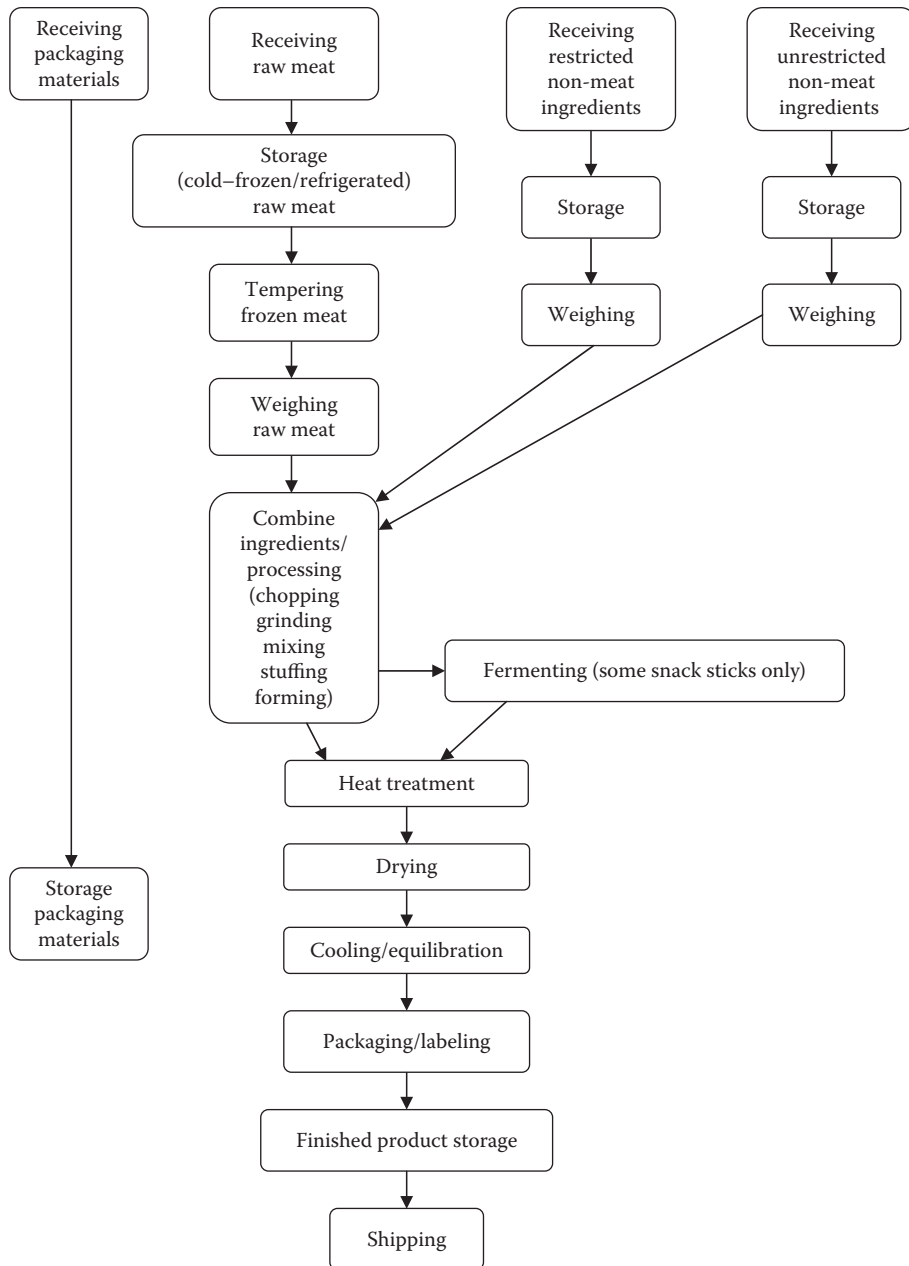


FIGURE 1.12 Flow chart for jerky or snack sticks. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

- The *thermal processing* step, which involves cooking the product for a specified amount of time, is designed to control bacterial growth, both spoilage and pathogens. The cooking process also helps to shorten the drying time.
- The *drying* step helps reduce the moisture in the product to the desired level. Reducing the moisture helps control *Trichinae* and enteric pathogens such as *E. coli* O157:H7 and *Salmonella*.

FSIS has established *product standards* for certain dried products that specify an MPR needed to achieve shelf stability and ensure the product meets the standard established for that product. For example, a “dry sausage” must have an MPR of 1.9:1 or less, in order to be labeled as a “dry sausage” and also ensure shelf stability. Likewise, a nonrefrigerated, semidry, shelf stable sausage must have an MPR of 3.1:1 or less and a pH of 5.0 or less to ensure shelf stability. However, for most type of salt-cured dried products, the water activity (available moisture) of the product is the primary factor affecting shelf stability and safety.

These products may be stored and shipped at frozen, refrigerated, or ambient temperatures. Storing product below ambient temperature is usually done for quality reasons.

1.12 Thermally Processed: Commercially Sterile

This processing category includes canned meat products, products in reportable pouches and semirigid containers. Common examples are stew, chili, soup, canned hams, Vienna sausage, hash, potted meat product, and pasta sauce with meat. Although there are several types of packaging options available, such as pouches, plastic cups and plastic pans, the metal can is still the most common package used. For this reason, these products are usually referred to as “*canned*.”

This category contains all products that have been thermally processed in order to achieve commercial sterility. The term *commercially sterile* does not mean that the product is completely sterile. Complete sterility is not achievable with canned products, because the *thermal processing* required to assure absolute sterility is so severe that the quality of the product would suffer. Certain types of microorganisms survive the thermal processing, but remain dormant or are inhibited from growth by some other factor. Thermally processed, commercially sterile product is RTE. It can be eaten directly from the container, although most consumers heat the product as a matter of personal taste.

1.12.1 Thermal Process

Our example product is pasta with meat sauce (Figure 1.13). You will notice some familiar processing steps, such as receiving, storage, assembling ingredients, and formulation. Because we have covered steps very similar to these earlier, we will not cover them now. Let us focus on some of the items that are unique to thermal processing:

- *Filling*: involves adding product, which has been mixed and formulated, to the product container. The typical container is a metal can. In most operations, filling is done with high-speed machines.
- *Types of cans*: There are several types of cans used most frequently in the meat industry. Three basic types of cans are round sanitary, drawn aluminum, and oblong. The *round sanitary* can holds a variety of meat-base products, such as pasta with meat sauce. *Drawn aluminum* cans are used most often for Vienna sausage and meat spreads. Their use is increasing because they use less metal and are more economical. *Oblong* cans are used for canned luncheon meats. To prevent an interaction between the meat product and the metal, cans are generally *coated* on the inside. The kind of coating that is typically used with meat products is sulfur-resistant. It prevents the sulfur released from meat proteins during retorting from staining the tinplate black.
- *Sealing*: Lids are placed on the filled cans and a *hermetic* seal is formed. This seal prevents air from getting in or out. It is this seal which will preserve the integrity of the can after it has been thermally processed, so it is critical that the seal is formed correctly.
- *Thermal processing*: involves placing the filled and sealed cans in a *retort* so that they can be thermally processed. A retort is a steel tank in which metal crates or baskets containing the cans are placed for subsequent cooking and cooling. The retort operates under pressures of 12–15 pounds per square inch (psi). This pressure cooking raises the cooking medium (usually steam) to temperatures above the normal boiling point of water, 212°F. For 3 min at 250°F is an example of a minimum process. The retort subjects product to a high temperature for a sufficient duration to destroy the organisms that might adversely affect consumer health, as well as more resistant organisms that

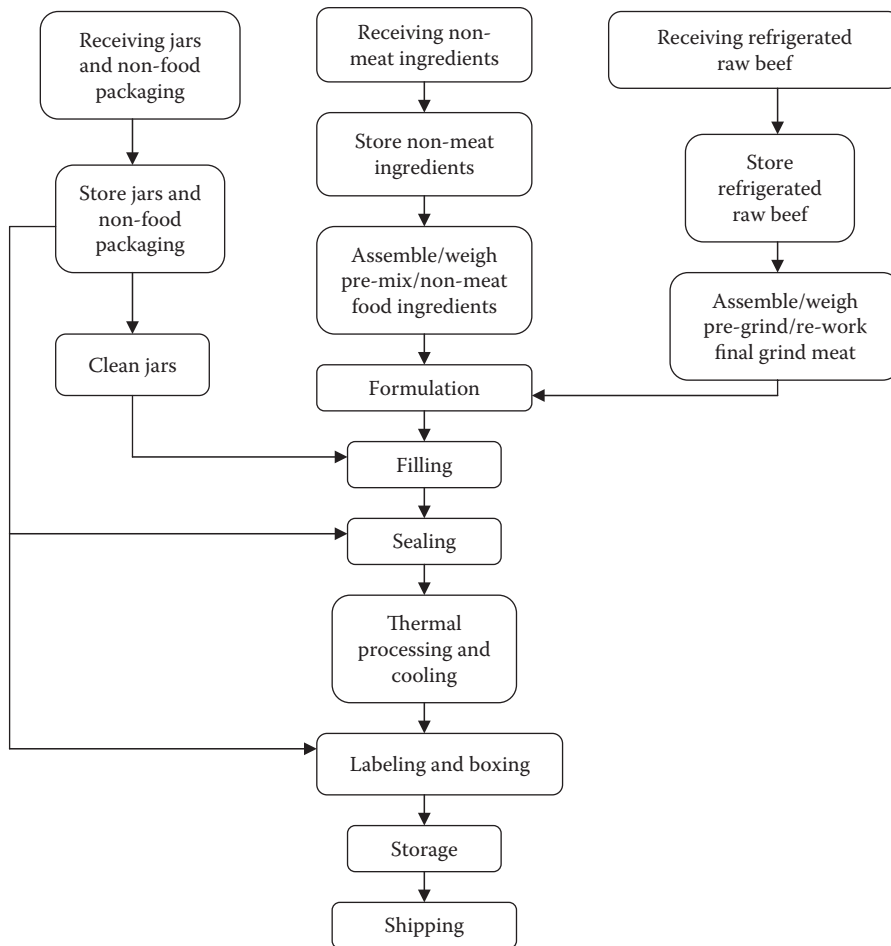


FIGURE 1.13 Flow chart for pasta sauce with meat. (Modified from *Meat and Poultry Safety Manual* published by Science Technology System (STS) of West Sacramento, California. Copyrighted 2011©. Used with permission.)

cause spoilage under normal storage conditions. These procedures are based on the destruction of all vegetative cells and all spores of the deadly, toxin-producing bacteria *Clostridium botulinum*.

- **Cooling:** After the heat process has been completed, all canned meat products should be cooled as quickly as possible to stop the cooking process and to lower the temperature below the range at which any heat-tolerant bacteria can grow. Because the cans are wet, it is best to permit some heat to remain to evaporate any remaining moisture, in order to prevent rust. When cans are being cooled, they contract and even well-made seams may permit some inward leakage. Therefore, the water used for cooling must be as near sterile as possible. Canning cooling water is chlorinated, and rust inhibitors may be added.
- **Incubation:** After product is cooled, some establishments will hold it for 10–30 days before the cans leave the establishment. This holding time is used to test a sample of the retort load. A sample of each processing lot is held at a temperature that would promote bacterial growth. This is done in an *incubator*. At the end of this incubation period, cans are examined for evidence of spoilage. If none is found, canned products are shipped.
- **Storage:** Canned meats should be stored in a cool, dry place because relative humidity and temperature influence their shelf life. Canned meat and poultry will maintain quality 2–5 years if the can remains in good condition and has been stored in a cool, dry place.

Here are some terms you commonly hear with regard to thermal processing:

- *Critical Factor*: Any characteristic, condition, or aspect of a product, container, or procedure that affects the adequacy of the process schedule. Critical factors are established by processing authorities.
- *Processing authority*: The person(s) or organization(s) having expert knowledge of thermal processing requirements of canned foods and utilizing procedures recognized by the scientific community as being adequate to properly calculate and assign thermal processes.
- *Process calculation*: Scientifically defined procedures that determine the process time and temperature as adequate under specific conditions of manufacture for a given product.
- *Process deviation*: Any change in a critical factor of the scheduled process that reduces the sterilizing value of the process.
- *Scheduled process*: The time, temperature, and critical factor controlled process, selected by the process authority and scientifically determined to yield commercial sterility under conditions of manufacture for a given product.
- *Sterilizing value*: This is normally expressed as " F_0 " and is the number of minutes required to destroy a given number of microorganisms at a given temperature. The F_0 value is used to compare the sterilizing values of different processes.
- *Vacuum*: Removal of air from the can to prevent oxidation of the product.

This category contains many different types of canned product. Although we discussed metal cans, you will also see plastic cups and flexible pouches on the grocery shelf. These items are processed in a similar fashion. This category groups all products that are thermally processed in order to achieve commercial sterility.

Part II

Meat Science

2

Muscle Biology

Amanda D. Weaver

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2.1 Introduction

Meat is comprised of numerous tissue types, including nervous, adipose, and connective tissue. Skeletal muscle tissue generally constitutes between 50% and 70% of carcass weight of meat animals, and subsequently, most of the value. Muscle, the living precursor to meat, is a uniquely complex tissue that assumes many forms in order to execute its various functions. Given that muscle is converted into meat postmortem, an understanding of the underlying biology that regulates this tissue antemortem is critical. This chapter will provide a brief introduction to muscle types and introduce the topics of muscle growth and development.

2.2 Types of Muscle

The majority of meat derived from carcasses of domestic livestock species is composed of striated skeletal muscle. However, the living animal depends on the functions of three distinct types of muscle to maintain life: smooth, cardiac, and skeletal muscles. These muscles can be distinguished by their primary functions, structures, and nervous control (Table 2.1). The following discussion will focus on the properties of skeletal muscle as it is used for food; however, a brief description of these three types of muscle will be given for comparison.

2.2.1 Smooth Muscle

Smooth muscle is a critical component of numerous animal tissues providing elasticity to the walls of arteries, linings of the gastrointestinal, reproductive, urinary and respiratory tracts, as well as the

TABLE 2.1

Characteristics of the Three Different Mammalian Muscle Types

Muscle Type	Striated	Nuclei per Muscle	
		Fiber or Cell	Nervous Control
Skeletal	Yes	Many	Voluntary
Cardiac	Yes	1–2	Involuntary
Smooth	No	1	Involuntary

lymphatic system. These cells vary widely in their size and shape, but a basic spindle form is common. Smooth muscle cells contain only a single nucleus that is typically located near the center of the cell. As in skeletal muscle, the proteins actin and myosin in smooth muscle are the basis for contraction and movement; however, these proteins are not organized into an ordered, striated appearance as in skeletal muscle. Contractile action of smooth muscle is controlled involuntarily via the autonomic nervous system and therefore conscious thought is not required to generate movement.

2.2.2 Cardiac Muscle

Cardiac muscle is located only in the heart where it provides rhythmic contractions throughout an animal's life. In a similar way to skeletal muscle, cardiac muscle appears striated when viewed under a microscope. However, cardiac muscle contains structural features not observed in skeletal muscle tissue, including branched fibers and intercalated disks, which separate adjacent cardiac muscle cells, and provide the ability to physically communicate contractile force directionally throughout the heart. Similar to smooth muscle, cardiac muscle cells generally contain a centrally located nucleus and are controlled involuntarily by the autonomic nervous system.

2.2.3 Skeletal Muscle

Skeletal muscle is the most abundant type of muscle in the animal body. Contractile, structural, and regulatory proteins in this muscle type are highly organized into a distinct striated pattern. Skeletal muscle is so named as it is attached to the skeletal framework of the animal, in various configurations, producing different types of levers and ultimately movement. Skeletal muscle fibers (cells) can range in length from 10 μm to more than a few centimeters. To provide genetic material along the whole length of skeletal muscle fibers, these specialized cells are multinucleated. Nuclei in mature skeletal muscle fibers are located along the periphery of the fiber. Skeletal muscle contraction is directly stimulated by somatic efferent nerves and therefore this muscle type is often referred to as voluntary muscle (Goll and others 1984; Gerrard and Grant 2006a).

2.3 Muscle Growth and Development

Although skeletal, smooth, and cardiac muscles take on widely different characteristics and functions in their mature form, early events leading to their development are similar. The development and subsequent growth of muscle is the product of complex cellular changes that accompany myogenesis and the growth of this dynamic tissue pre- and postnatally. To more efficiently produce muscle that results in high-quality meat products, it helps to have an understanding of these events and the processes by which they can be manipulated.

2.3.1 Myogenesis

The embryonic cells that ultimately become muscle, with few exceptions, are derived from the mesodermal layer of the developing embryo, a layer that also gives rise to fat and bone tissue. As mesodermal cells become more prominent, they begin to organize into cuboidal clusters known as somites. Further

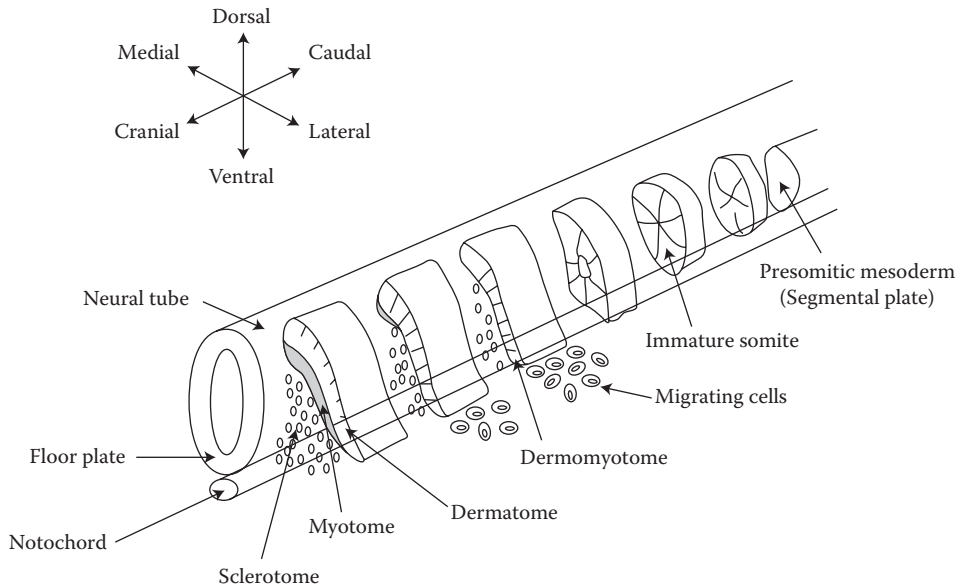


FIGURE 2.1 Formation and organization of somites begins as cells of the mesoderm organize into somites along the neural tube. Somites further mature into myotome, dermatome, and sclerotome layers which delaminate and migrate to become muscle, skin, and the vertebral column, respectively. This embryonic event occurs in a cranial to caudal fashion. (Reprinted from *Cell*, 78/1, Buckingham M., Molecular biology of muscle development, p. 16, Copyright (1994), with permission from Elsevier.)

organization of the somites gives rise to the dermomyotome and the sclerotome (Figure 2.1). Cells of the sclerotome eventually migrate and become the vertebral column while the dermomyotome cells separate to form the dermatome and the myotome, with cells of the dermatome becoming the dermis while cells of the myotome migrate to different regions of the body and give rise to muscle tissue.

The expression of one or more genes of a family of transcription factors, referred to as myogenic regulatory factors (MRFs) which are members of the basic helix–loop–helix (bHLH) family of proteins, initiates transcription of muscle-specific genes. This process ultimately results in committing mesodermal cells to the myogenic lineage. As cells express these key genes and become committed to the muscle cell lineage, they are termed myoblasts. Myoblasts are bipolar, spindle-shaped, mononucleated cells that are capable of progressing into the cell cycle and increasing in number through proliferation. However, to complete the developmental process and ultimately produce functioning muscle, cell differentiation must occur for myogenic cells to attain the level of specialization and organization required for contraction. At this point in development, myoblasts are signaled to cease dividing, align with one another, and begin to fuse together to form multinucleated myotubes (immature muscle fibers). Simultaneously muscle-specific genes, such as those encoding myosin and actin, are upregulated (Stromer and others 1974; Hawke and Garry 2001; Buckingham and others 2003; Biressi and others 2007). Figure 2.2 illustrates the progress of muscle fiber development.

Four key MRFs responsible for regulating myogenesis are Myf5, MyoD, myogenin, and MRF4 (Figure 2.2). Myf5 and MyoD possess key regulatory functions in the process of myoblast proliferation and differentiation and are more highly expressed in undifferentiated myoblasts. Myf5 is the initial MRF to be expressed in the developing embryo, and initiates early myoblast differentiation followed closely by expression of MyoD. Myogenin and MRF4 are considered secondary MRFs as they are responsible for terminal differentiation and maintenance of differentiated myofibrils. Differentiation must occur for the developing cell to become a specialized muscle cell. MRF4 expression occurs later during myogenesis and is the most predominant MRF to be identified in adult skeletal muscle (for reviews, see, Weintraub 1993; Perry and Rudnicki 2000; Pownall and others 2002; Sabourin and Rudnicki 2000; Houba and tePas 2004).

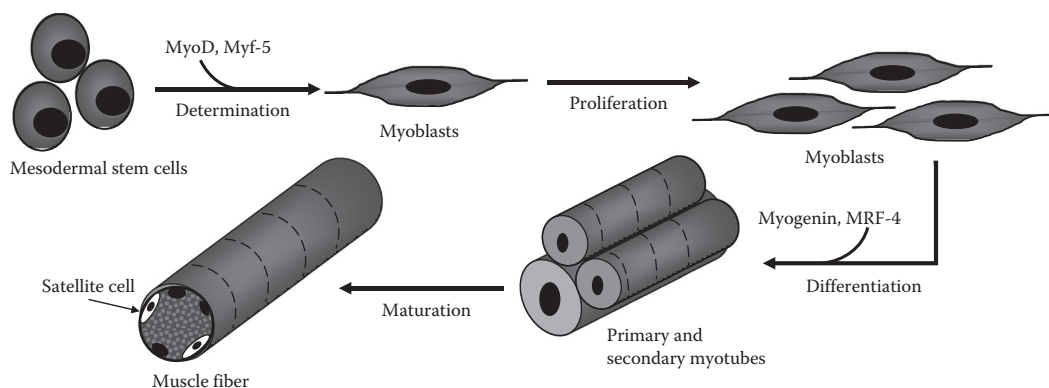


FIGURE 2.2 Events of myogenesis.

2.3.2 Muscle Fiber Formation

Muscle fibers express a biphasic developmental pattern in most domestic livestock species, resulting in two distinct populations of fibers, primary and secondary, in the developing fetus. Primary myofibers are generated first, via the fusion of myoblasts, into a structure known as a myotube. Primary myotubes then function as a structure for the organization, alignment, and fusion of other myoblasts into a larger population of secondary myofibers. Primary myofibers form during the initial stage of myogenesis (from approximately months 1–3 postconception in bovine fetuses) while secondary myofibers form during the second wave of myogenesis in the fetal stage (approximately months 3–7 postconception in bovine fetuses Du and others 2010). Histologically, primary myofibers are larger in diameter than secondary myofibers. Both structures initially possess centrally located nuclei; however, in contrast to secondary fibers, primary myofibers express slow myosin ATPase activity. Secondary myofibers maintain communication with their “parent” primary myotubes through membrane-associated proteins that form gap junctions between the structures. However, this association is lost following the innervation and contraction of primary myotubes which elicits the detachment of secondary fibers from the primary. Following separation, a new population of myoblasts can utilize the primary myotube as a template for fusion and development of additional secondary fibers (Figure 2.3). This well-orchestrated production of secondary myofibers gives rise to approximately 90% of the muscle fibers in most domestic livestock species. It is important to note the role of innervation in this process. Innervation is not required for the formation of primary myotubes. However, because contraction of primary myotubes is required for the development of secondary myofibers, innervation is a controlling factor modulating overall myogenesis (Ashmore and others 1972; Swatland and Cassens 1973; Swatland 1994; Gerrard and Grant 2006b; Du and others 2010).

Collectively, these biochemical and morphological events of fetal muscle development occur primarily during the first two-thirds of gestation and for most species are completed by birth. In other words, fiber number is essentially set at birth, with few exceptions. Therefore, events or conditions that enhance or impede embryonic or fetal development can dramatically impact fiber number and ultimate muscle mass of the individual.

2.3.3 Postnatal Muscle Growth

Muscle can increase in size by one of three processes: hypertrophy, hyperplasia, or accretion. Hyperplasia refers to an increase in fiber number resulting in an increase in muscle size, while hypertrophy is defined as an increase in fiber size, also resulting in an increase in muscle size. Muscle can also increase in size by the accretion of material, such as fat and connective tissue, between cells; however, this discussion will focus on the mechanisms of hypertrophy and hyperplasia.

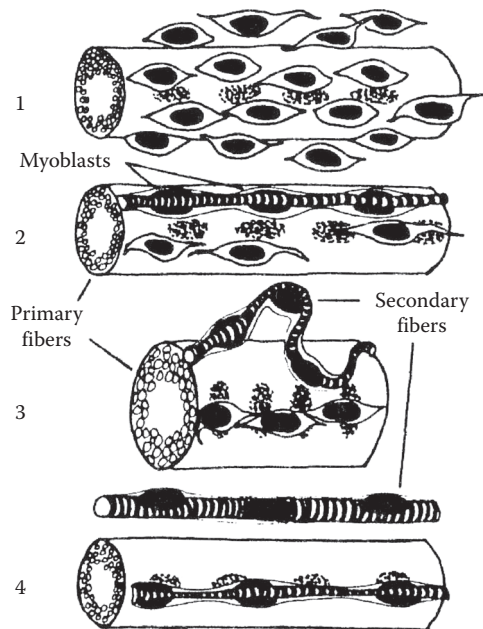


FIGURE 2.3 Stages in the development of secondary muscle fibers. (1) Proliferating myoblasts surrounding a primary fiber, (2) myoblasts fuse together to form a secondary fiber on the surface of the primary fiber, (3) contraction of the primary fiber causes detachment of the secondary fiber, and (4) secondary fiber is released and a new secondary fiber is forming on the surface of the primary fiber. (Reprinted from Swatland HJ. 1994. *Structure and Development of Meat Animals and Poultry*. Boca Raton, FL: CRC Press. pp. 287–363. With permission.)

In most domestic livestock species, muscle fiber formation is completed by birth or shortly thereafter and remains essentially stable. Therefore, achieving muscle growth postnatally is dependent on the enlargement of fibers developed prenatally. This hypertrophic growth of muscle fibers is mainly a function of protein synthesis and protein degradation. The difference between the amount of protein synthesized and the amount degraded is termed net protein accretion. During growth, the rate of protein synthesis must exceed the rate of degradation to generate a net accumulation of protein and ultimately an increase in muscle mass. In contrast, when protein degradation exceeds the amount of protein synthesized, muscle atrophy results. The relative rates of synthesis and degradation of muscle proteins influences the longitudinal and radial growth of muscle fibers (Beitz 1985; Young 1985; Trenkle 1986; Marple 2003).

Muscle fibers have a diameter of 10–20 μm soon after formation, and grow to a normal mature size of 50–80 μm in diameter. To increase in diameter, muscle fibers go through a process known as longitudinal splitting of myofibrils. Myofibrils are the filamentous organelles within muscle fibers, composed of contractile units known as sarcomeres. These organelles split longitudinally to form two new myofibrils which in turn enlarge and become functional via the synthesis of additional contractile, regulatory, and structural proteins. This process of producing new myofibrils is known as myofibrillogenesis. Radial growth of muscle fibers is stimulated by exercise or work and is commonly termed exercise- or work-induced hypertrophy. Conversely, longitudinal growth of muscle fibers is advanced by the stretching of muscles, generally through associated bone lengthening. This type of growth is referred to as stretch-induced hypertrophy and is accomplished through the process of sarcomere addition. Sarcomeres are the contractile units of muscle and are arranged end to end in a series or “box-car fashion.” Newly synthesized proteins are organized into new sarcomeres and generally added to the end of existing myofibrils resulting in an increase in fiber length (Williams and Goldspink 1971; Stromer and others 1974; Gerrard and Grant 2006a,c).

Both types of muscle fiber hypertrophy require synthesis of new proteins. This process involves transcribing DNA into mRNA within the nucleus, transporting mRNA out of the nucleus and into the

sarcoplasm for translation into protein, and finally arranging the newly synthesized protein into all the structures within the muscle fiber. It is important to note that this process not only occurs in the case of muscle hypertrophy. In fact, a critical and continuous balance exists between protein synthesis and protein degradation in muscle cells. Proteins are regularly assembled, positioned in the cell, and eventually replaced with a newly synthesized protein, in a process known as protein turnover.

The disassembly of proteins from muscle fibers is mediated by protein degradation pathways that involve the systematic proteolysis of proteins into peptides and ultimately into amino acids by proteases. The three main endogenous protease systems in skeletal muscle are the ubiquitin proteasome, the lysosomal system, and the calcium-dependent protease (calpain) system. Of these, the calpain system is responsible for a majority of protein turnover in living muscle and is also the most accepted candidate for postmortem tenderization (proteolysis) of meat. The best-characterized members of the calpain family include μ -calpain and m-calpain and their specific inhibitor calpastatin. The μ and m forms of the enzyme are cytosolic, ubiquitously expressed, and require micromolar and millimolar calcium concentrations, respectively, for activation (Goll and others 2003).

Lysosomes are cellular organelles containing proteolytic enzymes, such as the cathepsins, that function best in a low-pH environment such as that of a lysosome. These enzymes are capable of degrading proteins into small peptides and amino acids, but are responsible for relatively little protein turnover in muscle due, in part, to the limited number of lysosomes present in muscle. In addition, the role of this enzyme system in muscle is further hampered by the fact that cathepsins cannot degrade muscle proteins outside the lysosome and lysosomes cannot engulf intact myofibrils. Therefore, degradation by the lysosomal system in muscle is limited primarily to sarcoplasmic proteins (Goll and others 1983; Allen and Goll 2003).

The third protease system is the ubiquitin proteasome. The proteasome is a large protein complex that resembles a hollow cylinder with the catalytic site residing inside the cylinder. In contrast to the lysosomal system, the proteasome is highly concentrated within muscle cells, which means that its regulation is critical to prevent unintended degradation. The first line of regulation lies with the requirement of ubiquitination. In this process, ubiquitin is covalently attached to proteins indicating they are to be targeted for degradation. Polyubiquitinated proteins and/or polypeptides are then transported to the proteasome and preferentially degraded. The second control point for this system is the entrance to the cylinder itself, which is very small (1–1.3 nm) in diameter restricting the entrance of large proteins or protein complexes. Because of this restriction, large proteins or complexes must first be degraded into smaller polypeptides by another protease system, such as the calpain system. Research has shown that the proteasome and the calpain system work in cooperation to identify (target) and degrade myofibrillar proteins (Robert and others 1999; Allen and Goll 2003; Houbak and others 2008).

2.4 Regulation of Muscle Mass

In addition to understanding the processes involved in muscle fiber development and growth, researchers in the field of livestock production should also be aware of the numerous mechanisms by which these processes can be manipulated. Some of the most important factors affecting muscle fiber development and growth processes are discussed below.

2.4.1 Factors Influencing Muscle Fiber Hyperplasia

As aforementioned, muscle fiber number is essentially set at birth; therefore, manipulation of fiber number must occur prenatally. One exception to this rule is the case of muscle fiber death as new muscle fibers may be generated postnatally following severe injury resulting in fiber loss. This regeneration is achieved by the activation of satellite cells (quiescent, mononucleated muscle cells located between the basal lamina and sarcolemma), which proliferate and fuse with other satellite cells to form new fibers. These specialized cells can also contribute to muscle growth when a hypertrophic stimuli is applied such as resistance training. During such a challenge, satellite cells are activated, proliferate,

and fuse to existing myofibers resulting in growth. For a more detailed description and discussion of satellite cells, readers are referred to Mauro (1961), Hawke and Garry (2001), Cornelison (2008), and Relaix and Marcelle (2009). Although the ability to regenerate muscle fibers in this way is notable, as a mechanism for growth, this process does not contribute significantly to variation in the mass of muscle grown.

A number of intrinsic and extrinsic factors have been demonstrated to have an impact on the number of muscle fibers formed during the prenatal period. For example, larger species by nature express increased numbers of muscle fibers. Excessively large fibers pose challenges for nutrient and metabolite diffusion so that, by necessity, muscle fiber diameter is relatively constant between species and the difference in muscle mass between species is due primarily to variation in muscle fiber number. Research has also indicated that within species, different breeds express variation in fiber number, and gender differences sometimes exist for this trait as well (Table 2.2; Rehfeldt and others 1999; Wegner and others 2000; Gerrard and Grant 2006c). Also, gene mutations have been reported to result in hyperplasia and extreme muscularity.

The myostatin gene mutation, responsible for the “double-muscling” phenotype in cattle, is one such example. This member of the transforming growth factor family is expressed throughout myogenesis and is also detectable in postnatal skeletal muscle (Kambadur and others 1997; McPherron and others 1997). Mutations in the gene can vary, but ultimately result in a gene product (protein) with altered function. Myostatin appears to modulate myoblast proliferation and differentiation. Therefore, when myostatin expression is mutated in certain ways a substantial increase in muscle fiber number, and consequent muscle mass, is manifested. Belgian Blue cattle exhibit a high frequency of a myostatin deletion that results in an extremely muscular phenotype. In fact, Belgian Blue cattle with the mutated gene have an approximately 20% increase in skeletal muscle as compared with normal cattle (Shahin and Berg 1985; Muroya and others 2009).

Another key factor in determining muscle fiber number is maternal nutrition. This area of study was recognized after the Dutch Famine of World War II and is again receiving much attention. The concept of fetal programming asserts that nutritional status of the dam can influence the offspring during fetal development and throughout the subsequent postnatal period (Du and Zhu 2009). Nutrition during gestation programs fetal muscle development, along with other tissues and health parameters (Barker and others 2002). Furthermore, it has been demonstrated that the genetic potential for muscle growth cannot be fully realized if maternal nutrition is restricted beyond a certain level (Du and others 2009). Therefore, much of the current research in this area is focused on the use of this concept as a management tool for maximizing muscle yields of livestock species. The influence of maternal nutrition on muscle fiber number has been demonstrated in several livestock species (Dwyer and others 1994; Zhu and others 2004; Quigley and others 2005; Du and Zhu 2009) and while collectively these studies show a tendency for decreased muscle fiber number (specifically secondary muscle fiber number) more research is necessary to fully understand this area of growth biology.

TABLE 2.2

Comparison of Muscle Fiber Number between Genders of Various Species

Species	Muscle	Effect
Mouse	<i>Soleus, Tibialis anterior, Biceps brachii</i>	None
Rat	<i>Soleus</i>	None
Human	<i>Tibialis anterior</i>	Male > Female
Chicken	<i>Extensor longus</i>	Male > Female
Pig	<i>Longissimus</i>	None
Cattle	<i>Longissimus</i>	Male > Female

Source: From Rehfeldt C. et al. *Basic Appl Myol* 9:235–253. With permission.

2.4.2 Factors Influencing Muscle Fiber Hypertrophy

As discussed previously, muscle fiber size increases either by radial or longitudinal growth or both. Similar to muscle fiber hyperplasia, numerous factors can influence this type of growth, including sex, nutritional status, growth promoters, and genetic mutations.

Intact males demonstrate larger muscle fiber cross-sectional areas than females or castrated counterparts. This response is largely attributed to high circulating levels of the hormone testosterone which enhances both protein synthesis and degradation (Rehfeldt and others 2004; Therkildsen and Oksbjerg 2009). Although both processes are higher in intact males, protein synthesis differences are greater, resulting in net protein accretion. Specific muscles are also more sensitive to the influence of testosterone, such as the muscles of the crest of the neck in cattle.

Muscle fiber size is also responsive to nutritional status of the animal. Typically individuals placed on a high plane of nutrition develop muscles with larger diameters than those on a low or restricted plane of nutrition as nutrients are available for the synthesis of new proteins. It is also important to state the influence of nutrient partitioning as it influences the order of nutrient utilization by various body systems. The endocrine system coordinates the distribution of nutrients to physiological systems and tissues. Of the body systems, the highest priority during growth is given to the brain and central nervous system followed by the circulatory, respiratory, digestive, and reproductive systems. Hierarchy is also established for tissues, with the bone receiving first priority for nutrients during growth followed by muscle and adipose tissues (Owens and others 1993; Gerrard and Grant 2006d).

The callipyge mutation in sheep is a genetic condition resulting in extreme muscularity, primarily in the hind legs. This mutation is markedly different from the myostatin mutation in cattle. Sheep expressing this genotype develop fibers with much larger diameters than normal sheep after birth. Although these lambs demonstrate increased feed efficiency, muscle mass and lean product yield without increasing the risk of dystocia at birth, their use commercially is not widespread as the resulting meat is inherently tough when compared to normal lamb (Koohmaraie and others 1995; Jackson and others 1997a,b). Inadequate tenderness in this case is associated with elevated levels of calpastatin in the hypertrophied muscles. As indicated above, calpastatin is the endogenous inhibitor of the calpains which are credited with degrading structural proteins postmortem during the aging process. Therefore, increased levels of calpastatin limits the ability of the calpains to degrade proteins and improve tenderness during aging. However, in the living tissue, the increased protein synthesis and decreased degradation result in a net increase in protein accretion and ultimately larger muscle fibers and muscles (Koohmaraie and others 1995; Duckett and others 1998).

Finally, growth promoters such as β -adrenergic agonist and steroid implants have garnered much research focus since their introduction. These compounds work through various pathways to accomplish the common goal of increased muscle fiber size (Clancy and others 1986; Fritsche and others 2000; Gonzalez and others 2007; Kellermeier 2009) and will be further discussed in later chapters of this volume.

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3

Meat Composition

Robert G. Kauffman

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3.1 Preface

Anyone who has an interest in meat should know something about what it consists of, that is, what the pieces are, how much of the whole each piece represents, and how to measure each piece. Furthermore, the interested and educated person really should know from where meat originates and how to measure its composition and what causes it to vary. This is what this chapter is about. When you have finished reading it, I hope you will have a better understanding of where meat comes from, what it consists of, why its consistency varies, and how to assess it. This information should help provide a better understanding of how to use meat as a food, how to make it taste better and safer for

consumption, and how its properties can be best utilized for further processing, storage, and distribution. Specific references are given for further details on specific topics (Allen and others, 1976, Hedrick and others, 1993, Kauffman, 1971, Kauffman and St. Clair, 1978, Kauffman and others, 1986, Kinsman and others, 1994).

3.2 Introduction

3.2.1 Definitions and Measurements

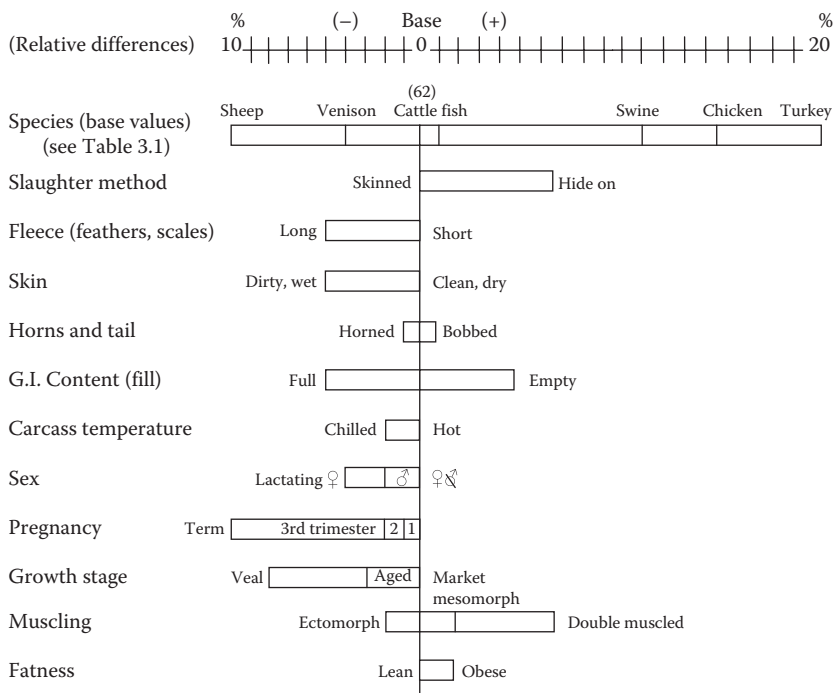
In the broadest sense, meat is the edible postmortem component originating from live animals. For the purposes of this text, these animals include domesticated cattle, hogs, sheep, goats, and poultry, as well as wildlife such as deer, rabbit, and fish. It is reasonable for the definition of meat to include such organs as heart and liver (often defined as variety meats), but the focus of this chapter is on meat defined as those tissues exclusively originating from an animal's carcass, a proportion amounting to about one-half to three-fourths of the animal's live weight. This carcass proportion of the live animal weight is classically calculated as dressing percentage and can vary considerably. Some species, such as the turkey, can yield a carcass weighing about 80% of the live weight, whereas a market lamb's yield is closer to 50%. Animals with small and empty gastrointestinal tracts (such as hogs or poultry rather than ruminants) that are not pregnant, that are more heavily muscled and fatter, that do not have long fleeces or dirty hides, and that have been slaughtered in a manner that leaves the skin and feet intact with the carcasses (hogs) will have higher dressing percentages (see Figure 3.1).

Excluding the skin, the carcass component of live animals basically consists of three parts: muscle, fat, and bone. Of these, muscle is the most important, constitutes the majority of the weight, and often is considered unequivocally synonymous with "meat." This can be a reasonable assumption, but fat deposits and some bones are often processed, merchandized, and used along with muscle and must be included in the broader definition of meat. Figure 3.2 is included as an example of the relative composition (in specific detail) of market animals and is representative of a live mature beef steer. From this information, one can calculate the proportions of any one part to the various larger component parts. For instance, the longissimus muscle represents (approximately) 51% of the back muscles, 12% of all carcass muscles, 7% of the carcass, and 4% of the live animal. These values can vary depending on species, degree of fatness, and other similar factors affecting dressing percentage. However, it provides a relative guide that reflects the composition of live animals and how it is related to the meat component. Furthermore, Figure 3.2 indicates that "meat" has its origin in many muscles of the carcass. In closer observation, one can deduce that some muscles contribute considerably more to meat than others, and that is because they vary in size and shape, dependent directly on the biological functionality.

Composition is defined as the aggregate of ingredients, their arrangement, and the integrated inter-relationship that forms a unified, harmonious whole. Figure 3.2 is an example of this. For market animals raised to produce meat for human consumption, the greatest emphasis is on the musculature and its relationship to everything else. The proportion of the animal's musculature is related to several criteria, the three most important being dressing yield, fatness, and muscling (expressed in terms of ratio of muscle to bone). Realistic averages of composition for most meat animals are included in Table 3.1. Muscle varies from 25% (lamb) to 50% (turkey) of the live weight and muscle to bone ratio varies from 1.8 (chicken) to 5.0 (venison).

There are several arithmetic approaches to expressing quantitative composition, but the one most commonly used is the weight of a part expressed as a percentage of a larger part, such as % muscle of a retail cut of meat, or % protein of a muscle. Another less-used technique is to express the part and the whole as logarithmic functions of each other.

Measuring composition can vary from subjective techniques to ones precisely objective. Even when a technique is considered objective, at least some subjectivity inadvertently prevails. Here are some of the more commonly used approaches for determining gross composition (% lean) of meat cuts.



¹ Some factors are interrelated such as species, slaughter method, and fleece.

² Condemnations (due to bruising, disease, contamination, etc.) and weighing errors are not included but could change yields slightly to excessively.

³ Interpreting Figure 3.1:

- A. For a skinned, hot, lean carcass of a market weight, polled heifer, the base dressing percent would be 62%. The heifer had an empty GI tract so add 5% to base. She was 8 months pregnant so subtract 8%. Her hide was wet and dirty so subtract 5%. She was heavy muscled (mesomorph) so add 2%.

$$\text{Approximate net dressing \%} = 62 + 5 - 8 - 5 + 2 = 56\%$$

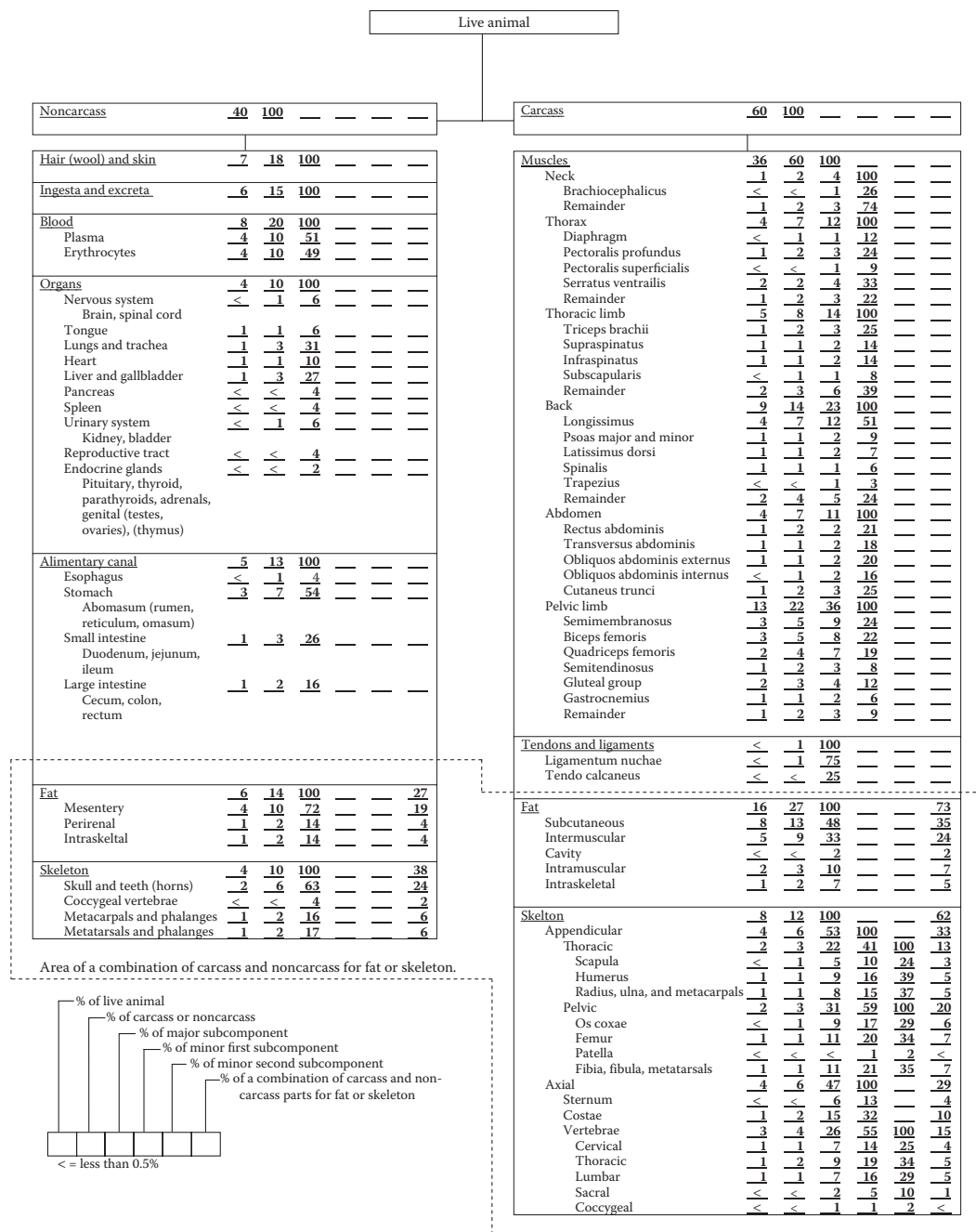
- B. For a lean carcass of a market weight wether lamb, the base dressing percent would be 52%. The wether had a full GI tract so subtract 5% from the base. He had a long fleece so subtract 5%. The wether was a double muscled Texel so add 7%. The carcass was chilled so subtract 2%.

$$\text{Approximate net dressing \%} = 52 - 5 - 5 + 7 - 2 = 47\%$$

FIGURE 3.1 Twelve factors affecting dressing percentage of meat animals.

3.2.1.1 Visual Appraisal

Every time consumers purchase cuts of meat, they often select them on the basis of their lean/fat/bone ratios as estimated by visual inspection. This is simply accomplished through visual comparisons. Quantitatively, the method lacks accuracy, but for practical purposes in meat selection, it is effective, especially when compositional variations are large. On a more detailed basis, visual scores can be established with photographs and then the meat cuts scored on proportions of lean. However, scoring is too subjective to reflect quantitative differences and is too difficult to standardize to be consistently applied over time. Perhaps the one greatest value of visual inspection is in the estimation of proportions of intramuscular fat (marbling) when determining gross composition by dissection (in which dissecting marbling is essentially impossible).



Key

The data represents cattle, swine, and sheep. Prototype for this figure was cattle and appropriate carcass proportion of all components for swine and sheep should be 70% and 50%, respectively, and then adjusted proportionately. Each number following a part is its percent of larger parts by weight. Example: scapula is <0.5% of live animal, 1% of carcass, 5% of carcass skeleton, 10% of appendicular skeleton, 24% of thoracic limb skeleton, and 3% of combined skeleton parts. Sum of percents within a group = 100%. Parts in parenthesis may be absent due to species, sex, or stage of growth.

FIGURE 3.2 Approximate proportion of components of meat animals.

TABLE 3.1

Gross Compositional Variations among Animal Species

Live Weight, kg	Beef 550	Veal 160	Pork 110	Venison 70	Lamb 50	TomTurkey 16	Broiler Chicken 2	Farm-Raised Catfish 0.7
<i>Average Proportions of Live Weight</i>								
Noncarcass (%)	38	46	27	42	48	18	23	37
Carcass skin (%)	^a	^a	5	^a	^a	9	9	^a
Carcass fat (%)	17	7	23	10	17	6	7	^a
Carcass bone (%)	10	15	9	8	10	17	22	12
Carcass muscle (%)	35	32	36	40	25	50	39	51
Total	100	100	100	100	100	100	100	100
Dressing yield (%)	62	54	73	58	52	82	77	63
Carcass muscle/ bone ratio	3.5	2.1	4.0	5.0	2.5	2.9	1.8	4.3

^a Included with noncarcass component.

3.2.1.2 Linear Measurements

A simple, inexpensive ruler can be used to measure subcutaneous fat thickness, muscle depth and width, and bone length and thickness. From these measurements, areas of each component can be estimated and then expressed as a proportion of the whole. Unfortunately, the areas are not exactly accurate, nor are unexposed bones and seam fat of different dimensions (as well as marbling) included in the estimate.

3.2.1.3 Area Measurements

By tracing the areas of the exposed muscles, bones, and fat on acetate paper and then measuring the exact areas of each component with a compensating polar planimeter, composition can be estimated. Also, such areas can be assessed using photometric, electronic, and computerized imaging techniques. Even though this is more accurate in determining the exposed areas, it has the same limitations as visual appraisal or linear measures because unexposed bone and seam fat as well as marbling cannot be accounted for.

3.2.1.4 Density

The Archimedean principle suggests that cuts of meat displace a volume equal to their own. Because the density of fat is less than that of muscle, such a technique, even though destructive if water displacement is used and expensive if gas displacement is used, would provide an accurate estimate of lean. However, for meat cuts containing bones, the technique would not be satisfactory. The density of bone is nearly twice that of muscle and would bias the estimate of muscle.

3.2.1.5 Anyl-Ray

This is based on x-ray attenuation as an index of tissue fatness. It generates electromagnetic waves of a character sensitive to absorption and reflection or back scatter by the elements in ground meat. The radiation is directed through a sample where size, shape, compaction, and weight must be constant while sample composition varies. When calibrated intensity of radiation is directed through a sample, this energy intensity is directly proportional to sample composition. A carefully mixed and selected ground meat sample is subjected to a minute amount of carefully controlled x-rays. Because lean absorbs more x-rays than fat, there is a difference in energy transmitted. The penetrating rays are collected by a radiation-measuring device, which in turn energizes a calibrated digital percentage fat meter. In the past,

it has been used to determine fat content in ground meat used for processing. The method is fast, requires a sample that can be reutilized for processing after measurement, and has a high degree of accuracy. The instrument correctly evaluates meat at any temperature, provided it can be properly compacted in the container. Further grinding or the addition of warm water may be necessary to achieve proper compaction for frozen samples. However, it is a relatively expensive method for determining the composition, and it cannot be used for small meat cuts or ones containing bone, and it estimates only fat and no other specific chemical components.

3.2.1.6 Dissection

This is the one most effective method of determining the gross composition of whole carcasses or individual wholesale or retail cuts. The method can be standardized and is highly repeatable in application. The method requires knowledge of anatomy and the patience and care to separate each component, preventing weight loss through evaporation and drip, and weighing and recording accurately. However, it does not account for variations of marbling, which would have to be assessed visually or subjected to chemical evaluation.

3.2.1.7 Proximate Analysis

For animal tissues, the primary chemical components used as a follow-up to or an alternative for physical dissection includes moisture, protein, lipid, and ash. The procedures for chemically analyzing each of these are described in the association of official analytical chemists (AOAC 1999). A major concern in using this method is adequate mixing and sampling of the tissues to be analyzed. Another limitation is in chemically analyzing bone because of the difficulty in grinding and sampling. (This does not apply for the ash determined in muscle.) Also, mixing ground components of soft tissues creates problems of fat collecting on the sides of the mixer. Finally, when moisture is being assessed, large errors occur when muscles are soft and watery or when excessive evaporation of moisture is lost from the surfaces of unprotected samples.

To determine the detailed chemical composition of muscle, fat, or bone, such as specific minerals, myofibrillar proteins, fatty acids, individual vitamins, and bound versus free water, numerous detailed and often extremely difficult, expensive, and sensitive chemical and spectrophotometric procedures are required. These procedures are not identified and described here because of their complexities and the need to maintain brevity.

3.3 Description and Composition of Muscle and Its Modifiers

3.3.1 Description

Meat animals contain, as a majority of their carcass weight, many muscles distributed in an unusually designed pattern to move the skeleton, for posture control, and for more specialized functions such as respiration, swallowing, and peristalsis. This musculature is categorized into two major types: striated and nonstriated. The less voluminous nonstriated or smooth muscles have some similar functions as striated muscles but possess different histological structures. Smooth muscles are primarily found in the linings of the gastrointestinal tract and the circulatory system as well as in specialized organs such as the gizzard of birds.

Striated muscles are categorized as either cardiac or skeletal. Cardiac muscles are confined to the heart and have the continuous responsibility of distributing and collecting blood throughout the body. Structurally, they are similar to skeletal muscles, except that they are more highly aerobic in their metabolic properties and therefore require higher concentrations of oxygen for their rhythmic contractions. Skeletal muscles are, as the name implies, associated with the skeleton; they either lie next to a bone or are attached to various bones, either closely or indirectly through their connective tissue fascia that may

attach directly or indirectly by tendons to distant bones. Depending on function and needs, skeletal muscles contract and relax and have very exacting cross-banding patterns.

Skeletal muscles play a major role in locomotion and posture control as well as in protecting vital organs. On average, the meat animal carcass contains about 100 bilaterally symmetrical pairs of individually structured muscles. There are large ones and small ones, depending on function and location. They have different shapes, colors, and concentrations of tendons. Many have a fusiform, multipennate shape, having a large middle portion that tapers at the ends. The attachments contain large quantities of tendinous connective tissue that attaches to bone. The long head of the triceps brachii would be an example of a fusiform-shaped muscle. Other shapes include flat or sheet-like muscles such as the cutaneous trunci, round-shaped muscles such as the quadriceps femoris, and irregular shapes such as the tensor fasciae latae, which has more than two attachments and is somewhat triangular shaped with thick and thin portions. In the more distal portions of the limbs, small muscles are uniquely attached to tendons for the specific purpose of either flexing or extending the feet and legs. In the more proximal locations, the muscles are larger and primarily serve as major sources of power. This is particularly true of the pelvic limb musculature. There are less than ten major pelvic muscles, whereas there are twice as many of smaller size in the thoracic limb. The longissimus thoracis et lumborum is the longest and largest muscle in many species and is located in the back to support the axial skeleton and to extend and erect the vertebral column. The flat muscles, generally located in the abdominal wall, support the abdominal cavity and its contents. Commercially, the flat muscles have less economic importance but they are the only ones found in bacon.

Skeletal muscles have a complex composition because they contain, in addition to muscle fibers, large quantities of supportive connective tissue, a complete vascular supply, and a nerve supply controlling each of the billions of muscle fibers. Also, skeletal muscles serve as storage depots for lipids and contain considerable quantities of extracellular fluids, primarily consisting of water.

Postmortem muscles vary in color, ranging from a dark purplish-red to a pale, light gray. This variation is primarily the result of myoglobin concentration as well as other biological factors such as pH. Myoglobin is a protein physiologically important in the transfer of oxygen and carbon dioxide to and from muscles during their normal metabolic activities. Breast muscles of poultry (pectorales superficiales) are very pale or white in color and contain low quantities of myoglobin, whereas leg muscles of venison are extremely dark purple and contain more than twice as much myoglobin. Striated muscles are multinucleated, distinguishing them from smooth muscles, which are mononucleated. These nuclei are near the sarcolemma cell wall whereas in smooth muscles, the nuclei are more centrally positioned. Skeletal muscles contain mitochondria, but not as many as are found in cardiac muscle. Other organelles such as ribosomes and the Golgi apparatus are also found in muscle fibers. Each fiber is surrounded by an intricate membrane, the sarcolemma, which surrounds the sarcoplasm that bathes the myofibrils, which are the contractile units of the fiber. Lipid particles in the form of neutral droplets and free fatty acids as well as glycogen granules are distributed throughout the sarcoplasm. (In postmortem muscles, glycogen is metabolized to lactic acid.) Enzymes are located in mitochondria and in other portions of the sarcoplasm. The sarcoplasmic reticulum and transverse tubules are responsible for the storage and transportation of calcium for contraction and relaxation.

To permit muscles to function properly as moving forces, they are harnessed to the skeleton through a unique set of connective tissue structures. This connective tissue “harness” circumvents the entire muscle and is called the epimysium; it winds its way through each muscle, dividing fibers into groups called fascicular bundles. The connective tissue at this level is called perimysium. The perimysium subdivides further into endomysium, which lines each fiber. The vascular system, which winds its way through muscles to supply the nutrients and remove toxic wastes, is closely related to individual fibers. In both the extracellular spaces and within fibers, there are fluids high in water content. In addition to the water, there are minerals, some water-soluble proteins, nonprotein nitrogenous materials, and other organic entities. Lipid in the form of neutral triglycerides is stored in the adipose tissue cells, which accumulate around venules and arterioles in the interfascicular spaces. This fat, when visible, is called marbling. Excluding water, the major components in muscle are the contractile proteins, which make up the myofibrils.

3.3.2 Gross Composition

A simpler approach to assessing the composition of muscles is to use proximate analyses to quantitate moisture, protein, lipid, ash, and carbohydrate. Muscle composition varies considerably and the accumulation of lipid is the most influential on this variation. On average, most muscles should contain about 1% ash (primarily represented by the elements potassium, phosphorus, sodium, chlorine, magnesium, calcium, and iron), 1% carbohydrate (primarily glycogen antemortem and lactic acid postmortem), 5% lipid, 21% nitrogenous compounds (predominantly proteins), and the rest (72%) as water. These values are compared to the composition of fat and bone as shown in Table 3.2. Some muscles may contain as much as 15% lipid (fresh weight basis), whereas others may contain <2%. Regardless of the lipid content, the protein/moisture ratio of about 0.3 remains quite constant for mature muscles. If time and expenses are limited, one may quickly, easily, and somewhat accurately assess proximate composition of muscles by making a few assumptions, using moisture analysis for the only determination. If it is assumed that ash and carbohydrate will not vary greatly and that their sum contribution is estimated at 2%, and if it is assumed that the protein/water relationship is 0.3, then if water is determined by homogenizing the sample and drying it, the only unknown left to be estimated is lipid content. This is calculated by difference. For example, if a sample (analyzed for moisture content) contained 70% moisture (M), then protein (P) content would be equal to $P/M = 0.3$, or $P/70 = 0.3$. Therefore, $P = 21$ or 21% protein. By subtracting the sum [2% (ash and carbohydrate) + 70% (M) + 21% (P)] from 100%, then lipid would be 7% or $[100 - (2 + 70 + 21)]$.

3.3.3 Molecular Composition

There are a host of chemical compounds in muscle. They include free fatty acids, glycerol, triglycerides, phospholipids, nonprotein nitrogenous components such as DNA, RNA, ammonia, amine groups, and vitamins. There are glycogen granules and ATP. Myoglobin is present. Several minerals are present in minute quantities. Most important from a quantitative perspective are the various proteins of each fiber. These proteins are classified into four groups, the largest of which is myofibrillar. Myofibrillar proteins represent about 60% of the total proteins, whereas sarcoplasmic proteins represent 29%, stroma proteins 6%, and granular proteins 5%. Figure 3.3 is included to provide a detailed overview of the complexity of muscle composition. It is not intended to be precisely accurate nor to be memorized, but to serve as a reference to identify the various components of muscle and their quantitative contributions to its mass. It is assumed that these values represent mature, postrigor muscles of various species. Of all information presented in this chapter, Figures 3.2 and 3.3 should receive special attention because they are a detailed summary of the most important features of meat composition. (It required more time and effort to construct these figures than everything else combined in this chapter!) It should be understood that the methods of analysis used to determine most of the components of Figure 3.3 greatly affect the quantities reported.

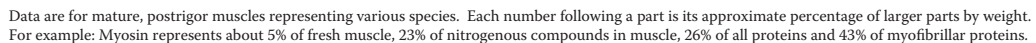
TABLE 3.2

Comparative and Approximate Gross Compositions of Muscle, Fat, and Bone^a

	Muscle	Fat	Bone
Moisture (%)	72	9	25
Nitrogenous compounds (%) (primarily proteins)	21	1	10
Lipid (%)	5	90	20
Ash (%)	1	< ^b	45
Carbohydrate (%)	1	< ^b	< ^b
Total	100	100	100

^a Proximate analysis expressed on a fresh basis for mature, postmortem tissues representing various anatomical locations.

^b <0.5%.



The myofibrillar proteins are responsible for the contractile mechanisms and thus shorten or lengthen the muscle for movement and support functions. Sarcoplasmic proteins are primarily represented by enzymes and myoglobin. Stroma proteins originate from the connective tissue structure found as a part of muscle, the most important quantitatively being collagen. Collagen is resistant to most enzymatic reactions except collagenase. When heated in water, collagen is converted into gelatin, which is readily hydrolyzed by several enzymes. About one-third of collagen's amino acid residues consist of glycine, whereas another one-fifth is proline and hydroxyproline. It is the only protein known which contains hydroxyproline, with the possible exception of reticulin. Hydroxyproline analysis is often used as a measure for determining the total connective tissue in muscles. Another stroma protein of less concentration is elastin. It is even more resistant to degradation: to degrade, it must be subjected to high temperatures in the presence of strong bases or acids. Elastin contains about one-third of its amino acid residues as

glycine and over one-tenth as proline. Reticulin is the other major stroma protein. Its amino acid composition is similar to that of collagen, and it is often considered a form of collagen that contains lipids and carbohydrates.

There are nine known major myofibrillar proteins, as illustrated in Figure 3.3. Quantitatively, the one most important protein is myosin. In referring to Figure 3.3, myosin represents 43% of the myofibrillar proteins, 26% of all muscle proteins, 23% of all nitrogenous compounds, and 5% of the fresh muscle mass. Myosin is the thick strand of protein that appears in the sarcomere structure. Actin represents about 22% of myofibrillar proteins and is the thin filament within this same contractile formation. The other seven proteins represent smaller compositional fractions, but play equally important roles in contraction. Titin represents 10% and has by far the largest molecular weight and is considered more structural than metabolic in function. Tropomyosin and troponin each contribute about 5% and can be found attached to the actin molecule and are primarily responsible for initiating contraction after calcium has been released by the sarcoplasmic reticulum. All the other proteins combined represent <20% of the weight.

All the above mentioned proteins are composed of the 22 amino acids shown in Figure 3.3. Each amino acid is different according to the molecular characteristics of its side chain. The 10 essential and 12 nonessential amino acids and their mole contributions to muscle mass are included in Figure 3.3.

In addition to the proteins, there are other important nitrogenous constituents in muscle. First are the vitamins, which are divided into two classes based on their solubility in either aqueous or nonaqueous solutions. The lipid-soluble vitamins are minimal because of the small quantities of fat normally deposited in most muscles. However, water-soluble vitamins, primarily the B vitamins, are present in substantive enough quantities to serve as appropriate sources to meet daily dietary requirements for humans. They include thiamin, riboflavin, niacin, pyridoxine, pantothenic acid, biotin, folic acid, and B12. Ascorbic acid (vitamin C) and calcium are essentially absent in muscles, and because of this, muscles are not considered a perfect food from a nutritional perspective. The nitrogenous, nonprotein extractives include creatine, nucleotides, ammonia, methylamines, free amino acids, and other derivatives of proteins. Two of the components in highest concentrations are carnosine and anserine. Other extractives include volatile organic carbonyls, such as acetyl aldehyde, acetone, carbon dioxide, and formaldehyde, all of which have been found in muscles. Various sulfur compounds include hydrogen sulfide, methylmercaptans, and methyl sulfides.

The elemental components include carbon, hydrogen, and oxygen in great abundance either because of their molecular weight or number of molecules and are listed in Figure 3.3. In addition, nitrogen is abundant because it is a component of all proteins. Some minute quantities of sulfur are present in the form of the amino acids cystine, cysteine, and methionine. Inorganic minerals include calcium, magnesium, sodium, potassium, chlorine, phosphorus, and iron, but their contributions to mass are minimal. In assessing the various elements in the various components of muscles, in most instances—whether proteins, lipids, carbohydrates, vitamins, or nucleic acids—the elements carbon, hydrogen, and oxygen are always present. The unique compositional difference among proteins, nucleic acids, and vitamins is that in proteins, nitrogen molecules are in the side chains; in the other two groups, nitrogen molecules are incorporated into the ring structures. The protein myoglobin is somewhat of an exception in structure in that it contains a heme group as well as a globular protein fraction and contains iron as its central ion in the heme ring. The iron element in myoglobin is paralleled by the cobalt element in vitamin B-12.

3.3.4 Modifiers of Muscle Composition

General fatness of the animal influences the composition of muscles. Individual muscle fibers remain constant in their composition, but fresh muscle may vary from 1% to 15% in lipid content. This variation is due to such factors as genetics, stage of growth, sex, and the amount of physical exercise. As animals mature and muscles stop growing, intramuscular fat may accumulate around the vascular system, thus decreasing the relative mass of other components. The nature of the connective tissue matrix also affects the accumulation of fat. Loosely arranged muscles such as the latissimus dorsi, having parallel connective tissue strands, contain more fat than tightly compacted muscles such as the peroneus longus. The latter's connective tissue strands are thicker and more tightly structured, thus physically preventing excess fat accumulation.

Nutrition affects muscle composition simply by controlling the total lipid accumulation, depending on the total caloric intake and expenditure. In submaintenance diets, fat is mobilized (rather than deposited) from muscles. Quality of nutrition can also affect the mineral and vitamin content of muscles, but not to the extent that fat deposition is affected.

Stage of growth affects the protein/moisture relationship of muscles. In very young animals, this ratio is low (~ 0.1), whereas at maturity, the relationship is about 0.3. As already indicated, this remains reasonably constant throughout the animal's life and serves as a reliable guide in estimating the composition.

In addition to the structural differences in connective tissue, anatomical location of muscles affects composition because some muscles contain higher concentrations of tendon and epimysial sheaths of connective tissue. Because of this, there is a difference in the quantity of stroma proteins as compared to myofibrillar, sarcoplasmic, and granular proteins. For example, lower limb muscles have higher concentrations of connective tissue proteins than do supportive back muscles. Even though the molecular nature of stroma proteins changes during growth, the absolute quantities do not change. Some muscles such as the gluteus medius and longissimus have proportionately more white fibers requiring less oxygen. Therefore, their energy needs for muscle contraction are more anaerobic than that of muscles containing more red fibers. Consequently myoglobin concentration is lower and this may be true for fat content as well. An exception to this is the trapezius. It contains over 60% red fibers but also contains high amounts of lipid. The semitendinosus contains two clearly defined portions, one having predominantly red fibers and the other predominantly white fibers. As a result, molecular composition within this muscle varies considerably. However, in this example, the white fiber portion contains considerably more lipid than the red fiber portion, suggesting that muscle location and function affect composition more than fiber type per se. Perhaps fiber type affects composition primarily by its effect on postmortem tissue characteristics. The postmortem musculature originating from short-term stressed animals (especially those genetically susceptible to stress) become soft and watery and are much more susceptible to exudation during processing. Therefore, composition is readily affected if processing is considered. Dark, firm, and dry (DFD) muscles that contain high concentrations of red fibers (often the result of long-term antemortem stress) are less susceptible to such abnormal postmortem shrinkage.

Disease influences composition. Portions of muscles may be eroded away by muscular dystrophy and replaced with fat. Certain inorganic elements are lost from the tissues during stressful conditions related to disease. Certain central nervous system diseases also affect the general composition of muscle, primarily affecting the fat component. Injury to muscles affects composition. When major nerves are severed (accidentally or experimentally) the muscle atrophies and fat accumulates in the vacated spaces.

Exercise stimulates fiber hypertrophy and mobilization of lipids within muscles. However, there is little evidence suggesting changes in other chemical components.

Genetics affects fatty accumulation in muscles because of its relation to rate of maturity. Certain species of animals, such as the domestic duck, deposit very little fat in muscles. The rabbit has similar tendencies, whereas certain breeds of pigs, cattle, and sheep deposit large quantities of intramuscular fat. Within species, some breeds have greater tendencies to deposit intramuscular fat. Duroc swine appear to contain more intramuscular fat for a given degree of body fatness and age. Hereford and Charolais cattle do not deposit as much intramuscular fat at a given physiological state of maturity as do Angus cattle.

Fat in muscles is related to the total fatness of the body. When carcasses from obese animals are examined, there is generally more intramuscular fat than from those possessing leaner carcasses as compared to total body fat per se (e.g., Japanese Wygue cattle producing highly marbled Kobe beef), whereas other animals deposit large quantities of subcutaneous fat but deposit very little intramuscular fat.

Muscles grow at different rates and mature at different physiological times. This in itself affects composition. These differences are small, but if a muscle matures earlier and also has the structural potential for accumulating fat, then it will have a higher fat content at a given age than another muscle that matures at a later stage. This variation is also responsible for differences in protein/moisture ratios among muscles.

Control of various body processes by the endocrine system affects fat deposition in muscles. Thus by the presence or absence of testosterone, fat deposition is regulated in muscles. Bulls, rams, and boars possess muscles containing less intramuscular fat than male castrates (steers, wethers, and barrows). When cattle are fed diethylstilbestrol (a synthetically produced hormone that is currently banned from

use), it changes the composition of muscles simply by slowing down the animal's physiological time clock. The supplementation of certain hormones will increase fat mobilization in muscles; however, most of these changes are small.

3.4 Description and Composition of Fat and Its Modifiers

3.4.1 Description

Fat is often associated with such words as obese, plump, oily, wasty, greasy, big, and thick. Also it has been defined as the excess "white stuff" that is trimmed from a piece of meat, or the grease that lubricates a skillet, or some of the juices that flow out of hamburgers during cooking, or the foodstuff that is responsible for making foods "rich" because of its high caloric content. Fat has probably received more attention from consumers, and has been referred to more often, than any other single biological substance.

Epithelial, nervous, muscular, and connective are the four basic tissues involved in the processes of postnatal growth. Connective tissue's primary function is structural support: it is responsible for the physical shape of such biological substances as bone, cartilage, muscle, and fat. Adipose tissue is a type of connective tissue that surrounds synthesized lipids, which serve as heat-cold insulators and as reserve supplies of body energy. Therefore, fat is defined as a collection of adipose cells suspended in a matrix of connective tissue distended with cytoplasmic lipids, water, and other constituents.

Often the words fat and lipid are used interchangeably. Generally this is appropriate, but specifically it is an incorrect concept. Adipose or fatty tissue contains lipids, but lipids per se do not contain connective tissue, water, enzymes, and other constituents present in fat. However, because lipids are the major components of fat, it is important to describe these lipids in greater detail. Lipids include that group of nonpolar compounds soluble in organic solvents but insoluble in water. Pure lipids are colorless, odorless, and flavorless and can be classified as follows.

3.4.1.1 Simple Lipids

Simple lipids are esters of fatty acids with certain alcohols such as glycerol. If lipids are solid at room temperature, they are called fats; if liquid, oils. Waxes are simple lipids that are esters of fatty acids with long-chain aliphatic alcohols or with cyclic alcohols. Examples of waxes include esters of cholesterol as well as the vitamins A and D.

3.4.1.2 Compound or Conjugate Lipids

Compound or conjugate lipids are esters of fatty acids that, on hydrolysis, yield such substances as phosphoric acid, amino acids, choline, carbohydrates, and sulfuric acid, in addition to fatty acids and an alcohol. Examples include phospholipids, glycolipids, sulfolipids, and lipoproteins [such as low-density lipoprotein (LDL) found in blood].

3.4.1.3 Derived Lipids

Derived lipids are formed in the hydrolysis of simple or compound lipids. Examples include saturated and unsaturated fatty acids, aliphatic alcohols, sterols, alcohols containing the β -ionone ring, aliphatic hydrocarbons, carotenoids, squalene, and the vitamins D, E, and K. Fat is found in nearly every anatomical location imaginable, but the great majority of it occurs subcutaneously, inter- and intramuscularly, in the mesentery, on the walls of the thoracic, abdominal and pelvic cavities, and in bone marrow (intraskel-etal). Fat is deposited in the udders of females and in the scrotal sacs of male castrates. Fat is deposited in brain, liver, and kidney, and the quantity may be excessive under abnormal conditions (diseases).

Lipids are found in some form in all body cells because phospholipids contribute to the structure of every cell wall. Blood and lymph contain lipids, the quantity varying greatly with time after an animal consumes a fatty meal. All dietary fats are transported to body tissues via one of these routes. Although

adipose tissue is ubiquitous, it is not evenly and universally distributed in obesity, but is deposited in certain preferential sites while other sites are spared. For example, feet, eyelids, nose, ears, and genitalia seldom accumulate excess fat.

3.4.2 Gross and Molecular Composition

Quantitatively, Table 3.3 represents an example of variations in proportionality of fat in different anatomical locations and it is expressed on both a dissectable and an extractable basis. For this example, as the animal matures, the total fat increases when compared to other body tissues, but for the most part the proportionate distribution remains reasonably constant. On an extractable basis, the internal cavity fats (cavity wall and mesentary) represent about one-tenth, intermuscular about one-fifth, intramuscular and intraskeletal about one-fourth, and subcutaneous over one-half of the total lipids deposited, regardless of stage of growth. Even though intramuscular and intraskeletal lipids are not included in the dissectable allocation, the total representation of extractable lipids (as a percentage of empty body weight) are lower than for the dissectable fat values because water is not included when extractable portions are expressed. One must remember that fat contains significant quantities of water. Table 3.2 includes the gross composition of fat and how it compares to that of muscle and bone.

Adipose cells vary in size depending on such factors as age, species, and state of nutrition. The increase in numbers of adipose cells does not necessarily dictate the quantitative amount of fat deposited. For mature pigs, about 45% of the total adipose cells are in intramuscular fat but this fat component represents <15% of the volume of extractable lipid. This is verification that cell hypertrophy contributes more to volume than does cell hyperplasia.

Lipids dominate in their contribution to adipose volume and weight. However, other constituents are also present. In immature tissues, there is a significant quantity of water. Also, because adipose tissue is structurally supported by a connective tissue matrix, the extracellular protein collagen is present. Other substances include enzymes responsible for lipogenesis and lipolysis, traces of certain minerals and minute quantities of glycerol, glucose, and glycogen as well as nervous tissue.

TABLE 3.3

Distribution of Pig Body Fat or Lipid During Growth

	Stage of Growth (Days of Age)		
	21–43	64–106	149–213
Dissectable Fat Basis			
Total fat as % of empty body	15	20	25
<i>Distribution by Location</i>			
Cavity wall (%)	5	5	7
Mesentary (%)	8	13	8
Intermuscular (%)	29	25	28
Subcutaneous (%)	58	57	57
Sum	100	100	100
Extractable Lipid Basis			
Total lipid as % of empty body	7	10	16
<i>Distribution by Location</i>			
Cavity wall (%)	2	3	6
Mesentery (%)	2	4	4
Intermuscular (%)	18	16	18
Subcutaneous (%)	51	54	52
Intramuscular (%)	20	14	11
Intraskeletal (%)	7	9	9
Sum	100	100	100

Adipose tissue lipids are primarily present as mono-, di-, or triglycerides. Each is composed of a molecule of glycerol bonded to one, two, or three fatty acids. These acids are either synthesized in the adipose cell or are synthesized in the liver and subsequently transported to the adipose cell via the circulatory system. Fatty acids in adipose tissue usually contain 16 or more carbon atoms, but there are a few that are shorter. The carbon chain may be completely saturated with hydrogen atoms, or there may be some double bonds, and these are called unsaturated fatty acids. The dietary origin of the fat dictates the variation expected and the iodine and saponification values reflect such variations. Oleic, palmitic, and stearic acids represent over 80% of the composition of meat animal lipids. However, the primary difference in physical properties of lard as compared to beef tallow is the higher quantity of linoleic acid that is more unsaturated and occurring in higher proportions in lard, giving it a softer structure at room temperature. Also, this higher degree of unsaturation results in a fat more susceptible to oxidative rancidity.

3.4.3 Modifiers of Fat Composition

The discussions that follow will parallel some of those already covered in the muscle section and thus will be limited to unique differences pertaining exclusively to fat. When compared with mature animals, young ones contain adipose tissue having considerably more water. Also, the phospholipid component is proportionally higher in young animals as compared with their triglyceride content.

For ruminants, most dietary fats are digested in the rumen where ingested unsaturated fatty acids are hydrogenated and then absorbed into the circulatory system. However, for monogastric animals, ingested unsaturated fatty acids are not hydrogenated and are absorbed and deposited in adipose tissue in their original structures. Therefore, this explains why pork fat is softer than beef tallow. The release of some hormones will stimulate mobilization of fatty acids. If this persists, the triglyceride fraction will be significantly reduced.

Anatomical location of fat is important when determining its composition. The number of intramuscular adipose cells represents nearly half the total adipose cell population of the body, yet the amount of extractable lipid is <10%. These cells are smaller and contain more water than do cells located subcutaneously. Also, within subcutaneous tissue, that portion located at the base of the pelvic limb contains less extractable lipid than the cells located over the back. Furthermore, the three distinct layers of subcutaneous fat over the back of pigs vary in fatty acid composition. The outer layer contains greater proportions of unsaturated fatty acids as compared with the other layers. Mesentary adipose tissue contains more saturated fatty acids than subcutaneous tissue; udder fatty tissues contain more fluid, nonlipid material, and less extractable lipids than other adipose tissues. Finally, certain organs such as the bovine kidney contains 30% oleic acid and 33% stearic acid as compared with bovine subcutaneous fat, which contains 40% oleic acid and <20% stearic acid.

Genetic variables influence the quantity of fat and its composition. Some examples include (a) more hydrogenated fatty acids in ruminants than monogastric species, (b) double-muscled cattle do not deposit fat as quickly as normal cattle, (c) some breeds of sheep accumulate greater quantities of fat over the rump, and (d) pigs contain more subcutaneous and less intermuscular fat than sheep or cattle. Females have the capacity to lactate, which is a unique process of fat accumulation in milk. Intact males of most species contain less fat than castrate males or females of similar chronological ages. Heifers contain more fat than steers at a given age whereas gilts contain less fat than barrows at a similar age. This observation may simply reflect variations in stages of compositional, physiological, and sexual maturity.

Atypical conditions such as obesity and steatosis (excessive fat deposition in muscles) increases the quantity of lipids deposited. Conversely, exercise and various environmental stresses reduce lipid deposition.

3.5 Description and Composition of Bone and Its Modifiers

3.5.1 Description

Bone is a complex tissue and subject to continual metabolic activity. A most obvious difference, when compared to muscle and fat, is its dense, hard, mineralized, cellular type tissue. The three cellular

components of bone are of one cell type and may change in morphological characteristics directly according to specific functional needs of the tissue. The cells involved include osteocytes which are responsible for maintenance; osteoplasts, which are involved in formation of new bone; and osteoclasts, which are responsible for mobilization and reabsorption of bone material. Histologically bone is characterized by its branching lacunae, which are cavity-like membranous materials, and by canaliculi, which are fine-structured canals. Bone contains a dense matrix of collagenous fibrous bundles in a ground substance encased with calcium and phosphorus. Bone is capable of structural alterations to accommodate stresses due to mechanical changes and biological demands incurred by pressure and by vascular, nerve, endocrine, and nutritional influences. Most of the rigid material in the skeletons of meat animals is either compact or cancellous bone. This indicates that there are different degrees of mineral density in the bone including available porous spaces which provide for the accumulation and maintenance of the marrow.

Another method of classifying bone is on the basis of bone formation. Some bones develop within mesenchymal tissue such as the skull, which is known as membrane bone. Other bones depend on prior cartilaginous scaffolding, such as the vertebrae, and this is called cartilage bone. This cartilage-type bone contains collagen and polysaccharides.

There are more than 200 individual bones in meat animals, and they are either on the axial skeleton or the appendicular skeletons (limbs). Figure 3.2, includes the major bones and their proportionate masses in the live animal. They all include bone marrow, which produces the majority of the red blood cells. They store minerals and mobilize them as needed for other body tissues. They repair themselves after an injury and are designed to provide the greatest support with a minimum amount of weight; this is why most bones have hollow structures.

3.5.2 Gross and Molecular Composition of Bone and Its Modifiers

Bone basically contains mineral deposited in an organic matrix. The matrix includes not only calcium, phosphorus, and carbonate, but also citrate, water, and small amounts of sodium, magnesium, potassium, fluorine, and chlorine. The crystals of bone mineral have a chemical composition similar to that of fluorapatite. Fibers of collagen run throughout the matrix. The remaining space in the bone “mortar” is filled with a semiliquid substance that exchanges materials to and from the bone mineral via the circulating blood.

More than 99% of the body calcium is in bones. Collagen is about 93% of the total organic portion of bone. There are small amounts of insoluble sclera proteins and ground substance that are composed of mucopolysaccharides and mucoproteins. In fat-free analyzed bone, the mineral content accounts for about two-thirds of the mass, whereas in fresh bone it is about two-fifths. As included in Table 3.2, water represents about one-fourth the mass, protein about one-tenth, and the remaining one-fifth portion (which is the most variable) is lipid. As indicated, type of bone, age of the animal, and species are three factors that most affect bone composition.

Bone ash is composed mostly of calcium and phosphorus and much lesser quantities of magnesium, sodium, potassium, chlorine, and fluorine. When comparing bone to fat and muscle as shown in Table 3.2, the average composition is considerably different. If bone is compared to the Achilles tendon (almost entirely connective tissue), the tendon consists of two-thirds water and one-third organic solids with very little inorganic material. This compositional profile is quite similar to muscle. For the ligamentum nuchae, which is slightly more similar to bone, water content is about 57% and organic solids make up most of the remainder, but the elastin content is considerably higher than that for tendon.

The factors modifying the composition of bone are quite similar to that of muscle and fat and will not be repeated here. As illustrated in Table 3.4, age, species, and type of bone are three major modifiers of bone composition. Other unique factors modifying bone composition are (a) absence of vitamins D and A from the diet, (b) abnormalities in endocrine secretions (both low and high quantities), (c) lack of mineral supplementation in the diet (especially calcium and phosphorus), and (d) wasting-type diseases that mobilize mineral content from the bone, creating brittle and friable structures that have been significantly altered in composition.

TABLE 3.4

Variation of Bone Composition When Comparing Species, Age, and Bone Type

		Moisture (%)	Lipid (%)	Ash (%)	Protein (%)	Sum (%)
Cattle	2–3 months	55	7	19	19	100
	48–96 months	27	17	47	9	100
	Costae	37	9	37	17	100
	Femur	34	21	39	6	100
Pigs	6–8 months	37	18	33	12	100
	24–48 months	25	23	43	9	100
	Costae	33	18	37	12	100
	Femur	26	28	43	3	100
Sheep	5–6 months	39	11	33	17	100
	48–96 months	23	16	49	12	100
	Costae	32	9	39	20	100
	Femur	24	23	47	6	100
Chickens	2–3 months	50	11	24	15	100
	12–13 months	39	9	34	18	100
	Costae	42	10	29	19	100
	Femur	44	12	30	14	100

Source: Data extrapolated from Field, RA. et al. 1974. *J Anim Sci* 39:493–499.

3.6 The Composition–Quality Paradox of Meat

When evaluating meat, both composition and quality are important. Leanness (as contrasted to fatness) is virtuous, but by itself, it fails to meet ultimate expectations of consumers. The nutrient density of muscle is higher in lean meat and nutritive value is a part of quality. Therefore, from this perspective, composition affects quality. However, quality is more than just nutrient density. Wholesomeness, appearance, water-holding capacity, and palatability are quality virtues too! Marbling contributes to juiciness and flavor, however, more marbling reduces nutrient density. Furthermore, the exterior fat covering of fresh meat cuts is related to marbling. The association is not strong, but fatter cuts usually exhibit muscles containing more marbling. The paradox is that meat animals are fed to heavier weights, for longer times, and to ultimately less favorable “feed-to-meat” ratios so that muscles will contain more marbling to ultimately satisfy the consumer demands. This negative relationship between marbling and leanness is difficult to compromise and is one of the reasons why beef and lamb cuts may be too fat.

In pork and turkey, trim and heavily muscled carcasses appear to be more susceptible to the pale, soft, and exudative (PSE) condition. Meat cuts from such carcasses possess lean, heavily muscled cuts containing minimum quantities of fat. Nevertheless, the muscles often shrink excessively during processing. Fresh cuts of pork (loin and ham) and turkey (breast) that are exceptionally lean may be pale in color, soft in texture, and watery, all of which detract from appearance and ultimately their acceptance by consumers.

Therefore, quality of all meat products must be considered along with composition when assessing the overall value. The conflict between composition and quality continues to challenge scientists to discover new genetic combinations, different feeding and management programs, and more satisfactory postmortem processing technologies to ensure an ideal meat product that meets consumer demands.

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4

Postmortem Muscle Chemistry

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4.1 Introduction

The conversion of muscle into meat is a complex process involving many biochemical and physical changes. Muscle tissue is converted from an extensible, metabolically active system into one that is inextensible and quiescent with regard to its biochemical reactions. The speed and extent of postmortem metabolism has a profound effect on the properties of the muscle and its subsequent use for food. Also the extent of shortening during the postmortem period affects the meat's textural properties. This chapter summarizes our understanding of muscle's structural components, its metabolic pathways, and its behavior postmortem. Several books¹⁻⁵ and reviews⁶⁻¹³ may be consulted for more extensive coverage of these topics.

4.2 Structure and Function of Living Muscle

4.2.1 Microstructure

4.2.1.1 Light Microscope Level

Skeletal muscle is the largest tissue component of meat animal species. A diagram showing the organization of muscle at various levels is shown in Figure 4.1. Whole muscle is usually connected to bone by tendons attached to a tough, nearly inextensible connective tissue layer called the epimysium. This layer is composed primarily of the protein collagen. The muscle is divided into bundles by thinner layers called the perimysium. Finally, each muscle cell or fiber is encased in a thin layer referred to as the endomysium.

The muscle cells when viewed longitudinally in the microscope have a striped or striated appearance. They were formed during embryonic development by the fusion of many precursor cells. The resulting muscle cells are typically long and cylindrical and contain numerous nuclei. The number of muscle cells remains relatively constant after birth. The increase in muscle mass during growth is due to large increases in cell length and diameter. Some of these precursor-type cells persist in the adult and are referred to as satellite cells. The satellite cells lie on the surface of the true muscle cells. A longitudinal view of a single muscle cell (double-headed arrow) is shown in Figure 4.2; this photograph is from muscle that has been fixed, paraffin embedded, sectioned, and stained. Note the nuclei (N) and the stripes that run perpendicular to the fiber long axis. Figure 4.2 also shows an adjacent capillary and blood cells (arrowhead).

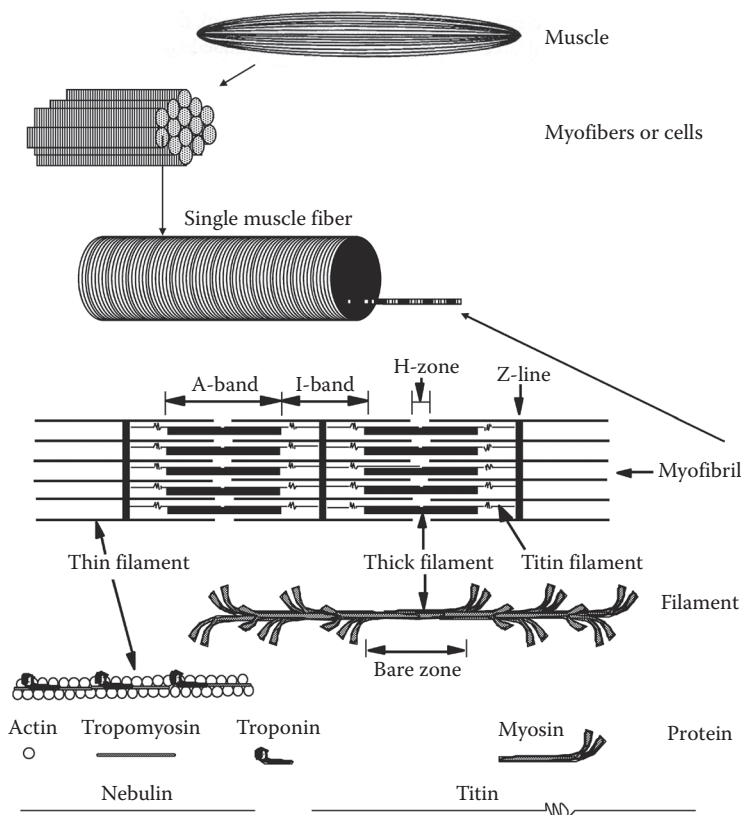


FIGURE 4.1 Diagram showing the levels of organization of muscle. (From Greaser, M.L. and A.M. Pearson. 1999. *Food Texture: Perception and Measurement*. pp. 228–258. Aspen Publishers, Inc., Gaithersburg, MD. With permission.)

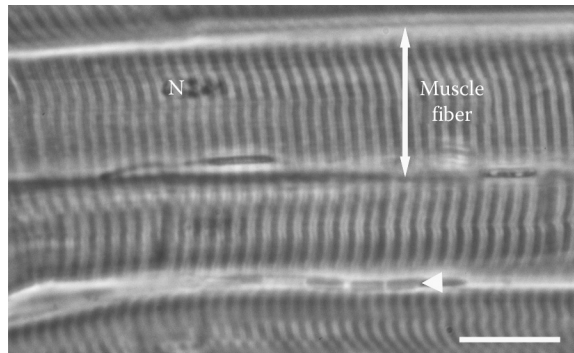


FIGURE 4.2 Light micrograph of a longitudinal view of several muscle fibers. This micrograph was prepared from muscle tissue that had been fixed, paraffin embedded, sectioned, and stained with hematoxylin and eosin. The double-headed arrow demarcates a single muscle fiber. The typical striped or striated appearance is visible. Each muscle fiber has multiple nuclei; one is shown with an N. Each muscle cell is adjacent to one or more capillaries (arrowhead) that supply oxygen and substrates for metabolism. Scale bar = 25 μm .

The alternating dark and light stripes are due to the presence of myofibrils inside the fiber. The individual myofibrils also have alternating stripes, and the striations in the fiber occur because the adjacent myofibrils have their respective light and dark bands aligned. The dark bands are called the A-bands and the light bands the I-bands (Figure 4.1). A thin perpendicular line referred to as the Z line bisects the I-bands. The banding structure is somewhat obscure in whole fibers and sections because the cell is too thick to have everything in focus. However, the myofibrils can be separately observed after disrupting muscle by homogenization and their band patterns are much clearer (Figure 4.3). The region between successive Z lines is called a sarcomere and it is the smallest functional unit of the myofibril. The lengths of the sarcomeres vary in suspensions of myofibrils due to their being derived from muscle cells which have varying degrees of shortening. The A-bands, if visible, always have the same length, but the I-bands decrease in length in myofibrils with shorter sarcomeres. Myofibrils with long sarcomeres have a zone in their middle that has somewhat lower intensity; this region is called the H zone.

4.2.1.2 Electron Microscope Level

An understanding of the filament arrangement that is responsible for these patterns became apparent with the advent of electron microscopy. The elegant work of Hugh Huxley showed that the sarcomere is composed of two major types of filaments.¹⁴ The thick filaments (about 1.6 μm in length) are found in the A-band and they interdigitate with thinner filaments (about 1 μm in length) that attach to the Z lines (Figures 4.1 and 4.4). Muscle contracts by a relative sliding of these two filaments over one another. The filaments are in turn composed of proteins; myosin is the major constituent of the thick filaments and actin, troponin, and tropomyosin make up the bulk of the thin filaments. Two other narrow, filamentous proteins are present in the sarcomere but are not typically visible by electron microscopy of intact muscle. Titin, a giant protein that extends from the middle of the sarcomere to the Z line (Figure 4.1), is elastic in nature and believed to be important for myofibril assembly and for protecting the muscle from over-stretch.

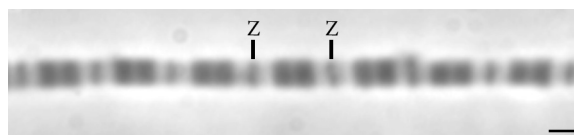


FIGURE 4.3 Light micrograph of a bovine psoas myofibril. The banding patterns visible in the intact cells are also seen in the myofibril. Alternating dark A-bands and light I-bands can be observed. The I-band is bisected by a thin line called the Z line. The section of a myofibril between a pair of Z lines is called a sarcomere. Scale bar = 1 μm .

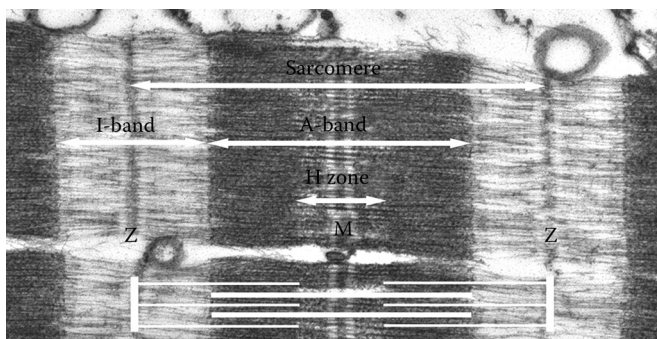


FIGURE 4.4 Electron micrograph of a longitudinal view of muscle. The banding pattern can be explained by the positions of the thick and thin filaments. The thin filament length is approximately 1 μm .

Nebulin is attached to the Z lines and is postulated to regulate the length of the thin filaments. These latter two proteins are sometime referred to as “cytoskeletal” proteins.

The sarcomere and its filaments are highly ordered in cross-section as well as longitudinally. The thick filaments are arranged in a hexagonal pattern. Six thin filaments surround each thick filament, and each thin filament is centered between three thick filaments.

4.2.1.3 Muscle Cell Heterogeneity

In spite of the high degree of filament order in muscle, there is considerable heterogeneity in the properties of the individual muscle cells. The striking color difference between the breast muscle of a chicken or turkey compared to the muscle of the thigh or leg is readily apparent. Early classification work grouped muscle cells into two groups, red and white.¹⁵ The red muscle cells have more myoglobin (responsible for the color), have slower contraction speed, and typically rely on oxidative metabolism for ATP generation. The white muscles have more glycolytic enzyme content, have a faster contraction speed, and have an energy metabolism that depends on glycolysis. However, it soon became apparent that this simple two-type system for muscle fiber classification was inadequate. The best current fiber-type classification system is based on myosin isoforms (either using type-specific antibodies or histochemistry after various acid or alkaline pretreatment of muscle cross-sections). Type I fibers belong to the red group; type IIA, type IIB, and type IIX (or IID) constitute the white group. However, even this nomenclature is muddled by the fact that many fibers contain more than one myosin type. In most mammalian muscles, the fiber types are mixed even down to the fiber bundle level.

An example of the fiber type heterogeneity is shown with cross-sections from pig muscle (Figure 4.5). In Figure 4.5a, a frozen muscle section has been stained for the mitochondrial enzyme succinic dehydrogenase. The red, oxidative type I fibers having a darker appearance while the type II fibers are more lightly stained. In Figure 4.5b, the muscle section has been incubated at pH 4.7 prior to exposure to ATP and a phosphate-precipitating agent. The lighter fibers (in this case also the type II fibers) have had their myosin inactivated by the low pH treatment, while the darker staining fibers (type I) have stronger ATPase activity remaining. The proportions of the different fiber types are different between muscles, between different regions of the same muscle, and different between species. These divergent properties need to be kept in mind in the ingredient mixtures used in meat processing.

4.2.2 Contraction Mechanism

The primary function of muscle is locomotion. The muscle cell is packed full of the contractile myofibrils. Contraction is activated by a nerve impulse that passes from the spinal cord to a specialized region on the muscle cell termed the motor end plate. A single axon connects with several hundred muscle fibers in a group called a motor unit. At the end plate, a small amount of acetyl choline is released from the end of the axon, and this acetyl choline diffuses to the cell surface and binds to acetyl choline receptors embedded in

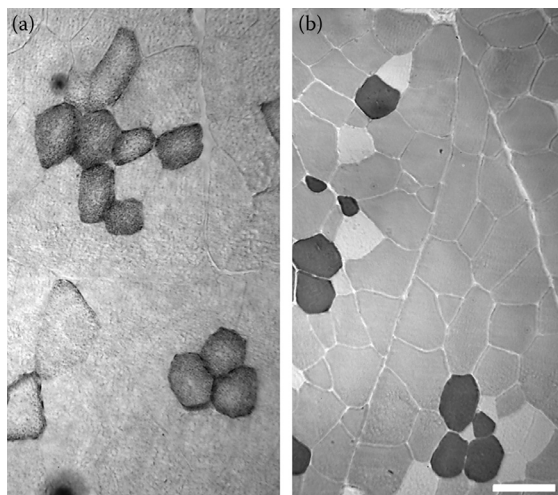


FIGURE 4.5 Light micrograph of pig muscle in cross-section. (a) Succinic dehydrogenase stained. (b) Myosin ATPase after preincubation at pH 4.7. In pig muscle, the dark type I fibers tend to be more clustered together than with muscles from other species. Scale bar = 100 μ m.

the muscle cell membrane. The receptors cause a local depolarization of the cell membrane, and this depolarization wave spreads over the surface of the muscle cell. In addition, the depolarization wave passes down special perpendicular invaginations called T tubules that penetrate to the center of the muscle cell. The T tubules are attached to a specialized intracellular membrane system termed the sarcoplasmic reticulum. The attachment involves a protein called the ryanodine receptor. (This protein is so named because of its affinity for the plant alkaloid ryanodine.) When the depolarization reaches the T tubule—sarcoplasmic reticulum junction—the ryanodine receptor opens and calcium is released into the cell cytosol. The calcium diffuses to the myofibrils and binds to troponin on the thin filaments. Calcium causes a shape change in the troponin; this in turn causes tropomyosin to move deeper into the groove of the actin. With the tropomyosin movement, a binding site for the heads of the myosin on the surface of the actin is exposed, and the myosin binds and pulls or pushes the actin a small distance (about 10 nm). The myosin head then releases and it can reattach to another actin. This sliding of the filaments requires energy that is provided by ATP. The terminal phosphate bond can store this energy and release it to do the mechanical work. The myosin head is believed to bend in its middle region, with the portion near the thick filament shaft acting as a lever arm. The calcium required for activation is pumped back inside the sarcoplasmic reticulum by an ATP powered process. The cell membrane polarity is reestablished by the sodium—potassium pump found in the outer cell membrane. It also requires ATP to move the sodium and potassium against their concentration gradients. When the calcium and ion gradients are reestablished, the muscle is relaxed. A single muscle contraction is called a twitch, and it only requires about 200 ms to complete.

4.2.3 Muscle Metabolism

An overview of the major metabolic pathways involved in muscle energy conversions is shown in Figure 4.6. The pathways illustrated all revolve around the production and utilization of ATP. The major fuels include glycogen, glucose, and fatty acids. In addition, small amounts of amino acids can be metabolized under certain conditions. Glucose and fatty acids enter muscle cells by escape from the capillaries, diffusion through the extracellular space, and active transport across the muscle cell membrane. Glycogen is a polymer of glucose that is stored in the muscle in preparation for ATP generation. Resting muscle or muscle undergoing mild activity relies primarily on fatty acid metabolism for ATP synthesis. However, intense work may require more ATP-generating capacity than can be provided by lipolysis and fatty acid transport from the blood stream.⁸ There is a small supply of triglycerides in the muscle cell that could liberate fatty acids, but this pathway does not appear to be significant for producing ATP.

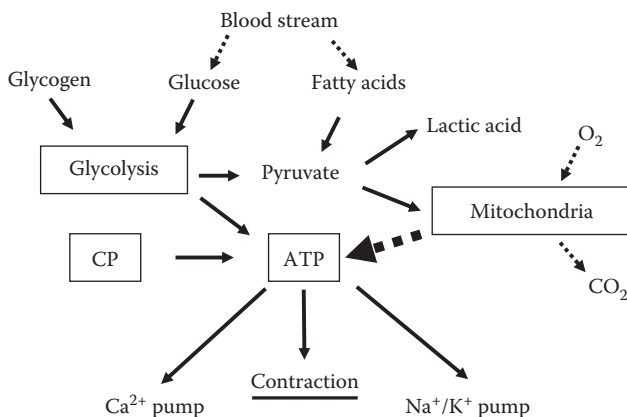


FIGURE 4.6 Overview of metabolism of muscle.

Both the glycolysis from glycogen and glucose as well as degradation of fatty acids result in the generation of pyruvate. In living muscle, pyruvate is usually transported to the mitochondria for further oxidation to carbon dioxide and water. The mitochondrial pathway yields a much larger amount of ATP than the glycolysis steps (indicated by the dense arrow in Figure 4.6). The greatest rate of ATP utilization in the muscle cell occurs during contraction. However, there is a continual need for ATP to power the calcium and sodium–potassium pumps in the cell.

An additional backup source of high-energy phosphate bonds can be provided by creatine phosphate (CP) (Figure 4.6). The enzyme creatine phosphokinase catalyzes the reaction of ADP plus CP to generate ATP plus creatine. When the ATP and C levels are high, the enzyme operates in the reverse direction to generate new CP. ATP levels are typically about 5 mM in the muscle cell, but CP levels may reach 20–30 mM in quiescent muscle. The transfer of the phosphate from CP to ATP is very rapid. Historically, there was much controversy about whether ATP really was required for muscle contraction because no decline in ATP levels could be detected during a twitch—the creatine phosphokinase reaction restores ATP to resting levels so quickly that the ATP level stays essentially constant. However, the limited quantity of CP (probably no more than fivefold greater than the ATP levels) means that it rapidly becomes depleted during short periods of intense work.

The steps in the glycolysis pathway are illustrated in Figure 4.7. The enzymes catalyzing the various steps are shown in italics. This is a remarkable pathway since the flux of metabolites can be varied nearly 100-fold with little or no change in the concentrations of the various intermediates.¹¹ Intense effort for many years has been exerted to determine what controls the rate of glycolysis.

It is now clear that stress can activate the glycolytic rate by a cascade of steps involving formation of cyclic AMP and ultimately the conversion of the relatively inactive phosphorylase *a* into the much more active phosphorylase *b* (Figure 4.8). The enzyme phosphofructokinase is also activated and inhibited by various metabolites that change in concentration between active and inactive cells. However, the key control of the glycolytic pathway is the supply of ADP and phosphate.⁶ Thus glycolysis will proceed rapidly when ATP breakdown rates are high and virtually stop when the ATP requirements are minimal.

4.3 Postmortem Changes in Muscle

4.3.1 Biochemical Alterations

4.3.1.1 Small Molecule Changes

The muscle does not cease to function at the time an animal dies. However, the metabolic functions are markedly altered. The dashed arrows in Figure 4.6 indicate the reaction pathways that become blocked within a few minutes postmortem. First, the cessation of the blood supply means that there is no longer

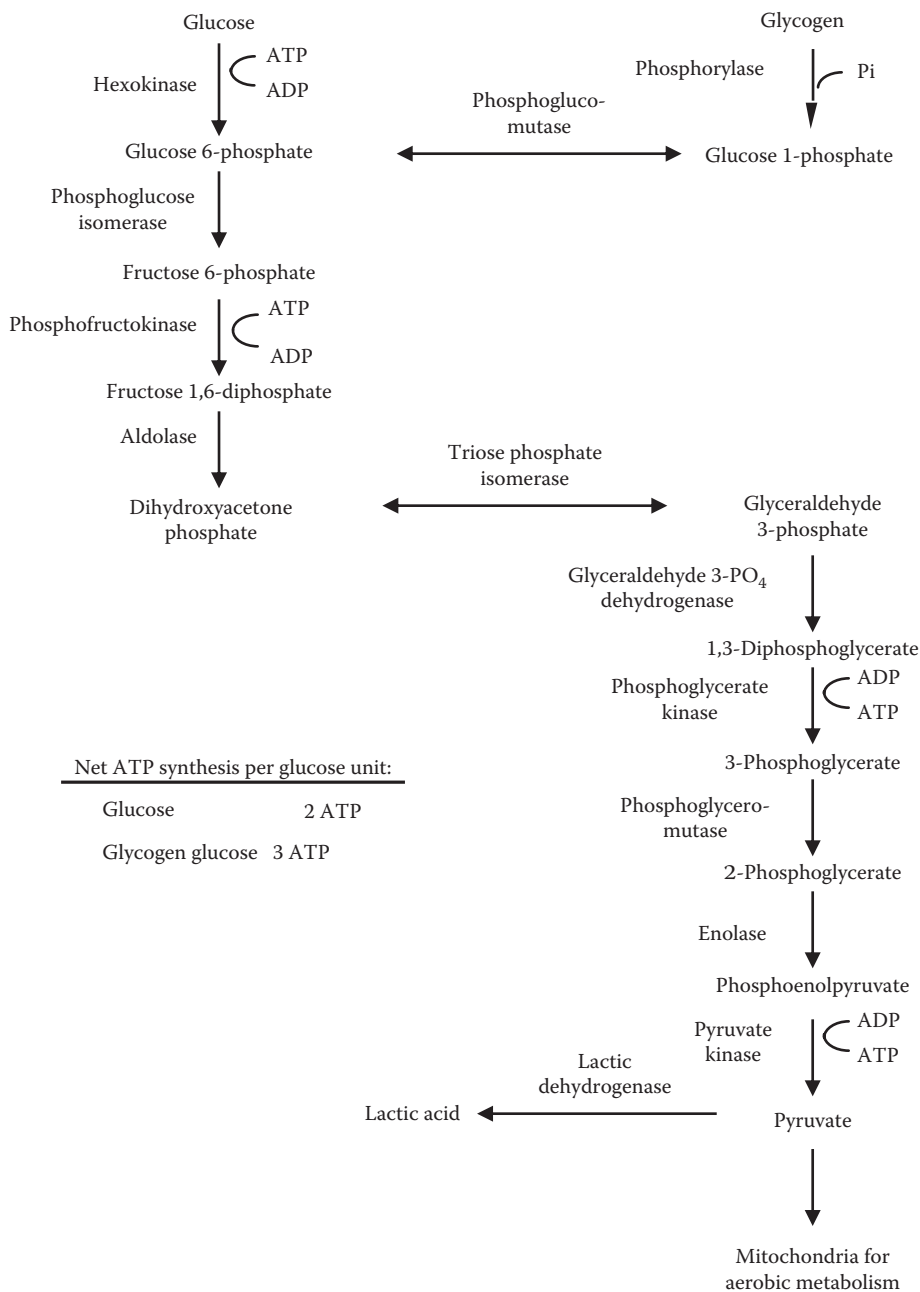


FIGURE 4.7 Glycolysis pathway.

a renewable supply of glucose or fatty acids from the blood stream. Second, the supply of oxygen is cut off. Thus, the major ATP-generating source for muscle is abolished after death. The pyruvate that is generated as an end product of glycolysis is converted into lactic acid, and, since metabolic waste products cannot be removed without a blood stream, the lactic acid accumulates in the muscle. Typically, there is a burst of muscular activity at the time of death due to trauma of the brain and spinal cord. Thus, the measured levels of creatine phosphate obtained within 5 min after death are usually depressed from those values found with biopsies or by nuclear magnetic resonance measurements on living muscle.

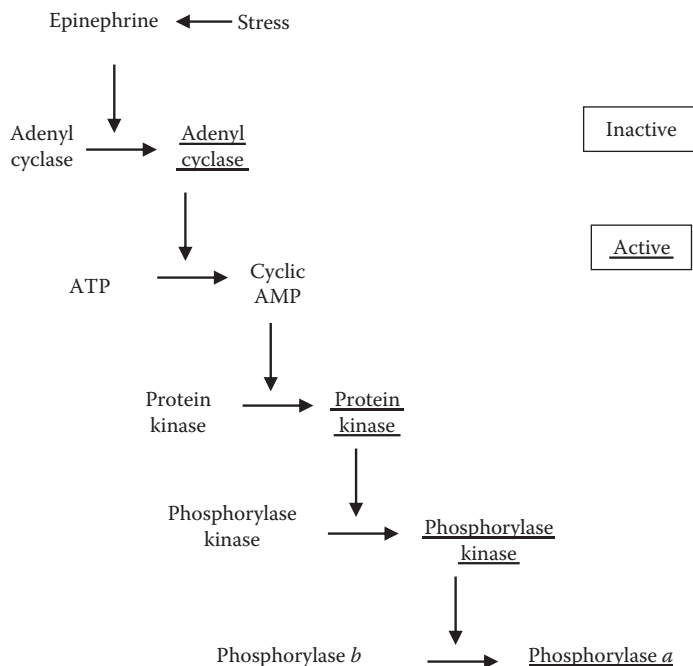


FIGURE 4.8 Stress activation of the glycolysis pathway.

Although the muscular activity rapidly subsides within a few minutes after death, the sarcoplasmic reticulum calcium pump and the cell membrane sodium–potassium pump continue to function to move their respective ions against the concentration gradients. The glycolysis pathway is the only source for this ongoing requirement of ATP. However, since there is a fixed supply of glycogen at the time of death, glycolysis can only continue for some limited time period postmortem. Usually glycolysis ceases before all the glycogen is depleted. Although the reasons for this cessation are not completely understood, possibilities include [1] the low pH that develops may inactivate one of the glycolytic enzymes and/or [2] the conversion of adenine nucleotides into inosine derivatives may halt the glycolytic flux. Some ATP is regenerated by the myokinase-catalyzed reaction of two moles of ADP to form one mole of ATP and one mole of AMP. The AMP is converted into IMP and ammonia by the enzyme AMP deaminase.

A graphical example of postmortem changes in a variety of muscle properties is shown in Figure 4.9. The time scale would be appropriate for pig muscle undergoing typical postmortem processing procedures (i.e., transfer of the dressed carcass to a 0–4°C chiller at 30 min after death). Note that the pH decline is fairly linear and is paralleled by the increase in lactic acid concentration. The latter often reaches values in excess of 100 mM. ATP concentrations remain fairly stable for the first couple of hours postmortem and then begin a linear decline. The commencement of this decline coincides with the depletion of the creatine phosphate. Creatine phosphate is usually depleted before the pH reaches 6.0. The overall shapes of the curves and relationships between the different parameters shown are consistent between different animals and species. The major differences will be in the time axis. The approximately 6 h time course shown for the pig would be more like 18–24 h in beef, 6–12 h in lamb, and <3 h for poultry. These are average times for muscles left on the carcass after slaughter. It should be noted that there is considerable variation in the cooling rate at different locations in the carcass. Thus, the surface will cool more rapidly than the deeper portions of the muscle tissue. One would therefore expect that the deeper portions of the round and chuck would reach the ultimate pH and go into rigor mortis before muscle that is closer to the surface. Likewise, in muscles susceptible to cold shortening, those regions within a couple centimeters of the surface that cool very rapidly may go into rigor earlier than muscle 5–10 cm below the surface. The size of the carcass, the amount of fat cover, the temperature of the chiller, and the air velocity will all have a profound effect on rate of postmortem metabolism and

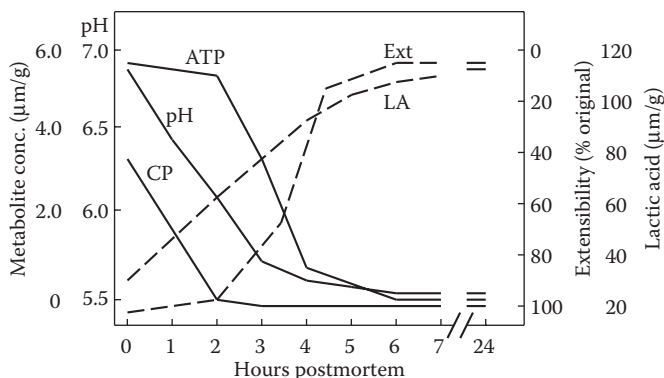


FIGURE 4.9 Chemical and physical changes in muscle postmortem. The pattern shown would be typical for pig muscle undergoing normal postmortem metabolism. Abbreviations: ATP, adenosine triphosphate; CP, creatine phosphate; LA, lactic acid; Ext, extensibility.

development of rigor mortis. In addition, the use of electrical stimulation to speed postmortem glycogen and high energy metabolite depletion will also markedly affect the time course of muscle metabolism.

4.3.1.2 Protein Changes

A number of postmortem changes in the muscle proteins have been identified. The myofibrillar proteins desmin, troponin T, titin, nebulin, and vinculin all become partially or completely degraded during the first week postmortem.^{12,16} Although the proteolytic enzymes responsible for this degradation have not been unequivocally identified, the patterns of fragments generated and the proteolytic susceptibility *in vitro* all suggest that the calpains are involved. Calpains are calcium-activated proteases originally described by Dayton and others.¹⁷ Three different isoforms of the calpains have been identified in muscle— μ -calpain (calpain 1), m-calpain (calpain 2), and P94-calpain (calpain 3). Calpain 1 requires micromolar concentrations for activity, while calpain 2 requires millimolar levels of calcium.^{18–20} Calpain 1 degrades several proteins in skeletal muscle which are related to drip loss (discussed below). The calpain 3 is slightly larger than the other two isoforms; it binds to titin but is unstable when released. The current hypothesis regarding postmortem protein degradation suggests that the calcium in the sarcoplasmic reticulum leaks into the cytosol and activates the calpains after the muscle ATP is depleted.²¹ A number of pieces of evidence support this hypothesis. First, soaking muscle strips in calcium solutions or injecting muscle with calcium results in increased proteolytic degradation. Second, animals such as the callipyge lamb that have higher muscle levels of calpastatin (the natural inhibitor of the calpains) have lower rates of postmortem protein degradation.²² Third, the proteins that are degraded by the calpains in the test tube are the same ones that are broken down in postmortem muscle. Fourth, an early decrease in pH is accompanied by an early increase in intracellular calcium concentration²³ which results in a higher calpain 1 activity.²⁴ Rapid postmortem pH decline and lower ultimate pH lead to greater activity of calpain 1 and reduced calpastatin activity.^{25–27} The calpains are maximally active near pH 7, so they would be expected to have much lower activity at the usual ultimate pH of around 5.5. The activity of calpain 1 declines rapidly postmortem; calpain 2 is more stable.¹⁶ Calpastatin activity also declines after death, so it remains unclear which of these components are most important in controlling postmortem protein degradation.

4.3.2 Physical Alterations

4.3.2.1 Rigor Mortis

The major physicochemical change that occurs in postmortem muscle is the development of rigor mortis which results in an increasing toughness of meat.²⁸ The term comes from Latin and means the stiffness

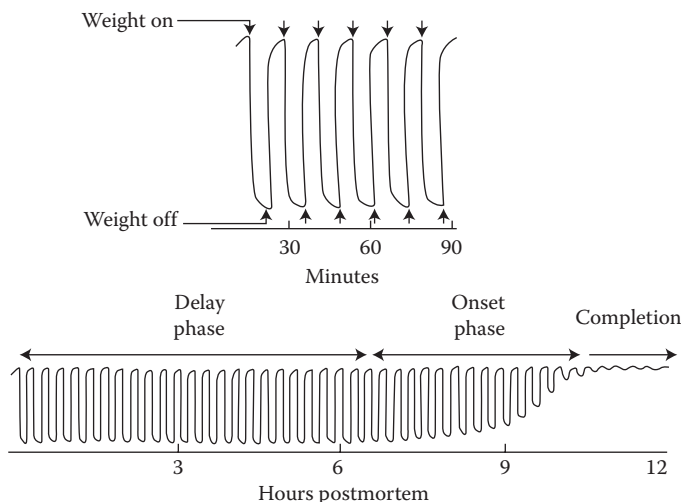


FIGURE 4.10 Rigor measurement and pattern. The upper trace demonstrates the pattern of weight addition and removal and the resulting changes in muscle length immediately postmortem. The lower trace shows the changes in extensibility from soon after death until the muscle is in full rigor. The pattern shown was obtained from a rabbit that had been anesthetized before death. (Modified from Bendall, J.R. 1973. *The Structure and Function of Muscle*, 2nd ed. Vol. II, pp. 243–309. Academic Press, New York, NY. With permission.)

of death. The time for rigor mortis development can be estimated by alternately loading and unloading a muscle strip and recording the changes over time.^{29,30} Such a trace is illustrated in the upper part of Figure 4.10. The extensibility remains relatively constant for some time postmortem (Figure 4.10 bottom); this period is called the “delay” phase.⁶ Subsequently, the extensibility declines rapidly during the “onset” phase. Finally, the muscle reaches a stage where there is no further decline in extensibility, and this is referred to as the “completion” of rigor.

The time course of rigor mortis is linked to metabolite changes in the muscle. The completion of rigor corresponds to the point where the ATP has been depleted (Figure 4.9). The onset period appears to start when the ATP levels begin to decline (note that the time courses shown in Figures 4.9 and 4.10 are not for the same species or muscle). The loss of extensibility is due to the firm attachment of the myosin heads to actin. In the normal contraction cycle, ATP is required to dissociate these two proteins and allow the filaments to slide over one another. In postrigor muscle, the supply of ATP is depleted leading to strong actin–myosin bonding so muscle shortening or extension is no longer allowed. The process is somewhat complicated by the fact that not all fibers in a muscle strip deplete their ATP at the same time because of biological variation and the difference in fiber types.

The pattern shown in Figure 4.10 (bottom) was produced from an animal that had been anesthetized at the time of death. Animals slaughtered under commercial conditions will typically have a much shorter delay phase or even the absence of delay. The time course of rigor mortis is extended two- to threefold at 10°C versus 37°C.³¹

4.3.2.2 Shortening

Unrestrained muscle may also shorten as it goes into rigor mortis. This shortening can be as high as 25–30% of the initial length when the muscle is maintained at 37°C.³² Minimum shortening occurs at temperatures near 15°C. Similarly if muscle is held isotonic, it will develop force during the onset of rigor mortis. The force developed is much weaker than that produced by living muscle during contraction (estimated to be <5%),⁶ but is sufficient to significantly change the sarcomere length and the tenderness properties of muscles still attached to the carcass. Force begins to develop at the beginning of the onset phase of rigor.³³ This force may be developing due to a rise in cytoplasmic calcium as

the ATP nears depletion or because of the activation of contraction due to binding of myosin heads to actin.³⁴

4.3.3 Unusual Patterns of Postmortem Metabolism

4.3.3.1 Thaw Rigor

The term “thaw rigor” is somewhat of a misnomer. The name refers to the shortening that occurs when muscle is rapidly frozen prerigor and then subsequently thawed. Muscle that has been treated in this way shortens markedly (as much as 70–80%) and loses large amounts of liquid (more than 25% of the initial weight) as drip.³⁵ The mechanism of this phenomenon is believed to be the disruption of the sarcoplasmic reticulum due to ice crystal formation followed by the release of calcium upon thawing.³⁶ The calcium then activates contraction since there is ATP remaining in the region of the myofibrils. The degree of shortening depends largely on the size of the muscle piece frozen and thawed. As the surface regions thaw, the inner parts provide a rigid scaffold to prevent shortening. The ATP is used up rapidly after thawing, so the outer portions will go into rigor before the core regions are thawed. Thus the degree of shortening may be minimal on a large muscle piece that has been frozen and thawed prerigor.

4.3.3.2 Cold Shortening

The usual dependence of postmortem metabolism rate on temperature does not hold in certain cases. The muscles that rely primarily on oxidative metabolism may undergo a slow but significant shortening if excised and held at temperatures below 10°C. Locker and Hagyard³² termed this phenomenon “cold shortening.” The muscle length can decline as much as 50% in unrestrained muscle.³⁷ Cold shortening is a slow process; the time course may be minutes to an hour and depends on the cooling rate. The shortening occurs before there is any reduction in muscle ATP levels.³⁸ This shortening can also occur on the carcass, particularly with muscles not placed under stretch when suspended from the rail.³⁷ However, the force is weak; Bendall⁶ estimates that it is only about 2–3% of the maximum produced in a muscle tetanus. Cold shortening is believed to be due to a gradual rise in the cytosolic calcium level by release from either mitochondria or the sarcoplasmic reticulum (the calcium pump operates more slowly at low temperature). The slightly elevated calcium causes a weak contractile response and the muscle shortens. The weak activation of the myofibrillar ATPase activity also increases the glycolysis rate compared to that at 15°C.⁶ Cold shortening is usually only a significant problem in beef and lamb. The rapid cooling that occurs with high efficiency chillers exacerbates the problem. Since cold shortening leads to an increase in meat toughness,³⁷ improvements in processing efficiency and food safety are offset by deleterious effects on meat quality.

4.3.3.3 Pale Soft Exudative Condition

An unusual postmortem phenomenon (first described in pigs by Ludvigsen³⁹) is one in which the muscle becomes pale in color, develops a soft texture, and exudes large amounts of liquid (Figure 4.11). The postmortem metabolic rate is vastly increased, with ATP depletion, completion of rigor mortis, and pH values as low as 5.3 occurring within 10–15 min after death.⁴⁰ The low pH that develops while the muscle temperature is still high leads to a denaturation of some of the muscle proteins, notably myosin. This reduces the water-holding activity of the muscle and results in excess drip loss. Currently, the mechanisms controlling water-holding capacity are not fully understood, but the structural organization of the muscle does influence the water distribution. More drip loss appears with short sarcomeres, and a higher proportion of water is held in the I-band compared with A-band of myofibrils as determined by nuclear magnetic resonance (NMR).^{26,41–43} Degradation of some proteins such as the intermediate filament protein desmin^{44,45,25,46–48} and costameric proteins vinculin and talin^{48,49} improve water-holding capacity by disturbing the formation of drip channel between the myofibril and the sarcolemma.^{26,50} However, the degradation of the protein integrin increases the water loss by the formation of drip channels in pork.^{48,51}

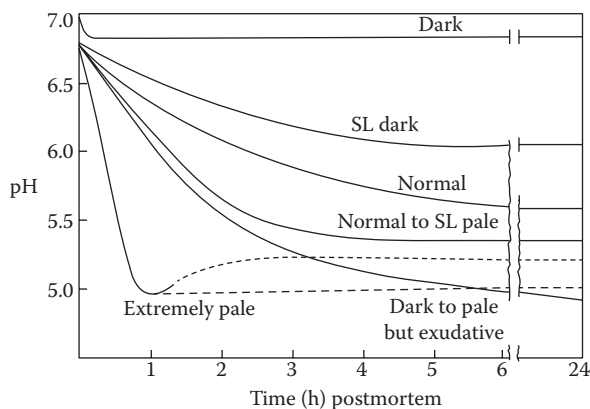


FIGURE 4.11 Variation in postmortem pH decline in pigs. The pH decline pattern is closely related to the resulting muscle color. (Modified from Briskey, E.J. 1964. *Adv. Food Res.* 13:89–168. With permission.)

The rapid postmortem glycolysis is often caused by a mutation in the ryanodine receptor (the calcium release channel, RyR) at the interface between the sarcoplasmic reticulum and T tubules.⁵² The mutation of the RyRs arginine 615 to a cysteine results in the protein-leaking calcium even when there is no nerve signal for contraction. The elevated cytosolic calcium postmortem causes activation of the myofibrillar ATPase and the resulting acceleration of glycolysis. Attempts to remove animals carrying the mutation are in progress using genetic tests, but not all animals that develop the PSE condition can be identified by this procedure. It may be that there are other mutations in the RyR that cannot be picked up by the current test. Such is the case with RyR mutations in humans where over 30 different mutation sites have now been described.^{53,54}

Attempts to change animal handling and postmortem processing procedures to reduce the incidence and severity of PSE development have been partially successful. The incidence of the PSE condition in the United States was approximately 15% for many years, but this has dropped to below 5% in more recent surveys. Gentle handling and allowing pigs a rest period before slaughter reduces the incidence of PSE.⁵⁵ Rapid cooling postmortem, such as briefly immersing a carcass in liquid nitrogen,⁵⁶ markedly reduces the PSE problem; however, this procedure has not been adopted commercially. PSE has been prevented by early postmortem injection of sodium bicarbonate.⁵⁷ The bicarbonate appears to slow the rate of pH decline as well as elevate the ultimate muscle pH.

It is now clear that a condition related to PSE also occurs in chicken and turkey muscle.^{54,58,59} The biochemical characteristics of PSE turkey are strikingly similar to those of PSE pork. With some turkeys the ATP and postmortem pH decline are accelerated, and the resulting muscle has a reduced cooked yield.⁶⁰ There is also a precipitation of phosphorylase and a reduction in the myosin solubility in muscles that have undergone this rapid postmortem glycolysis.⁶⁰ The incidence is greater as the result of stress from hot weather, transportation, and struggling of birds.^{61–63} Due to the obvious genetic differences between mammalian and avian species, Smith and Northcutt⁶⁴ suggested that the terms “pale chicken muscle” or “pale poultry muscle syndrome” as replacements for the use of PSE to describe pale poultry meat. No point mutation has been found in poultry RyR, but recent reports showed that alternative splicing in turkey RyR could lead to PSE development.^{54,65,66}

4.3.3.4 Dark, Firm, Dry Condition

The DFD condition in pigs results from the same ryanodine receptor mutation that causes PSE. In this case, the glycogen has been largely depleted before death and lactic acid therefore does not accumulate in the muscle. The time to rigor mortis completion is very short and the resulting ultimate pH is high (it may be >6.5). The meat is dark in color and has a firm texture. The surface is dry and sticky to the touch. Such meat has excellent properties for use in processed meat products because of its high water-binding activity.

It's unusually dark color may deter purchase at the retail meat case. However, since it only occurs in animals that may alternatively become PSE, conditions that reduce its incidence are usually sought.

4.3.3.5 Dark Cutter

The dark cutter condition occurs in beef muscle having a high ultimate pH (see Tarrant⁶⁷ for review). Like the DFD condition in pigs, dark cutters result from the antemortem depletion of glycogen. However, this depletion is not due to a genetic condition but rather appears to result from a response to stress.⁶⁸ The incidence of the dark cutting is often high in bulls (as much as 15%) since they tend to fight. Dark cutting can be reduced by avoiding mixing unfamiliar animals prior to slaughter. Meat from dark cutters may cause difficulty in retailing, but it is a superior product for use in processing because of its high pH and resulting improved water-holding ability.

4.3.3.6 RN Phenotype in Pigs

A genetic condition in some pigs results in a different form of postmortem metabolism.⁶⁹ The RN condition is common in the Hampshire breed of pigs. The "RN" notation comes from the French "Rendement Napole" meaning "Yield Napole," a method for evaluating ham-processing yield. Certain pigs have a lower ultimate pH (typically about 5.3–5.4 versus the normal found in pigs of 5.5–5.6 in the Longissimus dorsi), and this results in a slightly paler fresh meat color and less water binding after processing. It is characterized by a significant increase (~70%) in the glycogen levels in the muscle of the live animal. The rate of postmortem pH decline is normal. Meat from pigs carrying the RN allele often is more tender than that from noncarriers.⁷⁰ Animals with this condition can be determined using estimates of the "glycolytic potential" (GP).⁷¹ The GP equals $2 ([\text{glycogen glucose units}] + [\text{glucose}] + [\text{glucose-6-phosphate}]) + [\text{lactate}]$. Since the glycolytic potential is the sum of glycogen and its major postmortem breakdown products, it should give similar results whether determined on biopsy samples or from muscle at any time point postmortem.

The first RN gene description localized the mutation to the protein kinase AMP-activated $\gamma 3$ subunit gene (PRKAG3) where separate Arg200 $\rightarrow$ Gln and Thr30 $\rightarrow$ Asn changes were found, the former being identified as the causative alteration.⁷² Subsequent work identified additional mutations in the same gene (Val199 $\rightarrow$ Ile and Gly52 $\rightarrow$ Ser).^{73,74} In contrast to the RN Gln200 mutation, the Ile199 mutation was associated with low glycogen content, low lean meat percentage, and high ultimate pH.^{73,74} It appears that the higher glycogen content in RN pigs is not due to a defect in glycogen degradation, but instead is caused by a higher glucose uptake.⁷⁴

4.4 Summary

Muscle tissue is complex structurally and is nonhomogeneous in its protein composition and metabolic emphasis. The rate and extent of metabolic changes postmortem have important effects on the color of meat, its texture, and its usefulness for inclusion in processed meat products. In addition, there are profound differences in the rates of postmortem metabolism between muscles from different species. For example, electrical stimulation may be beneficial to speed postmortem metabolism in beef and therefore prevent cold shortening, but acceleration of metabolism in pigs may lead to greater incidence of pale, soft, exudative meat. Thus an understanding of the biochemical and physical processes that muscle undergoes postmortem is necessary for optimum design of slaughter and meat-processing procedures and the use of this tissue for food.

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